

Become Data Driven with IBM Z Infused Data Fabric

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 **Analytics**

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Become Data Driven with IBM Z Infused Data Fabric

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Note: Before using this information and the product it supports, read the information in “Notices” on page vii.

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This edition applies to the data fabric capabilities that are enabled by the IBM zSystems family of mainframes, including z14, z15, and z16, unless otherwise noted.

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Preface

The expansion of digital transformation initiatives resulted in the exponential rise of data generation. It is now apparent that for organizations to meet their rapidly changing business needs, they must use their most current data to fuel their insights.

However, how to do so efficiently and effectively is still a challenge for most organizations that are trying to solve today's complex business problems. Contributing to this challenge is that organizations are turning to a hybrid cloud approach of their IT infrastructure, scaling to include on-premises and cloud environments, in response to the data growth. This approach results in the requirement to integrate various data formats from disparate data sources into a central and unified view for use in analytics and AI, with security and privacy as a top priority.

As a solution, organizations are adopting the emerging *data fabric* architectural approach. This approach enables efficient data integration across environments, simplifies the delivery of disparate data throughout an organization, and enables self-service data consumption and governance.

Organizations with investments in the IBM Z® platform have a unique opportunity to integrate them into their data fabric strategy. A data fabric strategy makes mission-critical data that is on Z readily available for real-time decision making by using data where it originates, which achieves faster time-to-insight.

This IBM® Redbooks® publication introduces the emerging concept of a data fabric and how integration of valuable data that is stored on IBM Z enables deeper insights and improved AI capabilities.

The intended audience for this publication includes data science professionals, such as Data Stewards, who want to develop and implement practices for building out a data governance framework that is facilitated by data fabric for AI workloads on IBM Z. The audience also can include anyone who wants to use the rich capabilities of the IBM Z environment by using a data fabric that is infused with their business-critical data and applications that are on the platform.

In this IBM Redpaper, we explore the following key aspects and considerations when looking to use a secure data fabric strategy that uses IBM Z data and platform technology and capabilities:

- ▶ Chapter 1, “The challenge of digital transformation” on page 1 discusses the concepts, technologies, and components that comprise a data fabric.
- ▶ Chapter 2, “Using IBM Z strengths in a Data Fabric” on page 15 explores the key advantages in incorporating IBM Z with your data fabric strategy, and recommendations for unlocking those opportunities.
- ▶ Chapter 3, “Securely connect a Data Fabric to any data source, anywhere” on page 31 discusses secure practices for establishing connections between your data fabric and various data sources and enabling tools and technology, such as IBM Cloud® Pak for Data. Also discussed are considerations and risks for moving data and alternative approaches to data access.
- ▶ Chapter 4, “Data governance and discovery” on page 51 discusses the advantages and importance of incorporating a knowledge catalog and data discovery capabilities into your data fabric for better understanding and trust in the data. The focus is on these components in relation to IBM Z source data.

- ▶ Chapter 5, “Security and compliance with automated data governance” on page 67 discusses considerations for security and compliance across a data fabric, including recommended practices around governance and enabling tools for unified data governance.
- ▶ Chapter 6, “Consuming data from a Data Fabric” on page 85 describes various methods to make data available for use by business users and applications from your data fabric and provides recommended practices on data consumption implementations with considerations for use cases, people, and applications.

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The challenge of digital transformation

This chapter discusses the growing need for an efficient and integrated Data Fabric solution to mitigate the data complexity challenges that were introduced by the exponential growth of big data from digital transformation innovations.

A Data Fabric enables more holistic, data-centric decision making by simplifying data access and facilitating self-service data consumption within an enterprise through the use of intelligent and automated systems. A Data Fabric that integrates IBM Z allows businesses to deliver trusted and governed mission-critical data on the Z platform to the right users at the right time for business-ready AI.

This chapter includes the following topics:

- ▶ 1.1, “Introduction” on page 2
- ▶ 1.2, “What is a Data Fabric?” on page 2
- ▶ 1.3, “Business benefits of a Data Fabric” on page 3
- ▶ 1.4, “Data Fabric capabilities” on page 4
- ▶ 1.5, “Who uses a Data Fabric?” on page 5
- ▶ 1.6, “Data Fabric technical and functional architecture” on page 6
- ▶ 1.7, “Data fabric versus data mesh” on page 11
- ▶ 1.8, “Role of a Data Fabric in AI computing” on page 12
- ▶ 1.9, “Facilitating a Data Fabric by using IBM Cloud Pak for Data” on page 13

1.1 Introduction

Organizations are seeing their digital transformation accelerated by the advancement of hybrid cloud architectures. However, with this newfound agility and opportunity for business differentiation comes challenges in the form of data integration.

Over the last decade, developments within hybrid cloud, Artificial Intelligence (AI), the Internet-of-Things (IoT), and edge computing led to an exponential growth of big data, which created even more complexity for enterprises to manage. This issue has made the unification and governance of data assets and data environments a top priority because this growth led to significant challenges, such as siloed, complex data, an increase in security risks, and general bottlenecks to decision making.

As a solution, companies are turning to a Data Fabric approach to address most of these complex challenges. They use a Data Fabric to connect various, separate data systems, embed governance, strengthen security, and provide greater data access to business users.

As the Data Fabric approach grows in popularity, serving as an ideal architecture to address data complexity challenges, so does the lack of understanding of what is meant by the term. Throughout this IBM Redpaper, we use the term *Data Fabric* to mean an architectural approach for simplifying data access in an organization. This simplification effort is done through intelligent and automated integration of various data pipelines and data management capabilities, and results in a model of self-service data consumption.

A Data Fabric approach helps enterprises elevate the value of their enterprise data by providing the secret formula of getting the right data to the right people at the right time, regardless of where it is stored.

1.2 What is a Data Fabric?

A Data Fabric is an architectural pattern for managing highly distributed and disparate data. Because it is designed for hybrid and multi-cloud data environments, a Data Fabric supports the decoupling of data storage, data processing, and data use. It also enables the integration of data from various sources, such as mainframes, data centers, multiple clouds, and edge environments, which allows your enterprise to use all available data and gain competitive business advantages.

With the intelligent knowledge catalog capabilities support a Data Fabric architecture, you can elevate data into enterprise assets that are governed globally, regardless of where the data is stored, processed, or used.

Catalog assets are automatically assigned metadata that describes logical connections between data sources and enriches them with semantics (data quality analysis, profiling, classification, governance policy, and so on) so that you can provide easy to understand, business-ready data that is accessible across the enterprise for your applications, services, and users.

A Data Fabric provides a unified, consistent, and flexible view of enterprise data to business users when they need it, irrespective of the location of that data. Although this flexibility offers great benefits, it is also important to note that some applications and use cases place more stringent requirements on data location and organization that might require specific design considerations.

A Data Fabric is independent of deployment platforms, location, architectural design, or data processes. The objective is to simplify data complexity through automating data integration, governance, and consumption, which provides the ability to generate analytic insight from real-time data, at the transactional source, without moving that data off platform.

1.3 Business benefits of a Data Fabric

A key challenge that is faced by many companies is how to effectively use large volumes of data across many disparate data sources, when business users need it, to deliver real-time insights. A correctly designed, deployed, and implemented Data Fabric can help address these data complexity challenges and enable a data-driven enterprise strategy.

Generally, the following concerns are addressed by a Data Fabric architecture and associated business benefits:

- ▶ Self-service marketplace for easy data consumption and collaboration:
 - Provides a single point of access (marketplace) for users and applications to find, collaborate, and access high-quality data throughout the enterprise.
 - Enables improved user understanding of the meaning of data, its origin or lineage, and relationships to other data assets through centralized data governance and lineage.
 - Provides customizable metadata management capabilities that are easily scalable and accessible by way of modern APIs.
 - Enables enhanced collaboration through self-service access to trusted and governed data.
- ▶ Unified data governance to achieve a high level of data protection and compliance:
 - Quickly apply industry-specific governance rules with AI and automation, which enables automatic policy enforcement for any data access, ethical use of data where it resides, and improvements in agility, security, and productivity for data science professionals and business analysts.
 - Reduces compliance and regulatory risks by providing trust and transparency through an active governance layer for all data initiatives.
 - Multiple distributed data sources appear as one data entity for unified visibility.
 - Improvements in data discovery capabilities to help identify personally identifiable information (PII) and other critical data elements with massive scaling possibilities.
- ▶ Automating data engineering tasks and augmenting data integration:
 - Accelerates data delivery with optimized data integration processes and pipelines.
 - Reduces the amount of time that is spent on cleaning and preparing data, and coding data delivery and access processes.
 - Deliver quality data for business processes through the automation of real-time change capture processes.
 - Machine learning capabilities automate and extend custom data discovery, classification, and curation processes, which results in faster time-to-value.
 - Enables automation of real-time continuous analysis at the source of data, which supports generating real-time insights.

Organizations can also unlock greater value from their AI applications through a well-designed and managed Data Fabric, including the following examples:

- ▶ Enables businesses to harness AI for business and operational efficiency by accessing current enterprise data and embedding AI capabilities into production applications (within the context of a transaction) to enrich business-critical initiatives. AI functionals also can be embedded into operational processes, which results in optimization and IT efficiency improvements in middleware and application performance.
- ▶ Delivers trustworthy AI through automated management, monitoring, retraining, and redeploying AI models across the entire AI life cycle when model bias, drift, or degradation is observed. As a result, model accuracy, transparency, and trust are improved.
- ▶ Provides the ability to incorporate IBM Z originating data in analytics to generate analytic insight from real-time data, at the transactional source and without moving the data off platform, for more accurate and timely AI insights.

However, a Data Fabric that spans multiple data sources, such as cloud, distributed, and mainframe environments, is not a common characteristic of many of today's production environments. Thus, solution design and architecture matter.

Computation is more economical and performant when run near the data and moving computation is less expensive than moving data and poses less security and privacy risks. It ensures a solution design that supports data accuracy. Low latency access and use of data are critical design points for effective use in selected AI use cases. Elements of these key factors are discussed throughout this publication.

1.4 Data Fabric capabilities

A Data Fabric combines the power of emergent technologies with established data management concepts to enable dynamic and intelligent data orchestration across a hybrid data landscape. A network of instantly available information is created to power business insights.

It also provides the components that are necessary to facilitate the use of data as an enterprise asset and address the needs that arise from extreme diversity and distribution of data assets. Generally, a Data Fabric architecture can be logically partitioned into the following components:

- ▶ Knowledge, insights, and semantics

This component includes the process of enriching data assets with knowledge and semantics, such as business vocabulary, and performing analysis to extract insight for more automation. These capabilities provide a common business understanding of the data. It also supports collaboration and self-service data consumption for users to find, collaborate, and access high-quality data and other assets from across the data landscape of the enterprise.

- ▶ Unified governance and compliance

This component enables automated data governance, protection, and security tasks by using active metadata and automated data asset classification and curation.

- ▶ Intelligent integration

Data integration is augmented across distributed data sources to maximize performance while minimizing storage and costs, provide optimized workload distribution, self-tuning, and correction of schema deviations, and accelerate data engineer tasks with automation.

- **Orchestration and lifecycle**

Enables a consistent data experience that spans the composition, testing, operation, and monitoring of data pipelines through end-to-end lifecycle management. It is infused with AI capabilities to automate tasks, self-tune, self-heal, and detect source data changes.

1.5 Who uses a Data Fabric?

The following key roles, personas, and responsibilities work together to use a Data Fabric architecture to provide governed, trusted, complete, and analytics-ready data to users and applications that require it:

- **Chief Data Officer**

The responsibilities of this role can include defining the enterprise data strategy and enabling easy access to data, which facilitates analytics-driven decision making. The CDO plays an integral role in understanding how a Data Fabric strategy can be used to optimize the enterprise data strategy and communicating its importance throughout the enterprise.

- **Data Steward**

This data management role is tasked with maintaining and implementing the data governance policies agreed upon by the company.

In the context of implementing a Data Fabric strategy, a data steward's responsibilities can include the following examples:

- Data enrichment with metadata
- Data curation into catalogs
- Creating and assigning custom governance artifacts
- Creating data protection rules to automatically enforce uniform data privacy
- Using ready to use collections of metadata for compliance and enrichment

Data stewards play an important role in achieving a metadata-based knowledge core and unified data governance, security, and compliance in a Data Fabric.

- **Data Engineer**

Data engineers build systems that collect, manage, and convert data into usable information for data scientists and business analysts to interpret.

In the context of implementing a Data Fabric strategy, a data engineer's responsibilities can include the following examples:

- Preparing data for consumption
- Providing access to enterprise data sources
- Automating data preparation tasks, such as bulk ingestion, cleansing, complex transformations, and virtualization of data to keep data assets up to date
- Enabling data processing where the data originates

These engineers play a key role in the automated intelligent integration of data in a Data Fabric by converting raw data into usable information for businesses to interpret.

- **Data Scientist**

The data scientist role combines skills from three disciplines of computer science, statistics, and mathematics to understand business goals and determine how data can be used to achieve those goals.

In the context of implementing a Data Fabric strategy, a data scientist’s responsibilities can include the following examples:

- Searching data catalogs across the entire data landscape to find the data that they need
- Copying data assets from catalogs to analytics projects
- Collaborating with other data science professionals to prepare, analyze, and model the data

Data scientists use this self-service data consumption model. They create valuable analytics assets, such as predictive models and notebooks that can be cataloged and governed the same as any other data asset in the enterprise by using a Data Fabric architecture.

By bringing together your data stewards, data engineers, and data scientists by way of a Data Fabric strategy, you are better positioned to equip your enterprise with data that is ready for analysis.

1.6 Data Fabric technical and functional architecture

Data delivers business value only when it is contextualized and accessible to the correct users or applications in the organization. Therefore, the core of the Data Fabric architecture is a data management platform that enables the full breadth of integrated data management capabilities, including discovery, governance, curation, and orchestration.

When implemented correctly, a Data Fabric helps ensure that those values are available throughout the organization in the most efficient and automated way possible.

Figure 1-1 shows the underlying capabilities that can be used in a Data Fabric to enable certain business benefits. It also shows the hybrid multi-cloud nature of a Data Fabric architecture to help deliver the business-ready data that users and applications demand.

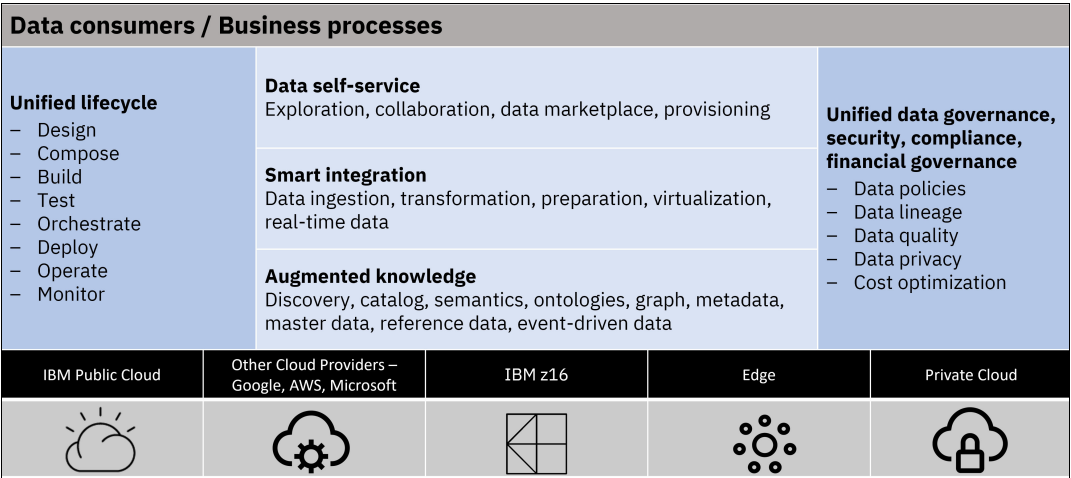


Figure 1-1 Capabilities and business benefits of a Data Fabric implementation

A Data Fabric enables residency of data in a wide range of platforms, including (but not limited to) on-premises systems of record environments, hybrid cloud environments, data lakes, data warehouses, and SQL databases.

A Data Fabric architecture abstracts away the technological complexities that are engaged for data movement, transformation, and integration. All data available is across the enterprise, and the infusion of AI throughout your business is accelerated.

1.6.1 Integrating IBM Z with your Data Fabric architecture

Some of your organization's most important data likely originates on IBM Z, so it makes sense to want to integrate IBM Z and its data into your Data Fabric architecture. By incorporating IBM Z into your Data Fabric, you unlock the unique advantage of querying data at its source and deliver real-time insight with little to no latency for real-time business decisions.

Such real-time insights are not achievable without this capability of querying data at its source. Also, a Data Fabric that uses IBM Z can deliver in-transaction insights by incorporating AI models directly into modern business and analytics applications. You can help protect your organization's sensitive data and workloads by keeping them in the security-rich environment of Z. You also can use automated policy enforcement capabilities through the Data Fabric to govern access, which enables secure self-service data consumption throughout the enterprise.

IBM Z with IBM Cloud Pak for Data

A Data Fabric can use IBM Z with IBM Cloud Pak® for Data (CP4D) to facilitate the transformation of becoming a data-driven enterprise. Combining the unique capabilities of IBM Z and CP4D creates a highly flexible and performant Data Fabric environment. It facilitates a design and processing ability that uses the benefits of performing analytics where the data originates. As a result, the cost, complexity, and risk of moving data between platforms are reduced. This approach is referred to as a *data gravity approach*, which is discussed in 2.2.1, "Use the data gravity of the Z platform" on page 17.

This approach can enable the following benefits:

- ▶ Lower costs of infrastructure, software, and people
- ▶ Simplified infrastructure with increased resiliency by using transactional system-level performance and availability
- ▶ Low to no latency and higher throughput by bringing AI and analytics where data originates
- ▶ Stronger security for in-place analytics and AI on Z with 100% encryption of all data
- ▶ Improved data governance and reduced risk through gaining insights without moving data between platforms
- ▶ Improved analysis and AI effectiveness with access to real-time data and insights

Figure 1-2 shows how IBM Z can play an essential role in your Data Fabric to deliver the valuable enterprise data needed for business-ready AI, and the high performance, resilient platform that is required to deliver actionable, real-time insight at the point of a transaction. A Data Fabric that is integrated with IBM Z is accomplished by using CP4D capabilities and services that serve as key connectors to IBM Z and its originating data.

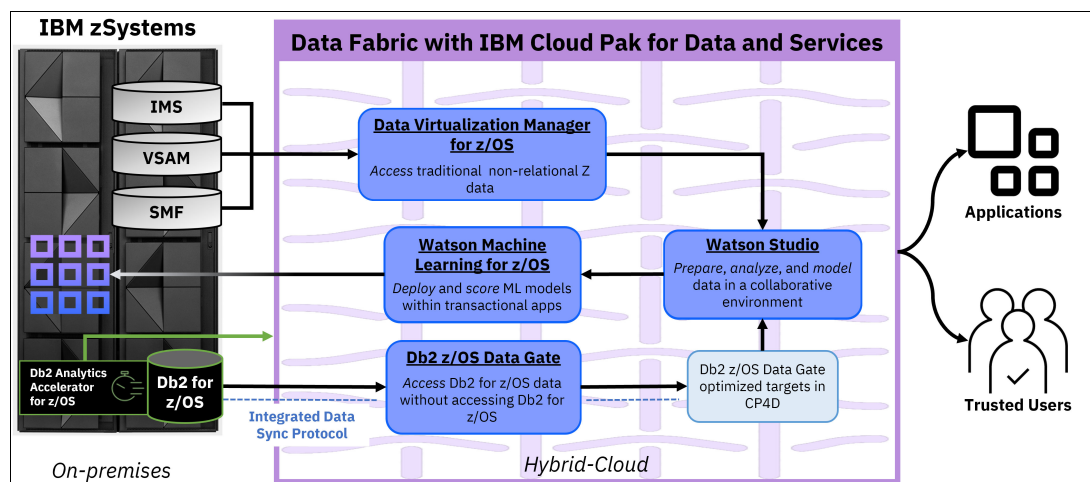


Figure 1-2 IBM Z-integrated Data Fabric architecture at a high level

Making this concept of a Data Fabric with IBM Z possible, CP4D provides many of the technology and tools to implement the core components of a Data Fabric. For example, CP4D enables a hybrid cloud architecture for data service modernization that uses self-service, virtualization, and unified data governance and protection.

CP4D provides data virtualization technology through the IBM Data Virtualization Manager for z/OS (DVM), which makes traditional nonrelational Z data accessible as a part of the intelligent integration component of a Data Fabric. It also provides a platform to facilitate the seamless integration of disparate enterprise data across a hybrid cloud landscape with an optimal balance of cost, performance, and intelligently optimized data processing (see 2.2.2, “Intelligent integration of IBM Z data” on page 18).

It also provides a suite of IBM and third-party services that span the data lifecycle, extends its functions, and supports your journey to AI.

Generating real-time data insights

With IBM Db2 for z/OS databases being home to massive volumes of systems of record data, it is critical that insight that is generated from its data is made available to the Data Fabric in real time.

The tightly integrated [IBM Db2 Analytics Accelerator for z/OS \(IDAA\)](#) and Db2 for z/OS provide the ability to process real-time analytics on live transactional data without affecting transactional workloads and without requiring any application changes or moving data off platform.

This product pairing also enables the processing acceleration of complex Db2 queries that are run by CP4D to support business-critical reporting and analytic workloads. It brings the processing to the data, which turns the Z platform into a hybrid transaction and analytic processing (HTAP) environment. This process is done without performance degradation because IDAA provides the extra power to support complex, resource-intensive queries against Db2.

For more information about IDAA and its capabilities, see 2.3.1, “Hybrid transaction and analytic processing on IBM Z with IBM Db2 Analytics Accelerator for z/OS” on page 22.

Accessing a synchronized copy of Db2 for z/OS

To gain even more benefits from your Db2 for z/OS subsystems and data, you can use the capabilities of CP4D combined with [IBM Db2 Data Gate](#) to quickly access a synchronized copy of Db2 for z/OS data by a Data Fabric. This pairing incorporates integrated synchronization protocols that maintain data currency and allows synchronized copies of Db2 for z/OS data to be hosted within any supported CP4D environment. In turn, it simplifies the delivery of read-only access to Db2 for z/OS data from the cloud without accessing Db2 for z/OS directly, which enables a hybrid cloud approach.

For more information about integrating IBM Z source data by using Db2 Data Gate, see 2.3.2, “Off-platform access to Db2 for z/OS data with IBM Db2 for z/OS Data Gate” on page 23.

For more information about the technical functions of specific Db2 Data Gate capabilities, see 3.2.4, “Tools for data movement and replication from IBM Z” on page 45.

Data virtualization technology in IBM Data Virtualization Manager for z/OS

A well-designed Data Fabric needs access to *all* data, even the seemingly difficult to access nonrelational data sources on Z. Therefore, when it comes to the integration of traditional nonrelational IBM Z data sources, such as Virtual Storage Access Method (VSAM) and Information Management System (IMS) into your Data Fabric, you can use the data virtualization technology in [IBM Data Virtualization Manager for z/OS](#) (DVM).

DVM simplifies access to these complex data sources through a structured query language (SQL) window, which enables scalable access to accurate, real-time data at its source. The simplified data access that is provided by DVM enables access to current, business-critical data by modern cloud applications by way of a Data Fabric, without the need to move data off-platform and no specialized database skills are required.

Also, DVM provides resource optimization, speeds up high-volume data retrieval, and automatically redirects processing of up to 99% of workloads to lower-cost IBM Z Integrated Information processors (zIIP) specialty engines.

For more information about data virtualization technology on IBM Z, see 2.3.3, “Integrating data from multiple sources with Data Virtualization for z/OS” on page 23.

Furthermore, enterprises that want to use a Data Fabric for the development and deployment of machine learning (ML) models can take advantage of IBM provided analytics and AI services in CP4D, such as [IBM Watson® Studio](#) and [IBM Watson Machine Learning for z/OS](#) (WMLz).

With IBM Watson® Studio, you can prepare and analyze data, and build, run, and manage AI models in a collaborative environment to solve business problems. Together with CP4D, Watson Studio provides the flexibility to build custom AI models, where your data is stored, and deploy them on any cloud to readily infuse into business processes and applications to optimize decisions anywhere on the unified data and AI platform that is provided by CP4D.

Regardless of the training platform and framework that is used to train them, machine learning models then can be readily deployed into production at scale with WMLz.

By collocating AI and core high-volume business applications that are running on IBM Z, WMLz works with the data gravity of the platform to achieve low latency and high performance inferencing capabilities. These capabilities support real-time scoring of every transaction and enable cost-effective scaling of AI by infusing it within transactional applications on IBM Z without affecting operational processing.

By using WMLz, you can uncover more opportunities to make more cost-effective, insights-driven business decisions, while ensuring that applications adhere to strict service level agreements (SLAs).

For more information about how to use WMLz in your Data Fabric, see 2.4, “Real-time integration of business-critical insight with IBM Watson Machine Learning for z/OS” on page 26.

This architectural approach is built around the following design points by using IBM Z capabilities to achieve superior AI application performance and business value:

- ▶ Collocate data with business applications and AI inferencing for minimal latency and exceptional performance.
- ▶ Modernize data infrastructure, develop, and deploy ML/AI models close to where data originates.
- ▶ Securely store and access your most valuable enterprise data within modern hybrid cloud applications.
- ▶ Use your investments in infrastructure and skills for a modern private cloud platform.
- ▶ Reduce cost and complexity by consolidating Db2 database applications without cloud specific rewrites.
- ▶ Modernize your database infrastructure for simpler administration, faster time to value and lower operational costs.

The results of deploying this approach to Data Fabric with IBM z Systems® is fast high-quality insight from transactional data without moving data off platform.

1.7 Data fabric versus data mesh

Conceptually, a Data Fabric and data mesh are similar, modern data management approaches that work to address the rising data complexities faced by enterprises. However, despite their similarities, they are fundamentally different approaches that are equally important to understand when making decisions about the data strategy of your organization.

1.7.1 Data Fabric

According to Gartner®, “A Data Fabric is a technology-enabled implementation capable of many outputs, only one of which is data products. A data mesh is a solution architecture for the specific goal of building business-focused data products.”¹

A Data Fabric uses a unique metadata-driven implementation that uses continuous analytics and enables augmented data management and cross data platform management.

It focuses on minimizing human involvement instead relying on a technology-centric approach that uses AI/ML solutions and automation to orchestrate data.

1.7.2 Data mesh

Conversely, a data mesh is a solution architecture that is designed with a technology-agnostic environment and focuses on services and people-centric solutions. It relies on business domain subject matter experts to create reusable data products, which enables a data ownership approach.

A data mesh focuses on data sources and must provide some type of discovery, analysis, and management principles to validate data objects and incorporate them into the design process. A Data Fabric and data mesh recognize the distributed aspect of data governance. Today, a data mesh uses a manual design and orchestration of systems with human intervention during design and maintenance.

When it comes to deciding which approach to use, it is important to understand that a Data Fabric and data mesh can coexist.

However, a Data Fabric provides the ability to use a use case-driven approach to achieve faster time-to-value. A Data Fabric provides the capabilities to facilitate and benefit from a data mesh through automation of tasks.

For more information about augmenting data management with a Data Fabric versus a data mesh, see the IBM blog [Augmented data management: Data fabric versus data mesh](#).

¹ Gartner: “Quick Answer: Are Data Fabric and Data Mesh the Same or Different?”, Published 1 November 2021 - ID G00749678 - By Mark Beyer, Ehtisham Zaidi

1.8 Role of a Data Fabric in AI computing

In the early stages of the current AI computing era, experimentation began by learning how to use modern computing capabilities to enable automated learning and improved decision making. Those early applications and projects often were designed and developed in a more or less stand-alone environment that enabled some new AI functional capability.

Some examples of these early AI projects include merging video captures and AI to enable visual recognition applications. In this example, AI can be used to alert a manufacturer that assembly line product quality was at risk, or to use video surveillance that is coupled with AI to recognize potential “threat situations” and automate “a call for help” or alert.

An important characteristic to many of these projects was they can be developed with a set of stand-alone systems, data, and components and run independently from core, well-orchestrated, end-to-end business processes.

As AI matures and advances, one of the key challenges is integrating and scaling AI into enterprise-class production environments. This overall challenge can be broken down into the following more specific challenges to consider:

- ▶ Integrating AI into systems and applications at scale.
- ▶ Meeting SLA needs for new AI use cases that have more stringent real-time requirements.
- ▶ Providing data accuracy, access, and governance for AI.

1.8.1 Core requirements demanded by AI

Many AI use cases have unique requirements for effective deployment and use. Specific design and use advantages are associated with an IBM Z-integrated Data Fabric.

Collocating data and compute for AI

Given the benefits of data gravity with core business systems, several advantages can be realized through the collocation of AI applications with critical data. Because large volumes of systems of record data are created, managed, and stored by using applications that are running on these same systems, it is important to consider Data Fabric design points that recognize these collocation requirements.

Fraud prevention, financial clearing and settlement, and loan approvals are some use case examples that require real-time, in transaction performance, and the ability to scale with the volume of underlying transactions for each use case. For example, it is arguably necessary to run fraud prevention capabilities on 100% of financial transactions, if possible, and ideally is done during the running of a specific transaction.

This low-latency, real-time, scalability is what IBM zSystems is designed to deliver. For such an example, it is advantageous to perform AI inferencing within the same platform that the core transaction is processing for optimum processing speed and resource consumption.

Design simplicity and performance

A well-designed Data Fabric works harmoniously with well-designed AI applications, and compute and inference environments.

Consider the following points as you design your Data Fabric:

- ▶ Mitigate complex environments by using data gravity to minimize data movement, limit data access latency, and reduce risk.
- ▶ Ensure that the design supports the required SLAs and low-latency response times for AI applications.
- ▶ Evaluate how to maximize the richness of data that is available through the Data Fabric.

1.9 Facilitating a Data Fabric by using IBM Cloud Pak for Data

The Data Fabric architecture that is facilitated by CP4D enables the acceleration of data analysis for better, faster insights. The key capabilities that are delivered by the CP4D architecture offer the following benefits:

- ▶ Simplifies and automates access to data across multi-cloud and on-premises data sources without moving data.
- ▶ Improves data governance and safeguarding, regardless of the source.
- ▶ Provides a self-service experience for finding and the use of data.
- ▶ Makes available AI-powered capabilities to manage the data lifecycle.

Figure 1-3 shows the five main capabilities of a Data Fabric and the connectivity that establishes a bidirectional flow between them, which is supported by CP4D.

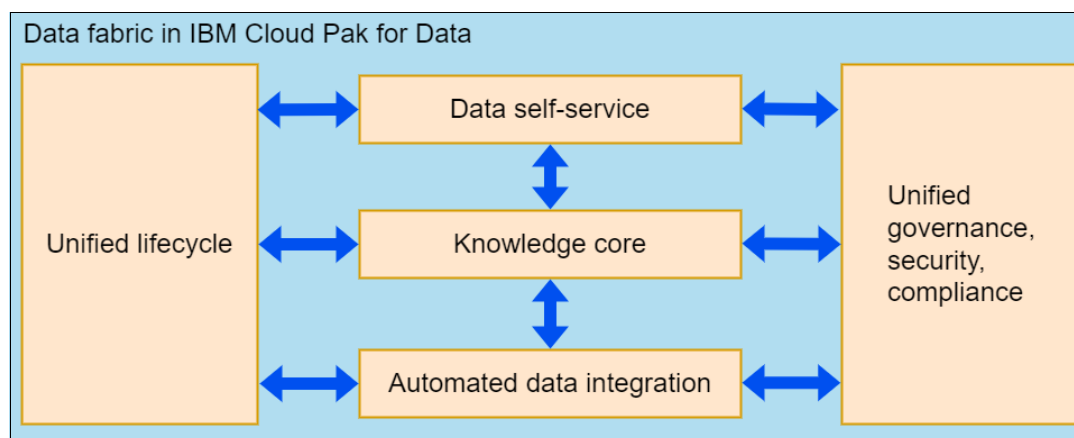


Figure 1-3 Main capabilities and connectivity of a Data Fabric

The following capabilities are shown in Figure 1-3 on page 13 and the IBM solution offerings that enable the delivery of business-ready data, including IBM Z data, that a Data Fabric demands are available:

- ▶ Metadata-based knowledge core describes the data and informs the semantic search for data. This feature enables auto-discovery, classification, and assignment of data governance artifacts and uses IBM Watson Knowledge Catalog service with the Knowledge Accelerators option.
- ▶ Data self-service in catalogs provides the capability to find data by using AI-powered semantic search and recommendations that use IBM Watson Knowledge Catalog service.
- ▶ Automated data integration enables automated bulk ingestion, cleansing, and complex transformations of data by using the Cloud Pak for Data platform, Data Refinery tool, Data Virtualization service, and IBM DataStage® service.
- ▶ Unified data governance, security, and compliance provides the ability to create data protection rules and automatically enforces uniform data privacy across the platform with Watson Knowledge Catalog service and Knowledge Accelerators.
- ▶ Unified lifecycle unifies and enables lifecycle management so that users can build, test, orchestrate, deploy to production, and monitor different types of data pipelines in a unified way. As a result, data asset search and use experiences with the Cloud Pak for Data platform are improved.

For more information about these CP4D capabilities, see 4.4.1, “Overview of CP4D” on page 58.



Using IBM Z strengths in a Data Fabric

In this chapter, we discuss how the integration of IBM Z and its data into your Data Fabric architecture enables your business to gain competitive advantages and become even more insights driven.

This chapter includes the following topics:

- ▶ 2.1, “Introduction” on page 16
- ▶ 2.2, “Real-time insight from business-critical Z data for your Data Fabric” on page 17
- ▶ 2.3, “In-place data access to IBM Z database systems” on page 21
- ▶ 2.4, “Real-time integration of business-critical insight with IBM Watson Machine Learning for z/OS” on page 26
- ▶ 2.5, “Use of security capabilities of IBM Z in a Data Fabric” on page 26
- ▶ 2.6, “Accelerate your AI journey with IBM Z” on page 27

2.1 Introduction

IBM Z is a proven platform for transaction processing, speed, reliability, and scalability that is highly available. It also features sysplex clustering technology that is supported by z/OS subsystems, such as IBM Db2, IBM CICS, and IBM MQ.

These features are one of the many reasons that leading companies around the world run their mission-critical applications on the IBM Z platform. The data that is collected from these enterprise applications became the primary source of business insights and the driving force behind new enterprise initiatives in the journey of digital transformation.

Data that originates on the IBM Z platform and users that access that data benefit from the low latency, resiliency, scalability and security that is embedded in the infrastructure architecture.

With the announcement of the IBM z16 system, which introduced the Telum processor—on-chip AI accelerator, customers can use a mainframe that is optimized for enterprise workloads with embedded, real-time AI insights. By using the software that is embedded in IBM Watson Machine Learning for z/OS (WMLz), including IBM supported open source tools, you can readily deploy AI models on the IBM Z platform and use the power of the Telum AI engine to help accelerate and enhance AI inferencing for deep learning, complex neural network models on the platform.

This unique advantage enables the ability to embed AI insight directly into high-performance transactional and operational applications and score every transaction in real time with minimal latency for actionable, real-time decisions with unmatched accuracy. WMLz enables data scientists to easily extract data from IBM Z sources and perform their analysis, which minimizes the need to move data around.

This solution is ideal when customers want to use the data gravity of the platform and access data from where it originates. The combination of transactional and analytical processing on the platform further supports the in-place use of data that is produced by data gravity in addition to the scalability and real-time performance requirements of most business applications today.

Also, IBM Db2 Analytics Accelerator (IDAA) provides a purpose-built analytics engine on IBM Z that is tightly coupled with IBM Db2 for z/OS. Thus, data engineers can perform their data analysis close to where the data originates, which reduces the need for data movement for analytics.

A current data management practice that is exercised by those who use the IBM Z platform is to make copies of IBM Z data and then, take those data copies off platform by using ETL or Data Replication techniques for data analysis. Although this pattern works, it has grown to be complex, difficult to manage, and exposes the organization to unnecessary risks with data privacy, compliance, and security.

Instead, a more secure technique is to use data virtualization technology on the IBM Z platform to access data at its source. Data virtualization enables the immediate use of that data within applications. It also improves flexibility and agility in your digital transformation.

Advancing from a traditional data management concept that is focused on data operationalization to a modern Data Fabric approach enables the integration of data management, data discovery, and governance. This advance provides a unified data experience across your organization.

When you include the IBM Z platform and its originating data in your Data Fabric approach, you add significant value through the untapped opportunity to better use the data where it originates, enabling faster time to insight and more accurate insights.

2.2 Real-time insight from business-critical Z data for your Data Fabric

A Data Fabric is an architectural approach that facilitates the end-to-end integration of various data pipelines. One of the primary goals with implementing a Data Fabric in your organization is to unify disparate data systems by eliminating silos; embed governance; strengthen security, privacy and compliance; and provide more accessible data pipelines to business users and application.

As a part of the digital transformation, many organizations use hybrid cloud innovations that make the most of investments in their IT infrastructure, such as IBM Z, to support Data Fabric initiatives with the promise of greater agility, abundant capacity, and low initial investments.

Through the integration of IBM Z and its corresponding data into your Data Fabric, you can securely access data across hybrid cloud and enable delivery of business-ready data for intelligent insights. Such an opportunity leads to lower IT costs, and more reusability of resources for scalability, and better decisions and results.

For organizations with IBM Z as part of their hybrid cloud architecture, the platform and the data that originates on it are essential to a Data Fabric by using data gravity for real-time integration of business-critical insights.

2.2.1 Use the data gravity of the Z platform

Data gravity is key to a Data Fabric, which you can use to simplify data management and governance to generate insight in a hybrid cloud landscape while addressing cost and risk.

Regarding the concept of data gravity, you must first understand the science behind gravitational pull. Gravitational pull relates to the mass of an object, where the greater the mass, the greater the gravitational pull of that object to other objects with mass.

In the context of *data gravity*, it is the idea that data, like any other object, has mass and thus attracts other objects with mass, such as more data, data types and applications. That is, the greater the amount (or mass) of data that is originating on the system, the greater that gravitational pull is of that data to other data objects.

By using the data gravity of your systems, you can bring the processing to the data, which is a key component in the efficiency and performance of your Data Fabric.

When assessing what is the best platform for analyzing and accessing IBM Z data from a data gravity perspective, the answer is always the IBM Z platform. The use of a data gravity approach to perform access and analytics to Z data sources enables the following benefits:

- ▶ Reduced data latency with low to no latency, which leads to faster-time-to-insight.
- ▶ Enhanced security by keeping sensitive data and information on the platform with 100% encryption of all data.
- ▶ Simplified infrastructure with reduced need to move data and enhanced resiliency with transactional system-level performance and availability to your analytics and AI workloads.
- ▶ More easily operationalize new use cases.

In-time analytics

In the context of IBM Z and data gravity, the Z platform is home to massive amounts of transactional and operational data and workloads that serve as the bloodline to many global enterprises. We can conclude that the vast amount of data on the Z platform implies a strong gravitational pull around that data.

With transactional servers being that core to any organization's enterprise infrastructure, it makes sense that organizations are moving their AI analytics and ML workloads to the world's most securable platform by using the data gravity Z provides.

Therefore, instead of taking the data that is generated on the platform and copying it to other platforms for processing (which exposes data security risks, data inconsistency, latency, and governance challenges) organizations can generate analytic insight from real-time data at the transactional source without moving that data off platform.

As organizations evolve and become more data-driven, accessing quality data at the right time on the right platform ensures better outcomes while preserving existing data management and recovery processes. This process is known as in-time analytics, or running advanced analytics on real-time transactional data at the point of origin. It allows you to use the high-value, time-sensitive data on IBM Z in real time.

When planning a Data Fabric architecture that integrates IBM Z to achieve in-time analytics, a unified data governance must be built that features compliance, intelligent data integration, and orchestration. In this way, you can enable easier access to enterprise data through various data pipelines.

For more information about IBM Data Replication Change Data Capture best practices, see this [IBM Support web page](#).

2.2.2 Intelligent integration of IBM Z data

IBM's approach to a Data Fabric includes the intelligent integration of data through the automation of data engineering tasks to augment data integration across hybrid cloud resources. This approach is fully supported by IBM Cloud Pak for Data and provides the functions and technology that are required to deploy a Data Fabric.

IBM Cloud Pak for Data (CP4D), is a cloud-native, enterprise data and AI platform that enables the facilitation of a Data Fabric for hybrid cloud. It allows you to connect to, govern, find, collaborate on, and use your data for AI and analytics.

Cloud Pak for Data can run on any cloud and anywhere you have an available Red Hat OpenShift cluster, even if that cluster is behind your firewall on an on-premises system.

Figure 2-1 shows the capability to connect any data, to any cloud, from anywhere it is needed, which is enabled by CP4D technology.

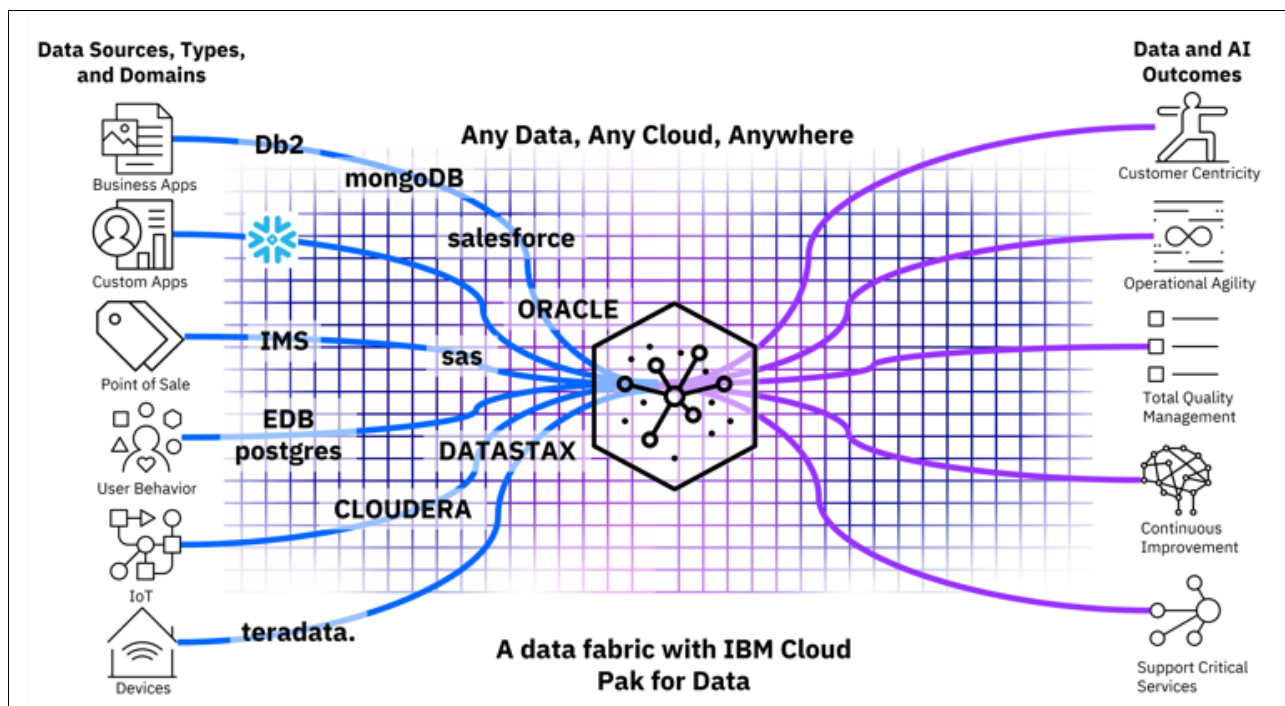


Figure 2-1 Data fabric approach using Cloud Pak for Data

Figure 2-1 shows various data sources that your enterprise can use, such as IBM IMS, IBM Db2, and MongoDB. You can imagine how complex data integration efforts can become with so many disparate data storage systems, applications, and environments, in various data formats.

The Data Fabric architecture that is offered by CP4D provides capabilities, such as virtualization, containers, and a microservices architecture, to simplify and modernize your data integration efforts. The architecture spans hybrid and multi-cloud environments, regardless of where that data lives for business-ready AI.

It also provides a unified platform for connecting and consuming data through a common layer. In doing so, CP4D helps enterprises efficiently and effectively bring enterprise data together and break down data silos so you can put your data to work quickly and use meaningful insights to reach your business goals.

Cloud Pak for Data provides the following capabilities, which add intelligence to the integration of your enterprise data and assets:

- ▶ Enables easy data consumption by extracting, virtualizing, transforming, and streaming data
- ▶ Integrated with a knowledge core to automate data integration
- ▶ Equipped with the intelligence to decide which integration approach is best suited based on workloads and data policies
- ▶ Can be used for data preparation as part of data engineering workloads or to create data products
- ▶ Provides the ability to publish updates to data products

By combining the technology that is offered by CP4D and IBM Z, you can realize the following benefits:

- ▶ Access and integrate relational and traditional IBM Z data within your hybrid cloud architecture, which enables the use of enterprise data in analytics and AI applications.
- ▶ Readily host IBM Db2 for z/OS data within CP4D by using IBM Db2 for z/OS Data Gate for read-only access by new applications (see 2.3.2, “Off-platform access to Db2 for z/OS data with IBM Db2 for z/OS Data Gate” on page 23).
- ▶ Achieve lowest latency and greatest application resiliency by deploying ML models in transactional applications that are running on IBM Z.

Precautions for existing data integration approaches

For some time, data replication and data transformation that used ETL techniques were the primary pipelines for data integration in an enterprise. These methods serve to copy data from platforms and bring it to distributed data warehouses and data lakes.

Although we do not recommend data movement as the most secure data integration practice, the following scenarios support the continued use of this pattern:

- ▶ Use of ETL methods when:
 - A need exists for data persistence. For example, moving data to enterprise data warehouses, or offloading enterprise data warehouse to the cloud.
 - Structured data with complex data transformations, repeatable data flows, and a requirement to move bulk data for AI are involved.
- ▶ Use of data replication methods when:
 - Change events occur in a data store.
 - Copying data between sources is log-driven and of low-impact, and a need for continuous availability and disaster recovery exists.

Consider the following precautionary risks when copying data from its source to other platforms:

- ▶ Increased risk for data security: The higher the number of storage locations of your valuable data sets, the higher the risk that unauthorized access can occur.
- ▶ Challenges with data consistency: Multiple copies of data can lead to inconsistent copies of that data, which can lead to untrusted data that can further leads to the lack of data use because of the lack of trust.
- ▶ Challenges with data latency: Source data is much more current than data that is copied to another system, which can lead to irrelevant, stale insights.
- ▶ Cost and complexity: Hard costs of data movement include hardware expenses, such as storage and compute power, replication and ETL software costs, and personnel to manage replication and ETL processes. Complex IT infrastructures result when you must manage multiple connections and processes to copy and move data between systems.

For more information about data movement techniques and considerations, see 3.2, “Protecting data copies from IBM Z for AI/ML” on page 43.

We estimate that these established patterns reduce over time because of a lack of scalability and the growing desire to reduce the need for data movement and obtain real-time insights.

Data virtualization as an alternative approach

Organizations that are adopting the Data Fabric approach are increasingly integrating data virtualization capabilities into their data integration architecture with the purpose of mitigating these issues of data movement.

Data virtualization provides a unified method to access data at the source in near real-time without moving data by abstracting the data from multiple sources and creating a virtual view for business users.

Data virtualization is a key technology when it comes to working with data gravity because it enables the performance of analytics on highly distributed data sets, which leaves the protected data at its source-of-origin and returns only the resulting query. This capability is useful in scenarios where the data and analytics results are time sensitive, and when you are working with protected data, such as personal information that prevents copying or movement of that data, using data virtualization and data gravity to gain in-place access to data at its source are key components to the efficiency and performance of your Data Fabric.

For more information about data virtualization capabilities on IBM Z, see 2.3.3, “Integrating data from multiple sources with Data Virtualization for z/OS” on page 23.

2.3 In-place data access to IBM Z database systems

For decades, many of the world’s critical systems across many industries run on IBM Z, including many different database systems (IBM database products and many third-party vendors).

Note: The IBM Z platform can be logically split into multiple partitions, called logical partitions (LPARs). These LPARs provide the ability to run one database, such as IBM Db2 for z/OS, on one LPAR, while running another database, such as Adaptable Database System (ADABAS), on a separate LPAR. Each of the LPARs can operate in isolation while simultaneously sharing resources.

Two IBM database management system (DBMS) products are available for the z/OS environment: IBM Db2 for z/OS and IBM IMS. Db2 for z/OS is a relational DBMS, whereas IMS is a hierarchical DBMS.

These database management systems can be made highly available through specific configurations, such as [data sharing](#) and [clustering](#). A vast amount of business-critical data is likely stored in these database systems on IBM Z, and exists in relational and nonrelational formats.

It is a common perception that IBM Z data sources, such as Db2 for z/OS, Virtual Storage Access Method (VSAM), ADABAS, and Integrated Database Management Systems (IDMS), are difficult to access, transform, and combine.

However, by using CP4D, you can readily access and combine an extensive and growing list of IBM Z data sources through modern APIs and virtualization capabilities and tools, such as the IBM Data Virtualization Manager for z/OS (DVM) integrated service offering in CP4D.

Allowing developers and data scientists to transparently access relational and nonrelational transactional data sources on the platform without the need for specialized skills with reduced time and resources translates directly to savings.

Exposing IBM Z data in a Data Fabric means exposing the metadata from that data, which is stored in a metadata repository in DVM where all lineage to the back-end data source can be referenced. This exposure enables business users and applications to provision data in-place, directly from the source databases.

2.3.1 Hybrid transaction and analytic processing on IBM Z with IBM Db2 Analytics Accelerator for z/OS

IBM Db2 Analytics Accelerator for z/OS (IDAA) is a purpose-built data analytics engine for IBM Z that is configured to Db2 for z/OS. It provides users the unique capability of running online transaction processing (OLTP) and analytics on the same Db2 database engine. It also is a virtual server that runs in a containerized form, which makes it logically invisible.

Delivering high-speed processing of complex Db2 queries in support of business-critical reporting and analytic workloads, it transforms the Z platform to a hybrid transaction and analytic processing (HTAP) environment. This type of transformation supports the critical ability to perform analytics on transactional data as it is generated, so you can gain deep insights quickly and support time sensitive business decisions.

IDAA runs in a containerized form as a virtual server with its own assigned disk volumes and processing engines, which enables it to process queries separately from Db2 for z/OS. As queries are directed to and received by the Db2 for z/OS subsystem, the Db2 for z/OS optimizer automatically decides where the query optimally is to run (that is, the front end of Db2 or on the Accelerator), routes the query accordingly, and returns the result to the requester.

This automated process, as shown in Figure 2-2, occurs transparently.

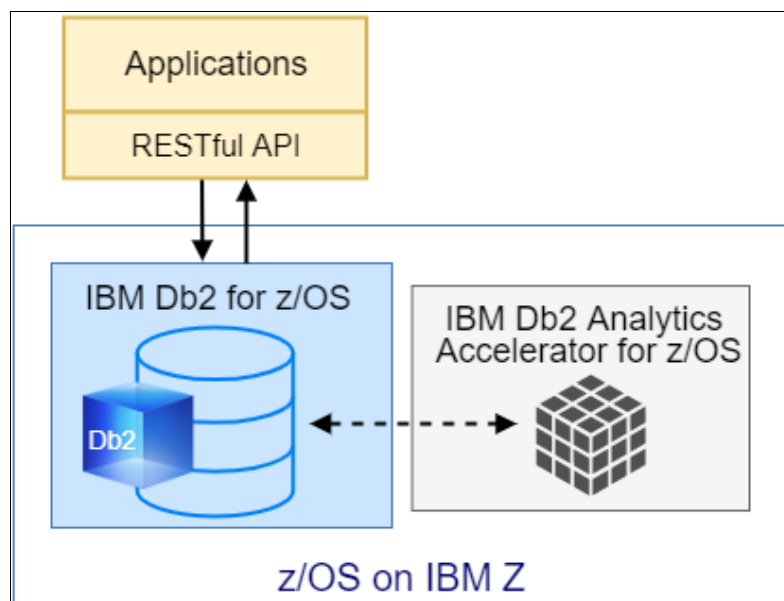


Figure 2-2 Query processing flow through Db2 for z/OS and IDAA

Data that is on the Accelerator is kept in sync with Db2 for z/OS by using integrated synchronization data replication protocols, which helps ensure transactionally consistent analytical queries against the latest data.

IDAA enables your enterprise to use the data gravity of z/OS-based transactional data as it is generated for business-critical reporting and analytics workloads under the control and security of IBM Z.

Note: All access to IDAA is managed through Db2 for z/OS.

Scenarios also exist in which you want to analyze z/OS-based data that is in a database subsystem other than Db2 for z/OS. It is at this point that [hIBM Db2 Analytics Accelerator Loader for z/OS](#) enters the picture.

The Loader can efficiently load non-Db2 mainframe and nonmainframe data into Db2 for z/OS and the Accelerator at the same time (or into the Accelerator only) while reducing ETL costs and CPU consumption, and maintaining data consistency.

The use of IDAA in your Data Fabric results in the ability to expose different, complex data sources to data analysts. The analysts can then run SQL queries and extract meaningful insights quickly from the most relevant data without affecting Db2 for z/OS applications and simultaneously offering the opportunity to reduce complexity and costs.

2.3.2 Off-platform access to Db2 for z/OS data with IBM Db2 for z/OS Data Gate

IBM Db2 for z/OS Data Gate (Db2 Data Gate) is a solution to help reduce the cost and complexities that surround off-platform access to Db2 for z/OS data. It offers a hybrid cloud approach to data delivery and removes the need to develop and maintain custom-built solutions for the same purpose.

It also synchronizes data sources between Db2 for z/OS and target databases on CP4D, such as Db2 for Linux, UNIX, or Windows (Db2 LUW) or Db2 Warehouse. This ability allows applications to access the synchronized copy of the Db2 for z/OS data instead of direct access at the data source. The synchronization is accomplished by using Db2 In-sync Replication technology, which results in a 1:1 copy of Db2 for z/OS data on a distributed Data Fabric that is integrated with IBM Z.

The purpose is to create a read-only copy of Db2 for z/OS data in Db2 LUW for new applications, or to have a copy of data in Db2 Warehouse for data analytics. Data Gate reduces the complexity of the data delivery process for new business initiatives, which allows you to use your infrastructure investments to accelerate your hybrid cloud journey.

Note: In a Data Fabric, Data Gate is directly linked to Db2 for z/OS only, and does not apply any transformations to the Db2 source data elements.

2.3.3 Integrating data from multiple sources with Data Virtualization for z/OS

Making IBM Z-based data available and easily accessible through data virtualization pipelines enables the simplification and democratization of your data landscape. IBM Data Virtualization technology extracts the complexity of natively accessing database systems, which breaks down data silos and provides a unified access layer to all data sources.

With the extensive volumes of data being generated and collected every minute, traditional ETL processes became too costly and slow to continue to rely on.

Data virtualization provides a solution to these problems. It offers the ability to put this data to use, regardless of its format, through a more secure, efficient, and dependable data integration process. It also removes the need for ETL methods and duplicate data storage by accelerating processing times, which helps to reduce storage costs.

Cloud Pak for Data includes data virtualization technology as a service that connects disparate data sources into a single collection, called a *constellation*, of data sources. All communication within the constellation and between the constellation and the application is encrypted with powerful IBM technology that is paired with Secure Sockets Layer (SSL) and Transport Layer Security (TLS) encryption with standard protocols.

From an application perspective, the data virtualization service node is one of the microservices within CP4D. It appears as a single Db2 instance that provides the ability for popular Db2 clients and applications to connect to and work with the service without modification.

When you use CP4D to facilitate the integration of data and eliminate data silos, you provide a unified access control and view layer to integrate with internal and external business intelligence and ML tools, which reduces ETL requests. The unified access layer serves as a collaborative data environment across your enterprise that provides ease of access and use, regardless of where the data is stored. It also promotes responsible data use with increased data accuracy through near real-time access.

Data virtualization technology in CP4D allows analytics queries to be processed on the server, where the data source exists, without requiring copies of data to be stored in a centralized location. In turn, data virtualization technology enables businesses to more effectively work with their data and accelerate time-to-value with quick, more accurate queries.

By design, the data virtualization engine uses advanced parallel processing and optimizations to rapidly deliver query results from multiple data sources. It also can automatically find databases and tables, which provides an even simpler way to query information from multiple sources.

Data virtualization capabilities in Cloud Pak for Data support the following data science initiatives:

- ▶ Self-service business intelligence: Virtual, reusable data assets are presented in a user-friendly format that enables use without the need to understand underlying data complexities. It also allows collaborative access to the data by business intelligence and reporting tools.
- ▶ Unified 360-degree view of data assets: This real-time, complete view of data assets helps users to understand and use the enterprise data more easily.
- ▶ Agile service-oriented-architecture (SOA) data services: Delivers data services layer to SOA applications and allows centralized access to the data by multiple business services with loose coupling between the services and physical data sources.
- ▶ Improvements in information control: Centralized access control improves data quality because of the reduction of data copies (lowers risk) and metadata repository catalog that enables transparency and visibility into your organizations data and relationships among the data.

For more information about IBM Data Virtualization in Cloud Pak for Data, see [Eliminate data silos: Query many systems as one](#).

Integrating nonrelational Z data sources with IBM Data Virtualization Manager for z/OS

Unlocking IBM Z data for use in modern API-based applications is a challenge for enterprises because of incompatible formats. The process often requires the use of ETL processes to enable access. Also, the ability to unlock IBM Z data for use in analytics and applications is a critical component in the success of your digital transformation because the platform is home to most business-critical data and workloads.

Enter, IBM Data Virtualization Manager for z/OS (DVM).

DVM is the only virtualization technology that takes advantage of the full breadth of benefits that are offered by the Z infrastructure. With DVM, you can make this data much more usable with virtual views of data that provides read/write access to IBM Z relational and traditional nonrelational data sources, such as Db2 for z/OS, IMS, ADABAS, VSAM, QSAM, IDMS, in place and without any specialized database skills.

Through integrated connectivity between DVM and CP4D, you can simplify the development of AI applications that are driven from IBM Z data sources. As a result, complex data structures are accessible by business users. You can then integrate IBM Z virtualized data sources with non-Z data in a single unified view for even more insight opportunities.

By enabling in place access to IBM Z data, you can keep the data within the highly securable infrastructure of the Z platform, and use Z security features, such as pervasive encryption and systems security protocols. The data access approaches that are provided by DVM help reduce overall mainframe processing usage and costs by using IBM Z Systems-Integrated Information Processor (zIIP) to efficiently redirect up to 99% of workloads that otherwise are meant for the General Central Processors (GCP).

For more information about DVM-supported data sources, see this [IBM Documentation web page](#).

The use of DVM in your Data Fabric strategy delivers the ability to expose metadata and user data in a securable manner. It supports data encryption technology to help you protect your critical enterprise data. It also maintains a metadata repository where it stores information that describes the data that can be accessed by the DVM server.

When implementing a Data Fabric strategy that uses CP4D, you can establish a connection to DVM-defining connections with remote data sources with the built-in connector framework in CP4D. This integrated connectivity through DVM makes creating data governance or data pipelines simpler because the data steward on CP4D can readily access data structures without being exposed to the complexities of IBM Z data sources.

For more information about DVM, see “Integrating nonrelational Z data sources with IBM Data Virtualization Manager for z/OS” on page 25.

For more information about key concepts, differentiators, and value propositions for the use of DVM, see the IBM Support white paper, [Introduction for IBM Data Virtualization Manager for z/OS](#).

2.4 Real-time integration of business-critical insight with IBM Watson Machine Learning for z/OS

IBM Watson Machine Learning for z/OS (WMLz) is an enterprise ML offering that provides a suite of tools for the development and deployment of predictive models, user management, and system administration. It runs on the IBM Z platform (and Red Hat OpenShift Container Platform) and provides the benefit of having a data science platform that can access data in place. This platform enables enterprise grade performance, high availability, and a highly securable environment for ML.

WMLz infuses AI directly into transactional applications for real-time, actionable insights. You also can benefit from WMLz transparently using the IBM Z Integrated Accelerator for AI in IBM z16 through the IBM Deep Learning Compiler (DLC) to inference every transaction, at scale, in real time, with low latency and unprecedented performance.

In terms of Data Fabric, WMLz works with the data gravity on the platform and integrates with application environments to provide developer-friendly APIs for applications on the same platform. This integration allows the business to use current data to maximize its value for improved real-time insights. Then, these insights can be infused into operational applications for various purposes, such as fraud detection.

When you use WMLz to keep the data where it originates and bring AI capabilities to where applications are running, you can reduce costs, security risk, accuracy, and time-to-value of your insights.

For more for information about Watson Machine Learning, see 4.5, “IBM Watson Machine Learning” on page 64.

2.5 Use of security capabilities of IBM Z in a Data Fabric

IBM Z offers secure access methods through IBM RACF®, Top Secret, or ACF2, in addition to access controls that are embedded into the DBMS. When IBM Z data is exposed for use by data engineers, it must adhere to the security controls, policies, and practices that are established on the platform. Such controls, policies, and practices can include user ID authentication, database access privileges, remote network connectivity, and SSL/AT/TLS.

In a Data Fabric architecture, Cloud Pak for Data enables a unified data governance approach by providing different layers of access control when connecting to IBM Z data sources, including (but not limited to) the following examples:

- ▶ Data governance metadata layer that uses role-based access methods to determine type of access.
- ▶ Network encryption access that uses SSL and Application Transparency (AT)-TLS.
- ▶ DVM access.
- ▶ Database access control, such as Db2 for z/OS or IMS.
- ▶ Other encryption facilities, such as pervasive encryption for data set encryption when accessing VSAM files.

To establish secure and trusted access from a Data Fabric to IBM Z, it is critical that the entire chain, or people responsible for each layer, must be consulted and an agreement to the security processes must be established and adhered to.

For more information about how to use the security and resiliency benefits of IBM Z to reduce compliance costs and risk, see this IBM Z and Cloud Modernization Center [web page](#).

2.6 Accelerate your AI journey with IBM Z

With IBM Z being the first on-premises platform to be ported with WMLz, data scientists can now have the same ML experience on the IBM Z platform as compared to modern Cloud platforms. Also serving as a runtime environment for ML models, WMLz includes the capability to automate deployment, monitoring, and management of the models and detects when it observes model degradation, which automatically retrains and redeploys the model for most accurate results.

With accelerated AI capabilities that were introduced by Telum on IBM z16, applications that are running on z/OS can take advantage of data-driven insights by using ML models. To remain competitive in today's digital transformation, developers and architects on IBM Z should be looking at data and how historical events and patterns can, and do, influence future decisions.

IBM Z continues to be a critical component in the overall data landscape of many enterprises worldwide. It also is becoming more clear how integral it is to integrate such a critical component into your enterprise Data Fabric strategy. A Data Fabric that integrates IBM Z enables data engineers and data architects to not only drive new insights from IBM Z data, but also discover the benefits of calling a model from every transaction, in a security-rich environment, and use AI at-scale.

For more information about AI and insights on IBM Z, see this IBM Z and Cloud Modernization Center [web page](#).

2.6.1 Supporting your journey up the AI Ladder with IBM Cloud Pak for Data

As your enterprise continues to fill with vast amounts of data that can produce meaningful business insights, you must be able to trust that data and put it to work efficiently and quickly.

Cloud Pak for Data provides a prescriptive, cost-effective approach to climbing the AI Ladder by providing a fully integrated data and AI platform and suite of services that modernizes how businesses connect, govern, find, and use data for use in analytics projects. It uses modern tools and technologies to facilitate analytics and collaboration, which helps foster productivity.

The IBM AI Ladder™ breaks an AI strategy into pieces, which are represented as the four rungs of the ladder. At each rung exists a unique advantage that is offered by the CP4D platform and its integrated services that can be used to expand the functions of the CP4D platform. Through use of the AI Ladder as a guiding framework and CP4D as an enabling solution, enterprises can establish the foundation for a robust Data Fabric.

Note: Developed by IBM, the AI Ladder provides organizations with an understanding of where they are in their AI journey and a framework for helping them determine where they need to focus. For more information, see [The AI Ladder: Demystifying AI Challenges](#).

Collect

At the first rung of the ladder, *Collect* refers to making data simple and accessible to be incorporated into any analytical process.

Cloud Pak for Data helps facilitate the connection of your data, regardless of where it is stored, which yields a rich data set for use in AI. Cloud Pak for Data simplifies connections to data sources that can be used by multiple services and is facilitated by using an easy-to-use web interface, called the Connections page.

This connection capability is important when considering the benefits of unlocking access to high-value, business-oriented data that is often created, stored, and managed on IBM Z. The platform uses data integration capabilities, such as, data virtualization and data refinery, to make data accessible for analytics, regardless of the source and type of that data.

Organize

The next rung of the ladder is, *Organize*, which represents the creation of a business-ready analytics foundation and is a critical requirement to becoming a data-centric enterprise. This step involves making data discoverable, cataloged, profiled, categorized, classified, secured, trusted, and reusable.

The IBM Watson Knowledge Catalog service that is offered in CP4D provides data classification and governance capabilities to help you organize your data by using metadata to add automated enforcement of governance policies, insight, and context about the data into the data catalog. The IBM Watson Knowledge Catalog service makes it easier to find relevant data in the catalog and enables the ability to develop an information architecture that can be optimized and scaled for AI applications.

Analyze

The third rung in the AI Ladder is *Analyze*; that is, build and scale AI with trust and explainability. To support this rung of the ladder, CP4D includes various analytics services that enable the on-demand generation of scalable insights by using various features, such as creating dashboards to visualize data by using IBM Cognos® Dashboards, and streams that enable you to build solutions that combine streaming and stored data with analytics.

Infuse

The last rung of the ladder is *Infuse*; that is, infuse and operationalize AI throughout the business with trust and transparency by using AI for tangible benefits.

With CP4D, you can strengthen AI capabilities and enable broader and richer data quality by providing access and visibility to both Z and distributed data, all through an integrated Data Fabric approach. This feature enables the use of data for automation and optimization in a cyclic approach of action and feedback, which allows it to be used for developing optimized insights and decision-making throughout the enterprise.

With CP4D, you also can use the seamless integration of Watson services, such as IBM Watson Machine Learning (WML), to deploy ML models into production at scale. As a result, you can build smarter applications or infuse AI with trust and transparency with IBM Watson OpenScale.

For more information about IBM and third-party provided services in Cloud Pak for Data, see this [IBM Cloud Pak for Data Documentation web page](#).

For more information about how CP4D supports the facilitation of a Data Fabric and enables your journey up the AI Ladder (including reference architecture), see this IBM Cloud [web page](#).

2.6.2 IBM z16 capabilities in a Data Fabric

IBM z16™ is the latest generation of the IBM Z mainframe family. It includes the Telum Processor that is embedded with an on-chip accelerator for AI inferencing.

Designed to enable real-time, AI embedded, in-transactional workloads with improvements for performance, security, and availability, Telum features innovations in security with transparent encryption of main memory, which is ideal for processing sensitive data in hybrid cloud architectures.

With WMLz providing the ability to call deep learning models directly in IBM Z, the on-chip AI accelerator empowers you to make informed business decisions in real time, without affecting response times for high volume transactional workloads. That is, data is used to understand historical behavior patterns, and models are created to make optimized decisions faster, which results in a better experience for your business users and clients.

When it comes to the value of integrating IBM z16 into your Data Fabric, the Telum Processor is available to help maximize the latency and data protection benefits in keeping data on the IBM Z platform. Through maximizing such benefits, you can use AI insights to achieve better control of outcomes of transactions before they complete processing; for example, detecting fraudulent behavior on a customer's credit card and taking remedial action to stop the fraudulent transaction from completing, which protects the customer's assets.



Securely connect a Data Fabric to any data source, anywhere

The goal of a Data Fabric architecture is to maximize the value of your enterprise data and accelerate your digital transformation. Weaving a Data Fabric with an integrated data layer is possible only if you can easily connect to all your available data sources. It is also essential to establish a well-connected pool of metadata that enables the identification, connection, analysis, trustworthiness, and governance of all kinds of data through a Data Fabric, regardless of where it originates.

In addition, contextualizing and categorizing all your data, including IBM Z data, with self-service access lays the foundation of a dynamic Data Fabric design. A dynamic Data Fabric design can enable the infusion of AI deep into enterprise systems and workloads, accelerating your organization's digital transformation journey.

This chapter discusses suggested secure practices around the use of a Data Fabric to establish data connections within an enterprise to eliminate data silos and achieve a holistic view of all the data available within the enterprise.

We also pay special attention to connectors and connecting processes to access IBM Z data sources from IBM Cloud Pak for Data that is running on an x86 machine, which enables self-service data consumption, collaboration, and deeper AI insights.

This chapter includes the following topics:

- ▶ 3.1, “Weaving a Data Fabric into hybrid cloud” on page 32
- ▶ 3.2, “Protecting data copies from IBM Z for AI/ML” on page 43
- ▶ 3.3, “Data gravity with AI on IBM Z through data virtualization” on page 48

3.1 Weaving a Data Fabric into hybrid cloud

As part of the digital transformation journey, cloud environments are becoming increasingly popular for enterprises to use because of their flexibility and low cost. However, these same enterprises spent vast amounts of time, money, and effort investing in and customizing transactional environments to accommodate their individual business processes. Because of this investment, it makes little sense to rewrite these systems for minimal to no benefit. Evolved from this dilemma is the hybrid cloud architecture.

Hybrid cloud architecture combines and unifies public and private cloud, and on-premises infrastructure to create a single, flexible, and cost-optimal IT infrastructure. With IBM Z integrated into your hybrid cloud, you get the best of both worlds: the agility and flexibility of cloud, with the AI-driven insights, security, quality of service, and resiliency that is offered by IBM Z platform infrastructure. By integrating systems of record with modern hybrid cloud and AI applications, enterprises can drive even smarter outcomes.

Hybrid cloud applications typically incorporate on-premises and on-cloud services and might also make use of different data sources across the enterprise, which sets the stage for the need of a Data Fabric architecture.

The Data Fabric design allows automated discovery of data on IBM Z and on the cloud from the same, easy-to-use user interface. Figure 3-1 shows examples of hybrid cloud infrastructures that can be easily integrated through a Data Fabric by using Cloud Pak for Data that is running on a Red Hat OpenShift Container Platform. Such infrastructures span from on-premises environments, such as Intel x86, Arm, IBM Power, IBM LinuxONE, and IBM Z, to private and public clouds.

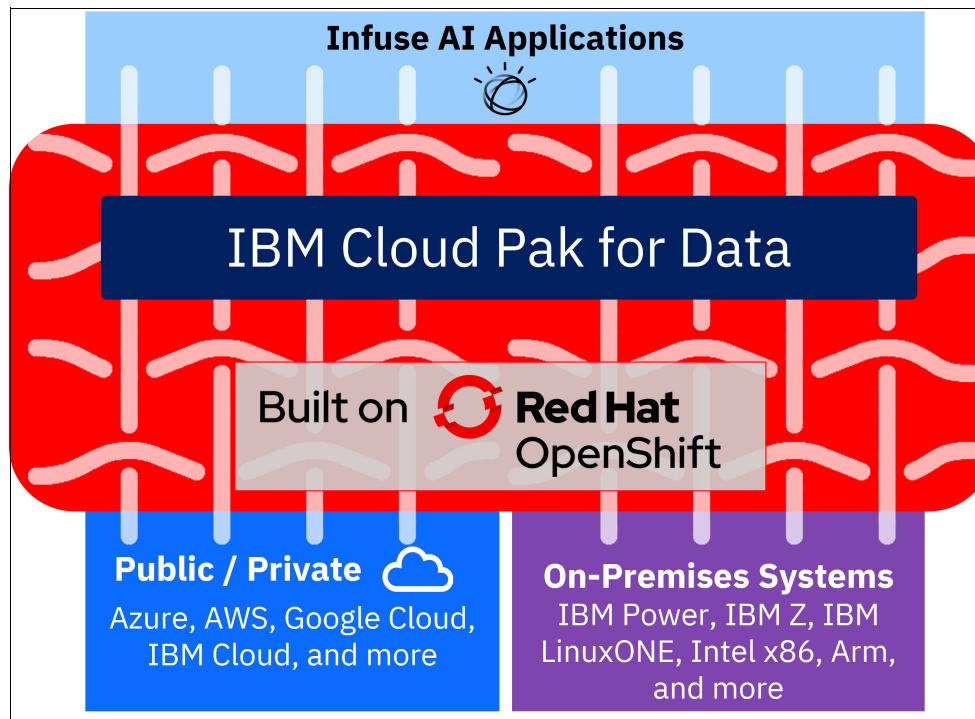


Figure 3-1 Data fabric across hybrid cloud architecture with Red Hat OpenShift Container Platform

Modern applications are interconnected, interact through APIs, and enable customers and enterprises to digitally run processes quickly. These applications need relational and nonrelational agile read/write access to IBM Z data in an online environment.

A Data Fabric architecture presents a single user experience across hybrid cloud to simplify the delivery of data that is required for analytics and AI, which enables you to unlock the value of your data in a hybrid multi-cloud data landscape. To help enterprises make business-critical data readily consumable by hybrid cloud applications in a way that maintains data currency and security, IBM is offering IBM Cloud Pak for Data as the preferred approach to deploying data fabric functions.

IBM Cloud Pak for Data can help lower your total cost of ownership, simplify data access, accelerate innovation that is based on Open Source technology, and fully support multi-cloud environments, such as Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform, IBM Cloud, private clouds, and on-premises infrastructures, including IBM Z.

Figure 3-2 shows the capabilities of a Data Fabric that is delivered in Cloud Pak for Data and the platform’s interaction with data sources.

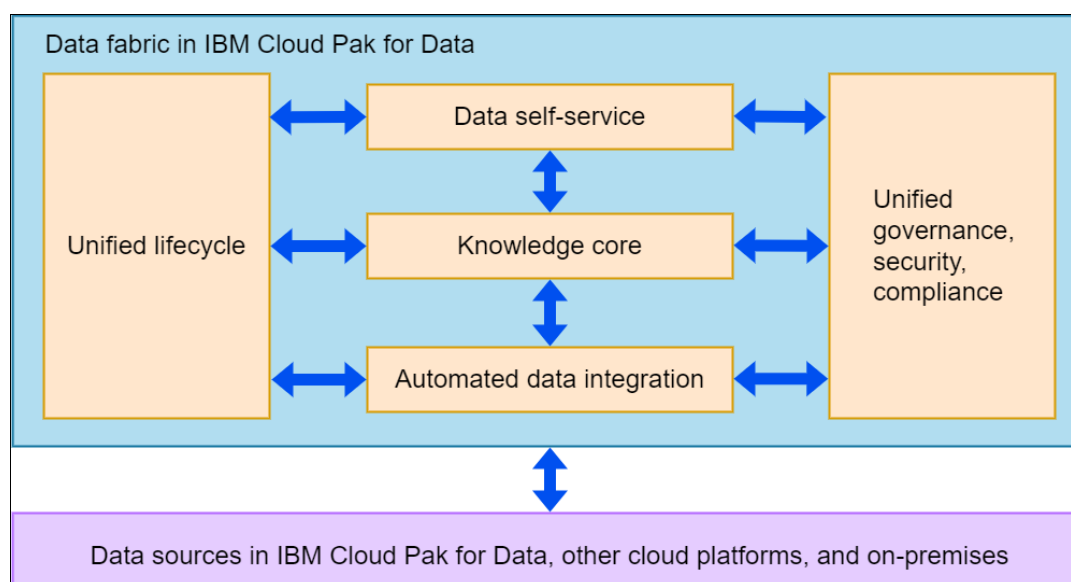


Figure 3-2 Main capabilities of a Data Fabric in CP4D

Cloud Pak for Data helps enterprises connect to their data, regardless of its location. Within CP4D, different types of assets are available for users to work with, where an asset is an item that contains metadata about data, other types of information, or operational code.

The following main asset types are used in CP4D:

- ▶ Data assets
- ▶ Operational assets
- ▶ Configuration assets
- ▶ Business intelligence assets
- ▶ Transformation scripts

To access data sources from CP4D, you must first create a connection asset before you can access the metadata or load data to or from the platform. *Connection assets* are a data asset subtype that contain the necessary information to establish a connection to a data source.

The process by which you connect to your data varies depending on the services that are installed on CP4D. Some services can use connections that are defined at the platform level, whereas others rely on connections that are specific to the service. For more information about supported connections in CP4D, see 3.1.2, “IBM Cloud Pak for Data supported connections” on page 35.

3.1.1 Centralized governance of connected data

Figure 3-3 shows various connections between different data sources and IBM Cloud Pak for Data (connected directly or through data virtualization technology), and are then made readily available for use by data consumers. This entire process benefits from centralized governance capabilities of CP4D and integrated services across the entire enterprise data in a uniform manner.

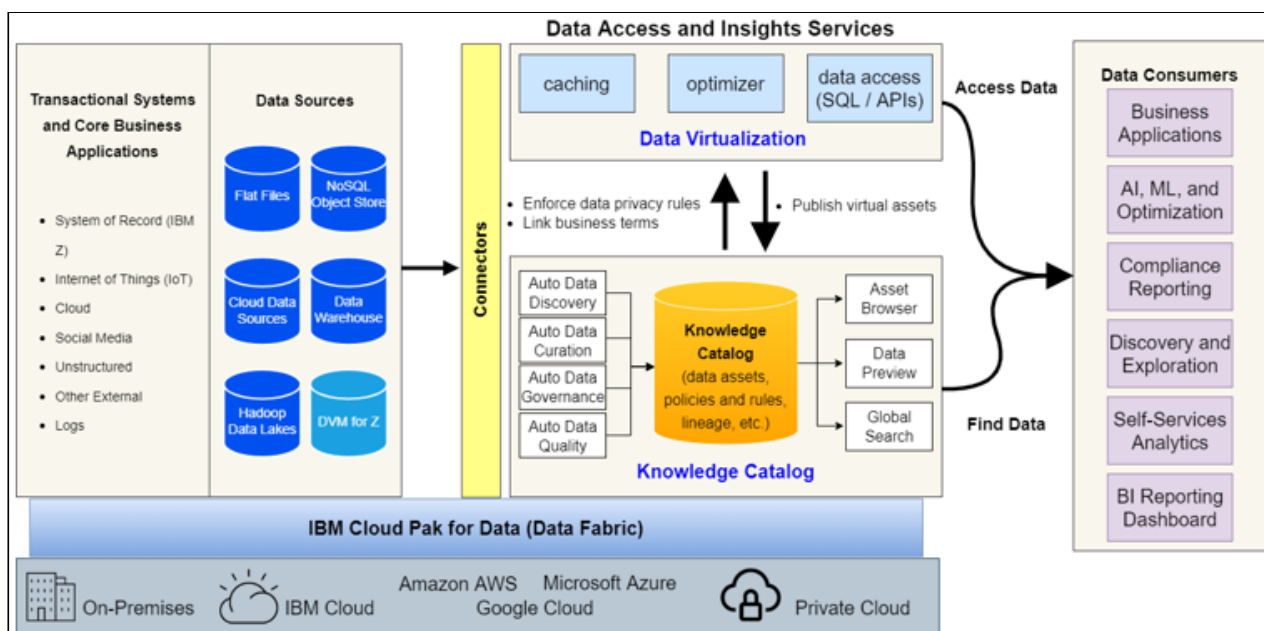


Figure 3-3 Data fabric enabled by Cloud Pak for Data

When data virtualization functions are used within CP4D, governance through centralized authentication and authorization is enforced for platform users to access data sources in a trusted environment. All communication between the environment and applications is securely encrypted by using IBM technology, and SSL/TLS encryption by using standard protocols. Through assigning specific data virtualization user roles that are based on a user’s job description, CP4D provides granular access management to the virtualized assets.

Also, CP4D includes IBM Watson Knowledge Catalog as a core service that serves as a secure enterprise catalog management platform that provides high-quality data assets that are easy to find. The platform is supported by a data governance framework that is used to ensure that data access is compliant with your business rules and standards by preventing access to sensitive information by unauthorized users.

Incorporating a knowledge catalog is key to enabling the self-service data consumption capability of your Data Fabric to easily find and share assets across your enterprise. These self-service capabilities enable pertinent users of IBM Z data within organizations to find quality data faster and spend more time exploring data to provide tangible insights that drive value for your business.

3.1.2 IBM Cloud Pak for Data supported connections

Connections can be added to a broad array of services (data sources) in projects and catalogs within CP4D. These connections can be source connections or target connections.

Source connections are used only to read data; target connections are used to load or save data only. Before you can read data from a data source or load data to a project, you must first create a connection to the data source from a project. Alternatively, before you can add data assets for tables accessible from a catalog, you must first add a connection asset.

Note: When you create a target connection, be sure to use credentials that have Write permission or you cannot save data to the target.

IBM Watson Query is another key service that supports your Data Fabric in CP4D. This support helps to optimize data access and deliver quality data with speed and simplicity. Watson Query and IBM Data Virtualization also help support self-service consumption of your data. It enables users to find, collaborate, and access high-quality data by using one view.

Through tight integration with IBM Watson Knowledge Catalog as part of a data fabric architecture, Watson Query connects governed data quickly to your applications for faster, quality results.

For more information about creating connections with Watson Query, see [Adding and connecting to data sources in Watson Query](#).

Various connections that are supported by Cloud Pak for Data can be classified into the following categories:

- ▶ [IBM services](#)
- ▶ [Third-party services](#)
- ▶ [Supported connection types by tool](#)

Connecting on-premises data sources with Data Virtualization Manager for z/OS

Data virtualization is a key technology to enable a Data Fabric architecture that integrates IBM Z and its originating data. Built into CP4D is data virtualization technology, which is driven by a modern microservice architecture that facilitates access to non-z/OS data sources without knowing technical details, format, or location.

However, this technology does not provide direct access to traditional IBM Z nonrelational data sources. It is this gap that IBM Data Virtualization Manager for z/OS (DVM) is useful.

Rather than physically moving the data from various on-premises and cloud sources by using the standard extract, transform, load (ETL) processes, DVM connects to different sources and integrates only the metadata that is required and creates a virtual data layer in a Data Fabric.

It also provides virtual, integrated views of z/OS-based data sources that are on the Z platform. This integration enables users and applications to use Z data in place, with read/write access without any data movement, replication, or transformation. This configuration significantly reduces the risk of unauthorized access and costly regulatory fines.

When the data on which you are building your machine learning models originates in IBM Z, such as Db2 for z/OS, IMS, IBM MQ, and VSAM, CP4D provides integrated connectivity capabilities through DVM. Its ability to federate these sources with virtually any other data brings the power of IBM Z to almost any application, whether it is mobile, analytic, or cloud. It does so as it uses zIIP specialty processors, with up to 98% zIIP offload.

As a result, minimal extra processing costs are incurred. It also is designed to use IBM Z hardware, and removes the need for IBM Z-specific skills. This connectivity significantly simplifies the development of modern applications that are driven from IBM Z data, which in turn allows developers and data scientists to readily access complex data structures by way of SQL.

DVM connection endpoints

DVM simplifies access to IBM Z data at its source, and provides a unified view of data across different data formats and sources without any need for data movement in a secure, self-governed fashion.

DVM technology provides various methods for client applications to access data in your z/OS environment, including SQL, JDBC, ODBC, DRDA, no-SQL, JSON, SOAP, and RESTful with IBM z/OS Connect. These access methods allow external applications to seamlessly make requests for READ or WRITE operations of underlying data, regardless of location or format.

Figure 3-4 shows the DVM endpoint architecture, which shows how various mainframe and nonmainframe on-premises data sources can be connected to a Data Fabric.

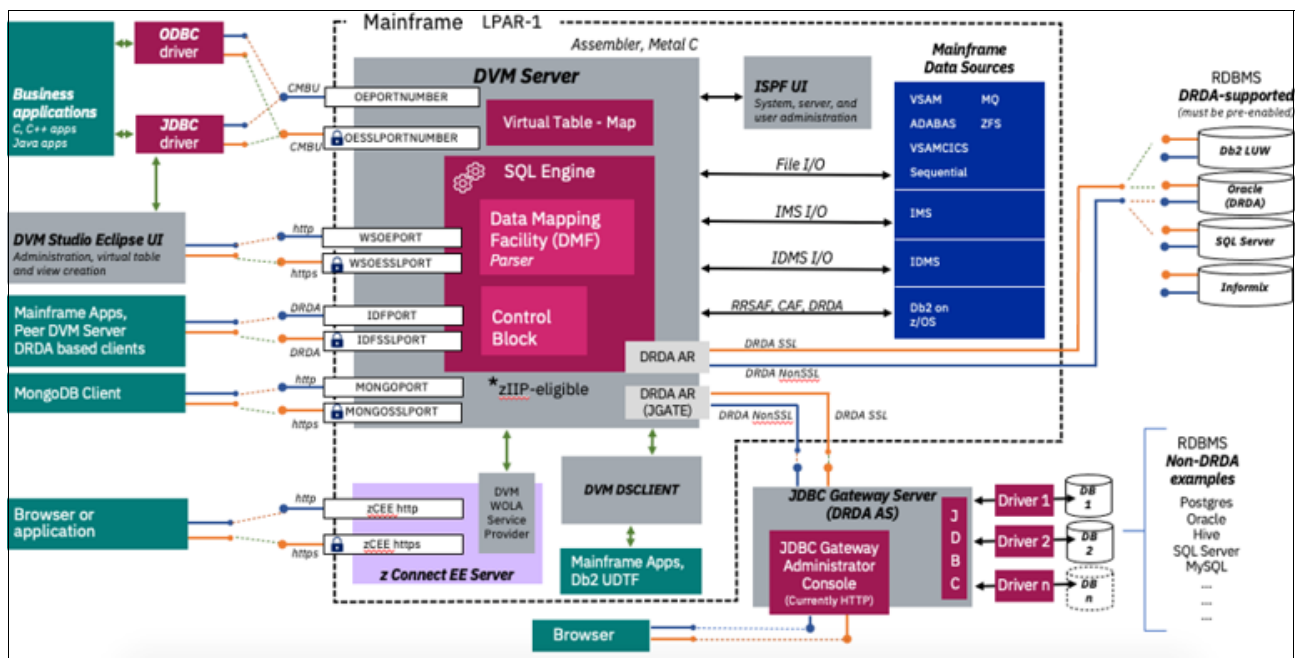


Figure 3-4 DVM endpoint architecture

Note: You must configure the IBM Data Virtualization Manager for z/OS as a data source before you can use it in data virtualization on your Data Fabric. For more information about the configuration process, see this [IBM Documentation web page](#).

Creating a DVM connection in IBM Cloud Pak for Data

To create a connection asset to DVM, the following connection information is required:

- ▶ Hostname or IP address
- ▶ Port number
- ▶ Username and password
- ▶ SSL certificate (if required)

Note: For Private connectivity, you must set up a [secure connection](#) to connect to a database that is not externalized to the internet (for example, behind a firewall).

Before beginning the process of creating a connection, you must first decide which method of creating a connection best supports your use case. The methods to choose from include establishing a connection in a project, catalog, deployment space, or the platform assets catalog.

Complete the following steps to set up a DVM connection from IBM Cloud Pak for Data:

1. Log in to an available IBM Cloud Pak for Data instance by using your username and password with admin privileges.
2. Click the Navigation Menu in the upper left corner of the Home page, expand **Data** and then, click **Platform connections** (see Figure 3-5)

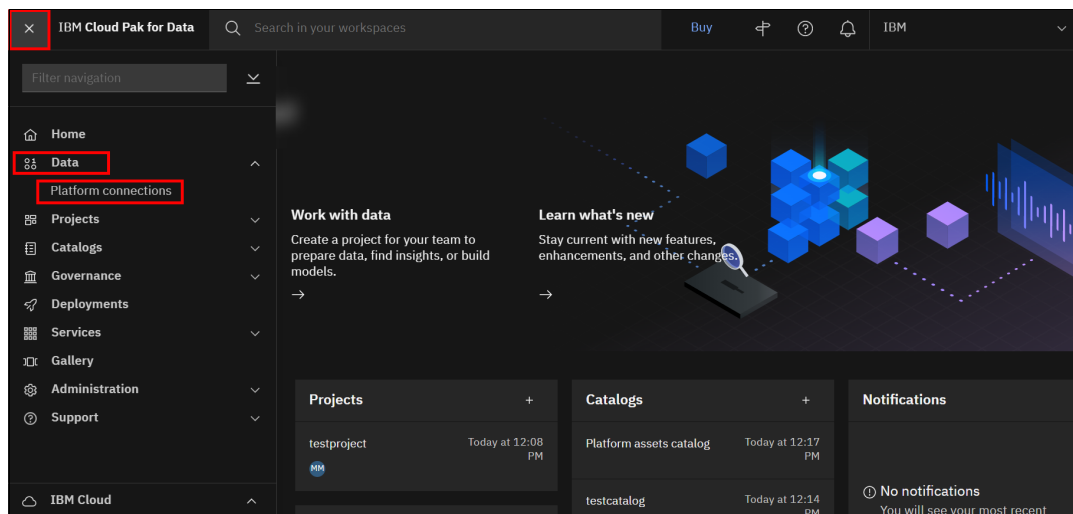


Figure 3-5 Navigate to Platform connections in CP4D

Note: If you encounter the No platform connections message (see Figure 3-6 on page 38) after navigating to the Platform connections page, select **Learn more** for more information about how to resolve the issue.

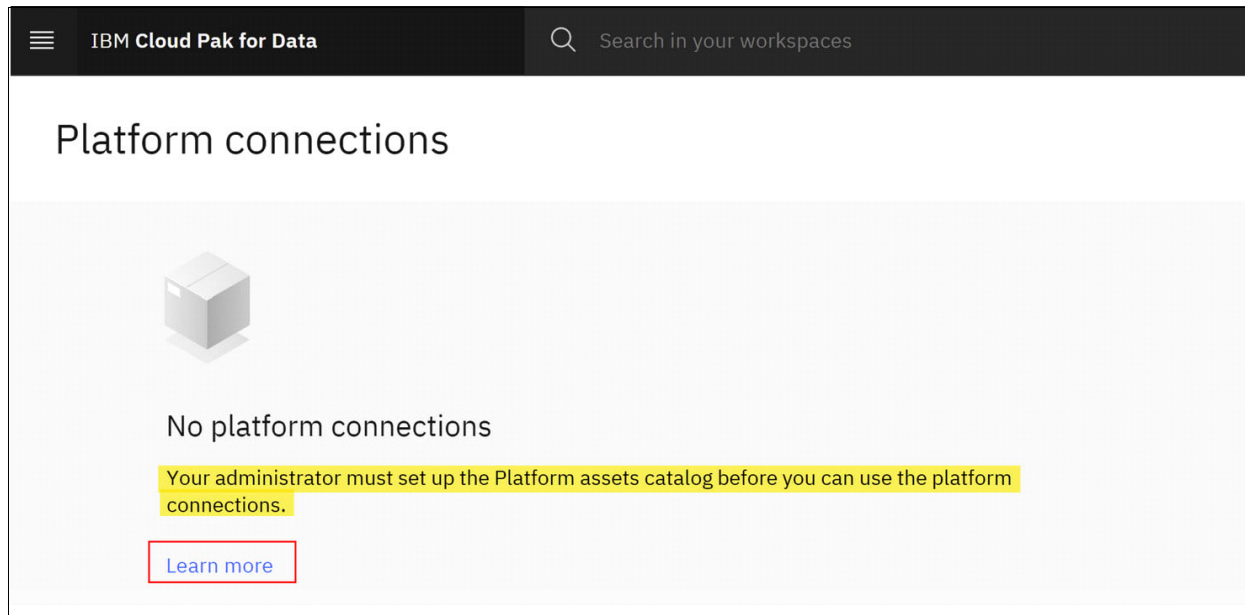


Figure 3-6 No platform connections message

- From the Platform connections page, select the **Connections** tab and then, click **New connection** (see Figure 3-7).

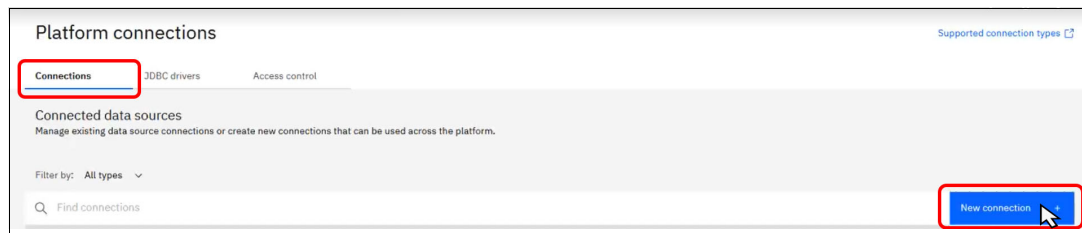


Figure 3-7 Create New connection

- From the Add connection page, click **Data Virtualization Manager for z/OS** as the connection type (see Figure 3-8) and then, click **Select** to open the Create connection page.

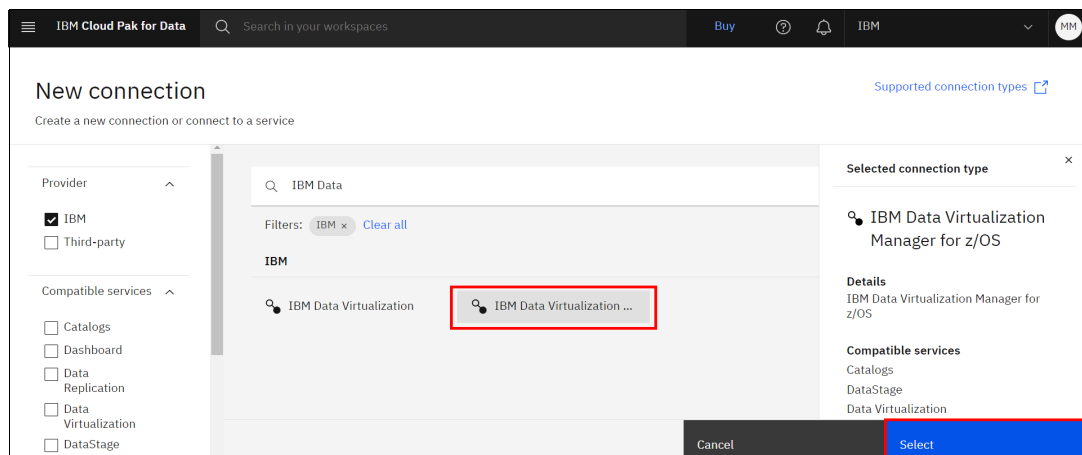


Figure 3-8 Add connection type

5. Complete the following steps in the Connection overview section of the Create connection page (see Figure 3-9 on page 39):
 - a. Enter a name and description (optional) for your Data Virtualization Manager for z/OS connection (names cannot include any white space).
 - b. Select the JDBC JAR files that are available for use in CP4D by using the JAR URL drop-down menu, or select **Upload new file** to upload the following [hJDBC driver JAR files for DVM](#):
 - Driver core implementation file:
dv-jdbc-[version #].jar
 - Logging framework API file:
log4j-api-[version #].jar
 - Logging framework implementation file:
log4j-core-[version #].jar

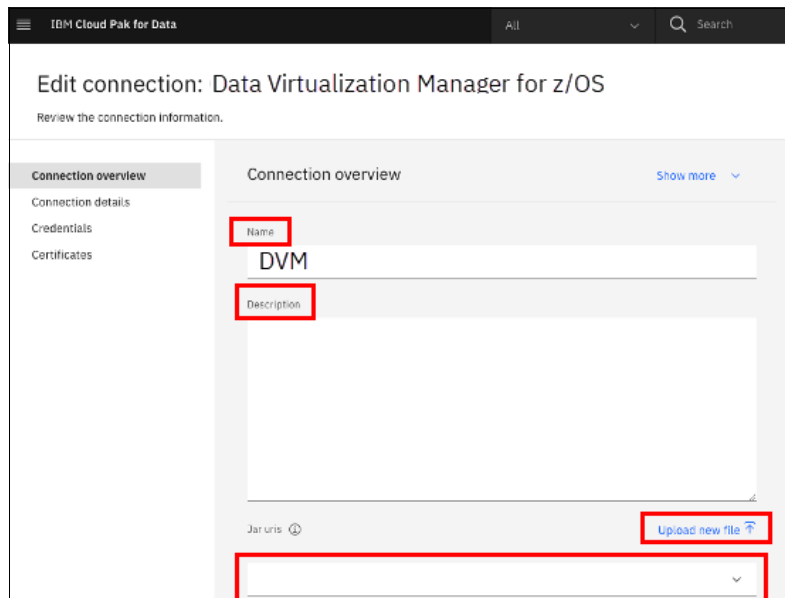


Figure 3-9 Edit connection overview in CP4D

6. Complete the following steps in the Connection details section (see Figure 3-10 on page 40):
 - a. Enter the JDBC URL that must use the following format:
jdbc:rs:dv://host:port;DBTY=value;Key=value;Key=value;...
Where host = network host-name or IP address, and DBTY = database name to connect to after connecting to Data Virtualization Manager.
 - b. For the JDBC driver class name, enter com.rs.jdbc.dv.DvDriver.
 - c. For the JDBC URL prefix, enter jdbc:rs:dv.

IBM Cloud Pak for Data

Create connection: Generic JDBC

Enter the connection information.

Connection overview

Connection details

Credentials

Certificates

JDBC url* ⓘ

jdbc:rs:dv://host:port;DBTY=value;Key=value;Key=value;...

JDBC driver* ⓘ

com.rs.jdbc.dv.DvDriver

JDBC properties ⓘ

Row limit support ⓘ

...

Row limit prefix ⓘ

Row limit suffix ⓘ

Figure 3-10 Edit connection details in CP4D

7. In the Credentials section, enter the username and password that allows access to the data source (see Figure 3-11)

IBM Cloud Pak for Data

Create connection: IBM Data Virtualization Manager for z/OS

Enter the connection information.

Connection overview

Connection details

Credentials

Certificates

Username* ⓘ

You must provide your Username.

Password* ⓘ

You must provide your Password.

Figure 3-11 Enter certificates and create connection

8. Click **Test connection** to ensure that a successful connection can be made (see Figure 3-12)

The test was successful.
Click Save to update the connection information.

Test connection

Figure 3-12 Test connection

9. If the test was successful, click **Create** to create the connection.

If the test was unsuccessful, depending on the error code, refer to the suitable [IBM Cloud Pak for Data Troubleshooting Documentation](#) to resolve the error and retry.

For more information about connecting to a data source by using a JDBC connection, see this [IBM Documentation web page](#).

3.1.3 Previewing and exploring IBM Z data from IBM Cloud Pak for Data

After a connection is established, you can easily discover the IBM Z data from Cloud Pak for Data by using the DVM Server ISPF interface. This connection provides access to persisted data on the mainframe for single queries or JOINS across sources, such as VSAM, IMS, ADABAS, Sequential Files, Syslogs, Logstream, SMF, and Tape.

By using popular, industry-standard APIs, Data Virtualization Manager for z/OS can simplify the development of AI applications by directly accessing IBM Z data with IBM Cloud Park for Data.

By using the SQL editor on the Cloud Pak for Data user console, you can preview, refine, and create virtual assets that map to the DVM information architecture.

Role of a Data Steward in setting up DVM connection

When all of the enterprise data is part of a Data Fabric, the Data Steward can adopt the following best practices about governing IBM Z data by using Cloud Pak for Data:

- ▶ Perform Unified governance across the enterprise.
- ▶ Create public or private projects that are typically used by data engineers or data scientists and request the publishing of virtual assets to the Watson Knowledge Catalog by the data curator responsible for data governance. Once remote Z data assets are published to the Watson Knowledge Catalog, they are accessible for machine learning, analytics, and AI-related activities.

3.1.4 Connect cloud data sources

You can integrate Cloud Pak for Data as a Service with other cloud platforms to configure access to the data source services on that platform. Then, users can easily create connections to those data source services and access the data in those data sources.

Note: Only the Account Owner or Administrator for the IBM Cloud account can configure integrations with other cloud platforms.

You must have the suitable permissions in your cloud platform subscription before you can configure an integration. These permissions are cloud platform-specific. For more information about delivering multicloud data integration by using Cloud Pak for Data as a Service data fabric solutions, see this IBM Cloud Pak for Data [web page](#).

If you use Amazon Web Services (AWS), Redshift (or other AWS data sources), or Microsoft Azure, you also must configure firewall access to allow Cloud Pak for Data as a Service to access data.

After you configure integration and firewall access with another cloud platform, you can access and connect to the services on that platform.

The service instances for that platform are shown on the Service Instances page. To view the platform-specific services instances page in Cloud Pak for Data, select **Navigation Menu** from the home page, expand **Services** and then, select **Services Instances** (see Figure 3-13). Each cloud platform that you integrate with has its own page.

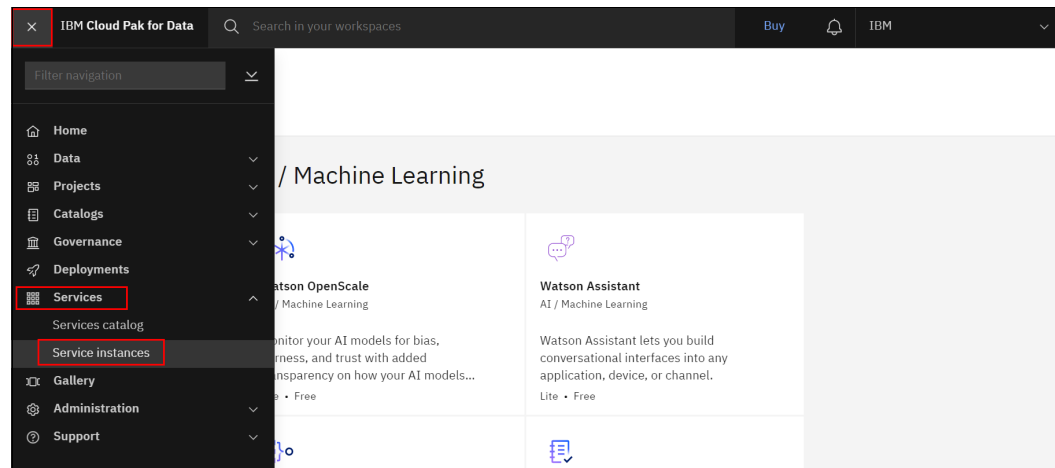


Figure 3-13 View Service instances in CP4D

The data source services in that platform are shown when you create a connection. After you start adding a connection in a project, catalog, or other workspace and the Add Connection page appears, select the **To service** tab to see the list of services, which are arranged by cloud platform (see Figure 3-14).

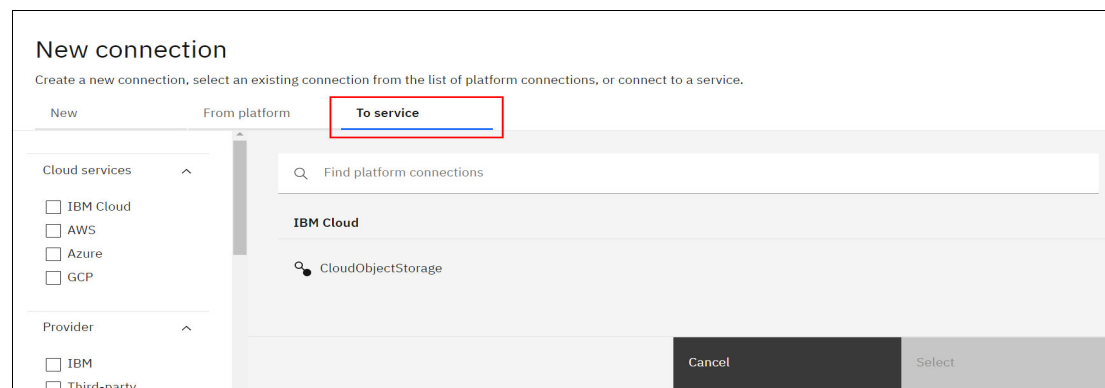


Figure 3-14 New connection To service in CP4D

You can configure integrations by using the following cloud platforms in addition to IBM Cloud:

- ▶ [Amazon Web Services \(AWS\)](#)
- ▶ [Microsoft Azure](#)
- ▶ [Google Cloud Platform \(GCP\)](#)

For more information about integrating with other cloud platforms, see [Integrations with other Cloud Platforms](#).

3.2 Protecting data copies from IBM Z for AI/ML

Suitable precautions must be taken to avoid compromises in data integrity, quality, context, and consistency of the copied data after it is moved from the original source. Most of the issues that are discussed in this section, including data consistency, can be addressed by using a Data Fabric architecture, such as the architecture that is facilitated by Cloud Pak for Data.

The following practices can help protect data copies that are made from IBM Z data sources:

- ▶ Extract metadata from the replication tool, provide data lineage information
- ▶ Ensure to correctly protect the IBM Z data end-to-end with same protection as that is enforced in the source environment.
- ▶ Consider the use of IBM Cloud Hyper Protect Services to protect the copied data. For more information, see the following web pages:
 - [Confidential computing](#)
 - [Go Beyond Confidential Computing with IBM Cloud Hyper Protect Services](#)

3.2.1 Refining data

To refine data, the data often is taken from one location, cleansed and shaped and then, the result is loaded into a different location. For example, you can cleanse and shape tabular data with a graphical flow editor service that is called Data Refinery on Cloud Pak for Data.

Note: The Data Refinery service is not available by default and is not separately installable. The service is installed when an administrator installs the Watson Studio service or the Watson Knowledge Catalog service on the IBM Cloud Pak for Data platform. To determine whether the service is installed, open the Services catalog and check whether the Data Refinery service is enabled and contact your CP4D administrator for assistance.

When you cleanse data, you fix or remove data that is incorrect, incomplete, incorrectly formatted, or duplicated. When you shape data, you customize it by filtering, sorting, combining, or removing columns.

You create a Data Refinery flow as a set of ordered operations on data. Data Refinery includes a graphical interface to profile your data to validate it and give you insights into your data.

To refine on-premises data sources, including IBM Z, you must add connections (see [this web page](#)) to those sources by using DVM. You also must add data assets (see [this web page](#)) from each connection. If you want to save refined data to cloud or on-premises data sources, you also must create connections for this purpose.

Note: If you are copying data from IBM Z source and storing it in cloud or on a different target, make sure to apply the same protection, data security, and governance policies as that of the source data.

3.2.2 Data curation and data quality

Data stewards can curate data into catalogs by using automated discovery and classification. Data quality can be greatly improved by connecting the Data Fabric to the original data source, rather than a copy or modified copy or out of context copy of the data.

Clarity is lost when you reinterpret the data outside the context from which it originally existed, particularly without the business rules from the applications. Every copy of data includes a cost of latency and risk that can result in loss of breadth of data. You must avoid the applicability limitations of replicated data that occur because of the resulting latency that inhibits real-time scoring of transactions (whenever the latent data is not helpful).

3.2.3 ETL/ELT considerations for data replication

Your organization might be performing real-time data integration through replication from IBM Z to the cloud. The replicated data can also be part of your Data Fabric.

Your organization also might transforming IBM Z data for use elsewhere through direct access with ETL tools, such as IBM DataStage. Again, this transformed data also can be brought into your Data Fabric securely without losing data consistency and breadth with the help of IBM Cloud Pak for Data.

For more information about data replication on the Z platform, see 3.2.4, “Tools for data movement and replication from IBM Z” on page 45.

Closing the data latency gap

Data virtualization connection on the Data Fabric is a cost-effective substitute for traditional data movement processes, such as incremental copy, data replication, and ETL.

With data virtualization, you can access data where it originates to reduce the time and resources that are used to combine data from multiple systems. Less time and resources use can translate into direct savings.

The use of data virtualization technology can support greater flexibility and agility, which is core to digital transformation. Also, it is critical that you have a good understanding of the integrated nature of your data. The following questions help facilitate this understanding:

- ▶ What happens to the governance rules when data is moved off from IBM Z?
- ▶ Who has access to that data after it is copied?
- ▶ How the copied data is being secured?
- ▶ Who is responsible for protecting the copied data?

Other obvious reasons for keeping data on IBM Z platform include the following examples:

- ▶ Data sovereignty requirements: Sensitive data cannot be moved by law.
- ▶ Data context: When you move partial data, the overall business context of the data is lost.
- ▶ The use of AI: Real-time inferencing from transactional data.
- ▶ Data gravity and latency benefits.

Moving data is never recommended when we are discussing best practices to maintain the security and privacy of that data. However, some workloads are not well-suited for virtualization; therefore, they must be replicated by using data movement practices.

Moving data might be permissible in the following scenarios:

- ▶ To avoid contention with the z/OS operational workload, you might need to move the data to an off-platform lake house or data lake.
- ▶ When no need exists for real-time inferencing.
- ▶ Historical data analysis.
- ▶ Latency is not an issue.

3.2.4 Tools for data movement and replication from IBM Z

The following sections describe the tools that can be used to securely move the data from the original source (IBM Z) to a target data store on within your Data Fabric.

Note: From a data protection, security, and governance perspective, avoid data movement (and replication) whenever possible and encourage data virtualization connections. This statement is especially true for IBM Z data, where the data can inherently benefit from world's most securable platform.

Replication by using IBM Data Replication tool

IBM Data Replication for Continuous Availability for CP4D supports highly available Db2 and Db2 Warehouse environments by synchronizing table and schema content, whether on the same platform, across the data center, or in a different part of the world. It packages the IBM Q® Replication technology for the CP4D platform, which extends replication facilities to organizations with Db2 and Db2 Warehouse on the CP4D platform.

It also provides containerized software (delivered as CP4D cartridges) so you can deploy data replication for continuous availability for CP4D to help modernize your platform at your own pace.

Note: Many business drivers are available for replicating IBM Z data in real time. Replication is often deployed as a component of a larger solution.

The IBM Data Replication tool feature for IBM Z data source includes the following highlights:

- ▶ Log-based change data capture ensures no effect on an application program's performance or availability.
- ▶ It can propagate thousands of data changes per second.
- ▶ Supports replicating changes that were made to Db2 for z/OS, IMS, and VSAM data.
- ▶ Supports various replication targets on-premises or in-cloud, such as the following examples:
 - Relational DBMS, such as Db2, Oracle, and SQL Server
 - Big data targets, such as Hadoop and Kafka
 - Cloud-based data targets

The following use cases are common for IBM Z data replication:

- ▶ The need for mission-critical, workload-level continuous availability at distance to support planned and unplanned outages.
- ▶ The need for a real-time data warehouse to provide the acceleration of analytical query processing (by using IDAA), offloading the System of Records (SOR).

- ▶ The need for advanced analytics processing on real-time data by using various modern analytics platforms, such as Hadoop and Apache Spark.
- ▶ The need to support various application modernization use cases in the cloud and on-premises.
- ▶ The need to support distributed query use cases, such as CQRS, offloading inquiry from the SOR.
- ▶ The need to support the aggregation of data from multiple sources into an operational data store (ODS) or a data lake.
- ▶ The need to support the offload of the SOR from directly supporting multiple consumers, evolving consumers, or both by replicating into a data lake.

For more information about replicating Z data, see the IBM Cloud Architecture Center for the [Replicate IBM Z Data pattern](#).

Synchronizing data from Db2 for z/OS to Db2 Data Gate on CP4D

The data on Db2 for z/OS can be synchronized with a Db2 LUW database on CP4D by using integrated synchronization technique. This technique is the same synchronization technique that is used on Db2 for z/OS data with IBM Db2 Analytics Accelerator for z/OS appliance, as shown in Figure 3-15.

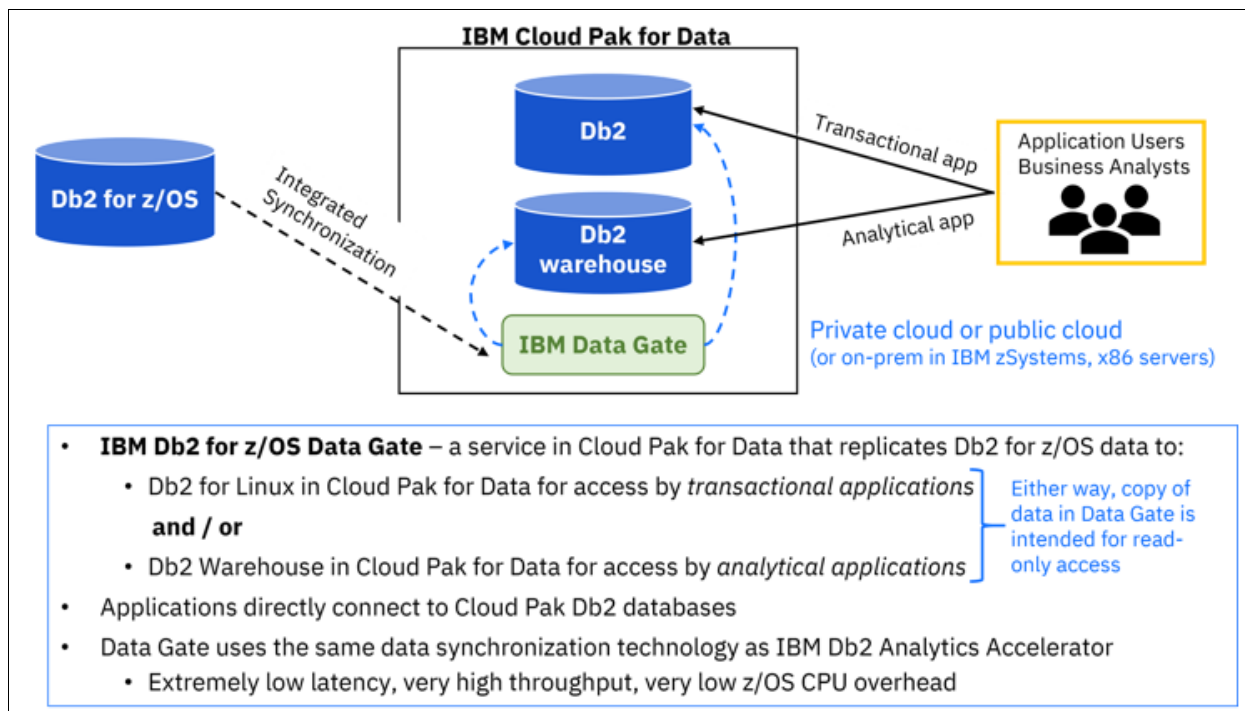


Figure 3-15 Optimized provisioning of Db2 for z/OS data in the cloud

Encrypting outbound network traffic from Db2 for z/OS to Db2 Data Gate

To encrypt network traffic between Db2 for z/OS and a Db2 Data Gate instance on CP4D, specific software components are required.

On CP4D, Db2 Data Gate defines an Red Hat OpenShift route when the service is provisioned. A Red Hat OpenShift route exposes a service with an externally reachable hostname, such as `dg01.apps.dgsvt2.os.fyre.ibm.com`.

For more information, see [Creating a Db2 Data Gate instance](#).

On z/OS, various components of the IBM z/OS Communications Server must be configured, which uses AT-TLS security protocols. In addition, a certificate and an RSA key pair are required.

For more information about encrypting outbound network traffic between Db2 for z/OS and Db2 Data Gate in CP4D, see this [hIBM Documentation web page](#).

The following IBM tools are available for data transformation:

- ▶ An industry-leading ETL tool, IBM DataStage is for your hybrid cloud environment. It also can be deployed as a service on CP4D.
- ▶ IBM Z Digital Integration Hub can be used with Data Virtualization Manager for z/OS to build in-memory caches of z/OS data for use on cloud-based processes.
- ▶ If it is available in your IBM Z environment, the IBM Db2 Analytics Accelerator Loader tool can be used to transform and load nonrelational data into the IBM Db2 Analytics Accelerator and accessed by way of Db2 for z/OS Connection. This tool also can speed up extracts of SQL-based transformation of z/OS data by effectively performing in-database transformation.

3.2.5 Data and system security

Security involves protecting systems and data through intrusion prevention, detection, and response to improper access from within and outside your enterprise. Such improper access can result in information being altered, destroyed, or misappropriated or it can result in misuse of your systems to attack others.

Without a comprehensive approach to security, no IT system or product can be considered completely secure, and no single product or security measure can be completely effective in preventing improper access. IBM systems and products are part of a regulatory compliant, comprehensive security approach, which necessarily involves more operational procedures, and might require other systems, products, or services to be most effective.

Note: IBM does not warrant that any systems, products, or services are immune from, or will make your enterprise immune from, the malicious or illegal conduct of any party.

3.3 Data gravity with AI on IBM Z through data virtualization

Data fabric architecture in CP4D enables connections directly to IBM Z data sources by using DVM. AI is enabled on the platform without moving the data out of the security-rich environment of IBM Z.

It is on IBM Z where unique data sources, transactional applications, data virtualization, and Open Source technologies combine to offer an incredible opportunity for machine learning initiatives.

The following tasks are performed before a machine learning model is built while using IBM Z data sources:

- ▶ A connection from CP4D to DVM is created (see “Creating a DVM connection in IBM Cloud Pak for Data” on page 37).
- ▶ Data assets from CP4D running on x86 are discovered (see “Discovering assets” on page 97).
- ▶ The IBM Z data is transformed by using Data Virtualization Manager for z/OS (see “IBM Data Virtualization Manager for z/OS” on page 101).

DVM offers organizations the potential to modernize the way they develop applications by using Red Hat OpenShift Container Platform, simplifying access to traditional nonrelational IBM Z data sources, and providing access to an extensive number of data sources by directly connecting to them. With this winning combination, confidential computing is closer to reality than ever before. Other problems are solved that are faced by businesses in the areas of skills shortage and DevSecOps challenges. Mainframe modernization also is enabled.

Connecting through DVM allows you to enjoy the benefits of Data Fabric capabilities, such as centralized data governance of connected data in CP4D. It also enables you to keep the data on IBM Z and avoid moving it off-platform for AI/ML workflow, as shown in Figure 3-16.

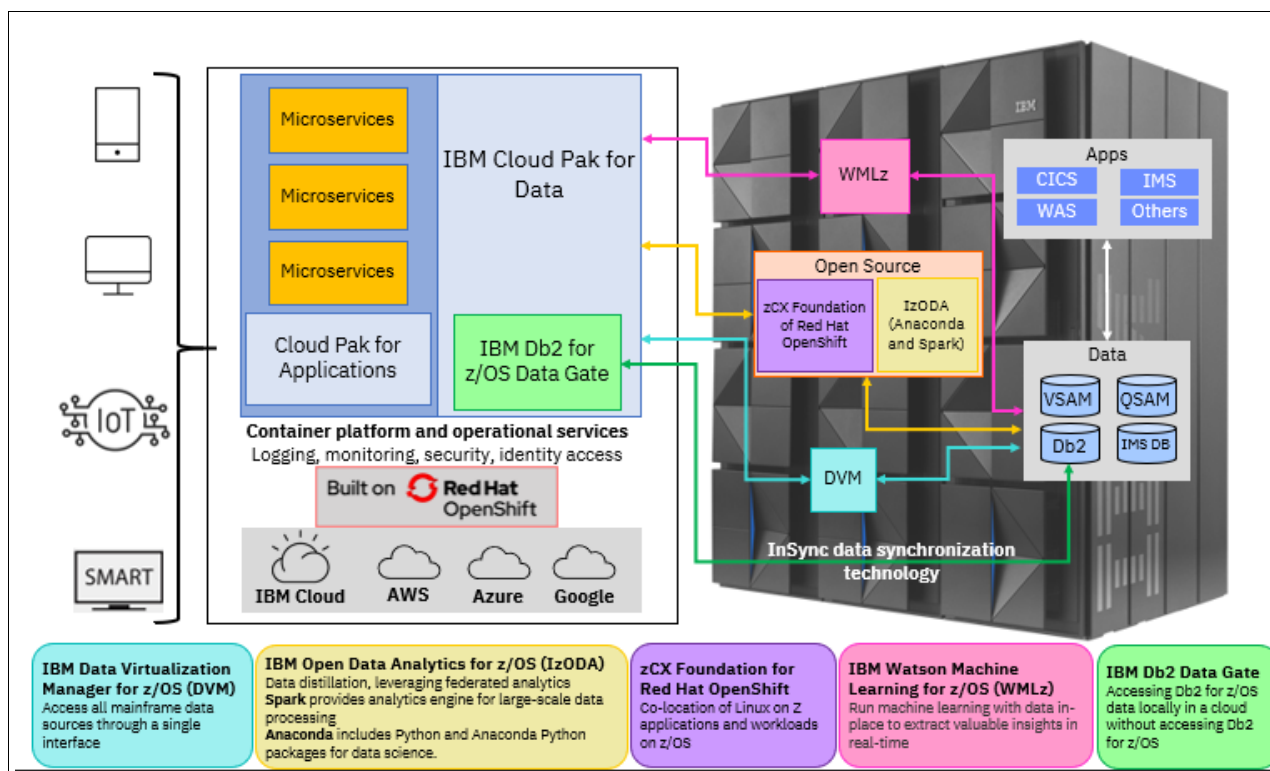


Figure 3-16 Data gravity enabling technology on IBM Z

3.3.1 Data gravity and latency considerations

Data gravity is about keeping the data where it is and moving the computation to the data. If most of the data that is used in building your machine learning models originates from Db2 for z/OS or other nonrelational IBM Z data sources, such as IMS, IBM MQ, and VSAM, we can virtualize the data in IBM Cloud Pak for Data through DVM connection. This connectivity significantly simplifies the development of analytics applications that are driven from IBM Z data, which allows developers and data scientists to readily access complex data structures by way of SQL, which avoids latency issues.

For more information, see “Closing the data latency gap” on page 44.

Watson Machine Learning for z/OS

Watson Machine Learning for z/OS (WMLz) is an end-to-end machine learning platform for AI on z/OS that enables model training, deployment, and monitoring on the platform. It also provides the flexibility to train models in any framework, on any platform, and then deploy those models on Z to benefit from the unique advantage of real-time, in-transaction inferencing.

After a model is deployed with WMLz, the platform provides continuous monitoring of the model’s accuracy to promote trust in your model results. For example, machine learning models can be developed by data scientists on Linux systems, imported, and deployed at point-of-transaction in a z/OS environment easily from CP4D. Providing the capability to run AI models in a cost-effective and efficient way that does not affect operational processing.

With Watson Machine Learning for z/OS, real-time scoring is feasible for high-volume, response time-sensitive z/OS-based transactional and batch applications. Predictive models that are developed by using machine learning technology offer the opportunity to embed AI models into operational applications, and score every transaction without significant impact to SLAs or cost.

Insights that are generated from these AI-infused applications can reveal business opportunities to be quickly acted upon by the business to increase revenue, decrease cost, and mitigate fraud in real-time. Examples of use cases that can benefit from real-time, in-transaction inferencing include customer cross-selling and up-selling, risk mitigation, and fraud detection among others.

IBM zCX Foundation for Red Hat OpenShift and IBM Open Data Analytics for z/OS (IzODA)

zCX Foundation for Red Hat OpenShift (zCX for Red Hat OpenShift) technology brings all the benefits that are associated with the Red Hat OpenShift Container Platform to z/OS application developers and data scientists.

It also reduces latency and allows you to bring the AI and analytics computations closer to the IBM Z data by using modern Open Source technology platform. In this context, zCX Foundation for Red Hat OpenShift can be considered another enabling technology for data gravity in a modernized IBM Z environment. IzODA brings the benefits that are associated with Apache Spark to z/OS application developers and data scientists.

What does this mean for IBM Z data?

We can use DVM connection to get to mainframe data from the cloud for AI model training on IBM z16™, which uses the on-chip AI Accelerator from CP4D. Then, we can deploy the model on zCX for Red Hat OpenShift Container Platform without ever moving the Z data off platform, all with unified data governance and no z/OS specific skills from CP4D on Red Hat OpenShift Container Platform.

3.3.2 Building AI models with IBM Cloud Pak for Data

CP4D is a single end-to-end platform that can provide all of the key capabilities of a Data Fabric architecture for data self-service, automated data integration and enrichment, data virtualization, data governance and protection, and AI.

It provides a single point for data scientists, data engineers, and data stewards to collaborate on a single platform to acquire, govern, and extract best insights from any data anywhere in your enterprise in the least amount of time. For more information about typical CP4D tasks for building an AI model that uses IBM Z data sources, see Appendix A., “Unlock IBM Z data for your AI journey” on page 95.

For more information about all supported connections for automated discovery by using IBM Watson Knowledge Catalog in CP4D, along with the corresponding connection types and whether quick scan of the data is supported, see [Discovering assets \(Watson Knowledge Catalog\)](#).

For more information about data discovery, see the following Cloud Pak for Data – Watson Knowledge Catalog documentation:

- ▶ [Running automated discovery](#)
- ▶ [Working with the discovered assets](#)



Data governance and discovery

This chapter discusses the effect of data discovery in enabling governance. Data discovery includes analyzing the metadata and transforming it into different forms for the greatest possible use.

Various tools provide data governance, such as Watson Knowledge Catalog, which is one of the core services in Cloud Pak for Data as a Service and an essential part of the [Data Fabric](#).

Watson Knowledge Catalog provides a metadata-based knowledge core, data self-service, and unified governance. Also, IBM InfoSphere® Information Server for System Z provides capability to integrate data directly on the mainframe.

This chapter includes the following topics:

- ▶ 4.1, “Data governance” on page 52
- ▶ 4.2, “Taking action with metadata” on page 55
- ▶ 4.3, “Data cataloging” on page 57
- ▶ 4.4, “Governance with IBM Watson Knowledge Catalog” on page 58
- ▶ 4.5, “IBM Watson Machine Learning” on page 64

4.1 Data governance

Most organizations are transforming their business priorities. One of the main priorities of any organization today is the prioritization of data and how to perform an efficient data analysis. This efficiency is enabled by using artificial intelligence (AI), which automates data access, use, management, movement between different platforms, and more. This transformation around data creates a necessity for automation, AI, and self-service across the organization to reduce or eliminate the manual processes and accelerate the business processes.

This transformation to an intelligent, data-centric culture requires a solid technological foundation where trusted and business-ready data is made available to the masses and leads to applying machine learning (ML) throughout the technology stack for the automation of manual tasks.

After organizations master an agile and trusted data pipeline to serve ML, they can then deploy AI initiatives that accelerate their paths to transformation. For more information about intelligent data governance and trusted data lake, see the ESG white paper [Intelligent Data Governance for a Trusted and Business-Ready Data Lake](#).

What is data governance?

Data governance refers to the process of managing the quality of data within an organization to ensure that data always is accurate, available, consistent, secure, and usable during its lifecycle.

Business analysts and data scientists search for information across the enterprise to gain insight and understanding of that information to support business needs. With the most up-to-date results, they can answer the following questions:

- ▶ Where is the data and can we find it?
- ▶ What is the meaning of the data?
- ▶ What are the requirements for its handling or usage?
- ▶ What shape is it in? Can we use it?
- ▶ Where has the data been?
- ▶ What other systems or users are accessing or using the data?
- ▶ Can we trust the data?
- ▶ Who is its owner?

For more information about data governance, see this IBM Documentation [web page](#).

4.1.1 Necessity of data governance

Data governance ensures that the organization's data is connected across systems and provides meaningful and trusted information to the business. When the data governance is done effectively, it supports the ease and the quality of data analytics. Preserving the quality of data in analytics accelerates the digitization of organizations by giving a better view for decision making and improving the support of different processes.

4.1.2 Data governance tools

Data governance solutions and tools provide understanding, security, and trust around an organization's data. As companies scale and accumulate more data sources and assets, they must determine the suitable big data environments for storage and access purposes.

They also need an architecture to govern sources, integrate them, and make them available across an organization. Data integration becomes increasingly important because it affects the workflows and decision making of various teams.

Use data governance tools in your overall strategy for data management and as part of a complete DataOps practice. Data governance solutions help you understand what data you have, where that data is stored, and how it can be used.

For more information about governance solutions, see this IBM Analytics [web page](#).

Data governance tools feature the following benefits:

- ▶ Better data security and compliance
- ▶ Improved data quality
- ▶ Faster automation

4.1.3 Data lakes

A *data lake* is a centralized data repository for management of large data volumes and serves as a foundation for collecting and analyzing structured, semi-structured, and unstructured data in its native format.

Data lake provides the following benefits:

- ▶ Drives new insights from the collected and analyzed data.
- ▶ Better predictions for enhanced the decision making.
- ▶ Improved optimization that leads to minimizing the costs.

Data warehouses versus data lakes

Although data lakes and data warehouses both store data, each repository has its own requirements for storage, which makes it an ideal choice for different scenarios.

Data warehouses require a defined schema to fit specific data analytics requirements for data outputs (dashboards, data visualizations, and other business intelligence tasks). These requirements often are specified by business users and other relevant stakeholders who use the reporting output regularly.

The underlying structure of a data warehouse often is organized as a relational system (that is, in a structured data format) and sources data from transactional databases.

Data lakes incorporate data from relational and nonrelational systems, which allows data scientists to incorporate structured and unstructured data into more data science projects.

For more information about data lakes and comparisons to other data storage methods, see this IBM Analytics [web page](#).

4.1.4 Data lake architecture

Figure 4-1 shows the base solutions architecture of the data lake. This architecture includes the following main components:

- ▶ Data science workbench is a set of tools that are used for data science and analytics. The relevant IBM product for these tools is IBM Watson Studio.
- ▶ Knowledge Catalog is a data catalog that is used to understand and use the needed data, and share data assets and models with other members. The relevant IBM product is IBM Watson Knowledge catalog.
- ▶ Data warehouse is a consolidated data repository from multiple data sources to support business analytics and reporting. The relevant IBM products is IBM Cloud Pak for Data System with IBM performance Server for PostgreSQL.

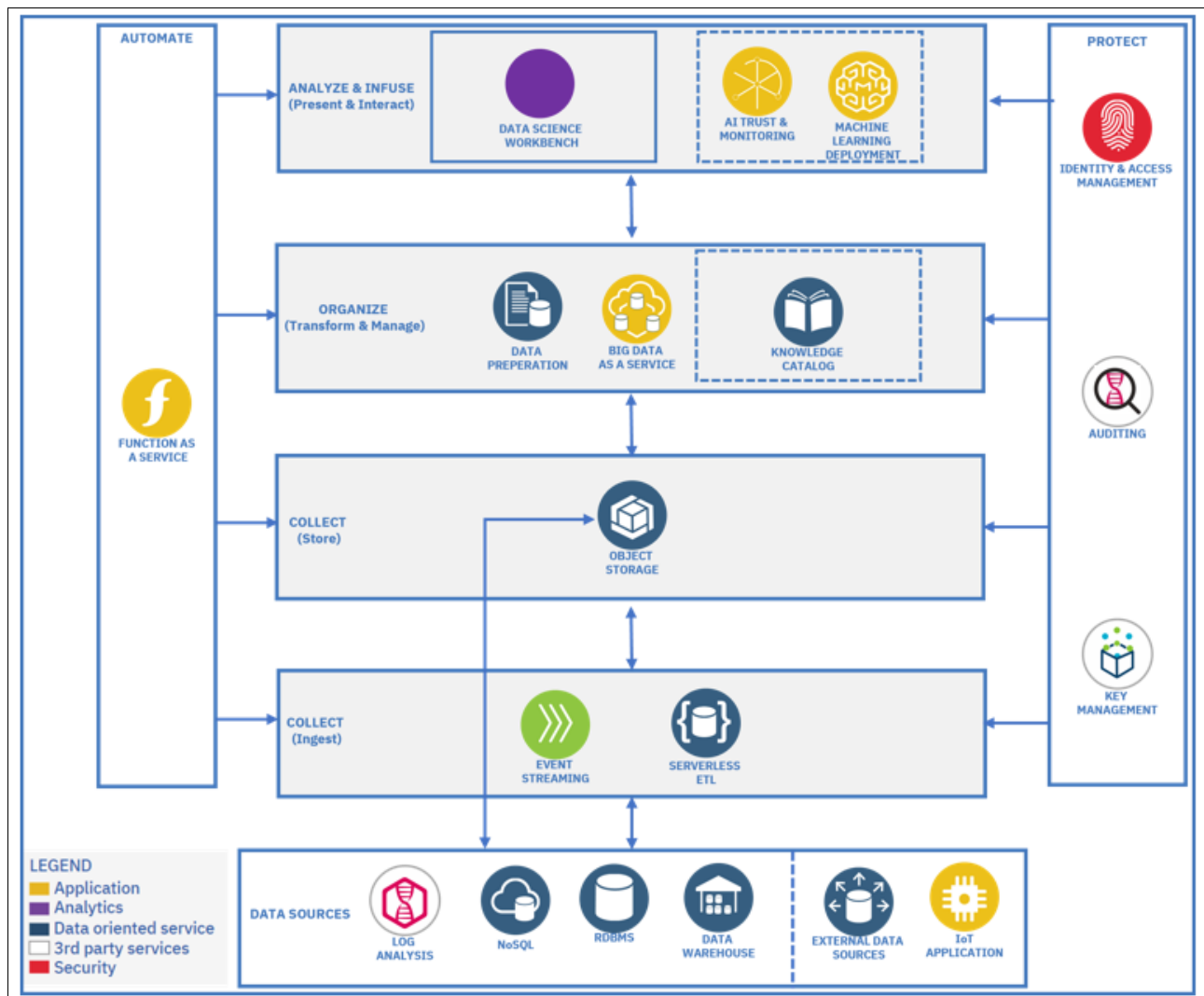


Figure 4-1 Cloud data lake architecture

For more information about the data lake architecture for big data modernization, logging, IoT, and the components that are shown in Figure 4-1, see the [IBM Architecture Center – Cloud data lake](#).

4.2 Taking action with metadata

This section provides an overview of metadata, its types, and how data changes can be extracted from it.

4.2.1 What is metadata?

Metadata is information about data; that is, data that describes or summarizes data. It describes the data that is flowing through your job in terms of column definitions, which describe each of the fields that make up a data record.

Metadata describes the data that is flowing in an organization in terms of column definitions, which describe each of the fields making up a data record. Metadata is handled through table definitions, or schema files.

For more information about metadata and IBM tools for handling metadata, see this IBM Documentation [web page](#).

4.2.2 Types of metadata

The following types of metadata are available:

- ▶ **Passive metadata:** Technical data that provides information about systems, databases, files, and processes.
- ▶ **Active metadata:** Defines all of the changes that occur to the data. The active metadata includes the business side, which provides a business context and a business name for assets that are created and managed by other applications. It also includes the operational side that describes the job runs, database tables, or data files that are affected by changes.

4.2.3 Metadata discovery

The discovery of metadata involves the use of automated tools for extracting and understanding meaningful insights from the metadata. To access a DVM metadata source, a JDBC wrapper server is created to form the connection in which the information about the data source and objects that must be accessed are provided.

Metadata discovery can be done by using different methods, including automation, linking, and semantic enrichment of your metadata and automated regeneration of metadata whenever data sources are updated.

In data linking, the data is connected to other pieces of data to become more useful and understandable. By using semantic enrichment, the missing information in the metadata can be acquired based on the information that is in the data source.

Automated discovery in IBM Watson Knowledge Catalog

IBM Watson Knowledge Catalog is an add-on service that is available in IBM Cloud Pak for Data. It enables the opportunity to perform metadata discovery in an automated way. When you run a discovery, metadata is imported into the default catalog.

With IBM Watson Knowledge Catalog, you can conduct the data discovery by using different approaches, including the following examples:

- ▶ **Analyze columns:** Runs a column analysis on the data assets to identify properties, such as data class, data type, and format.
- ▶ **Assign terms:** Assigns business terms to imported assets that are based on name similarity, data classification, and ML insights. In the discovery results, you can publish selected term assignments or all of them.
- ▶ **Analyze data quality:** Runs a quality analysis on the data assets to scan for common dimensions of data quality, such as missing values or data class violations.
- ▶ **Publish results to catalog:** Publishes discovery results to the default catalog to make them available to other users. Results include term assignments, quality score, and other quality-related statistics.

Note: The catalog settings for duplicate asset handling are *not* applied.

- ▶ **Use data sampling:** Runs column analysis or data quality analysis on a data sample. Sampling can be configured globally for all data quality projects, specific data quality projects, or at the discovery job level. Only users with the Administrator role can modify sampling settings at any of these levels. The global sampling settings are applied unless they are overwritten at the project or job level.

For more information about running automated discovery by using IBM Watson Knowledge Catalog, see this [IBM Documentation web page](#).

For more information about running data discovery by using IBM InfoSphere Information Server, see this [IBM Documentation web page](#).

4.2.4 Capturing changes in the data by using metadata

Capturing the changes in the data by using metadata throughout its lifecycle can be done through *metadata management*, which is the practice of managing knowledge about the information.

Metadata refers to a rich structure of knowledge. This structure captures the meaning of a term or data asset, its relationships to other assets, and rules that might apply to it to determine the quality, policies, and regulations that specify its use.

The following methods and data categorizations can be considered for easing the capture of changes:

- ▶ **Enrich the data through labeling and tagging with metadata** (by using automation and AI, which makes it scalable and repeatable).
- ▶ **Add governance, compliance, and security tags and flag data as private, confidential, or sensitive.**

- ▶ Identify associated compliance, governance, and security requirements by way of metadata tagging, which enables audit or logging of access and use, PII flag, association with any governance, or compliance regimes (such as HIPAA), classify within security classification schemes, turn on encryption, or other masking method.
- ▶ Program through export tools and APIs to access incoming data that can include rules, enforce policy, identify data, and apply related tags.
- ▶ Data fingerprinting for identifying and tracking the data across the system.
- ▶ Discover relationships between and among data items.

For more information about metadata management, see *Metadata Management with IBM InfoSphere Information Server*, [SG24-7939](#).

4.3 Data cataloging

A *data catalog* is a detailed inventory of all data assets in an organization. It is used to help data professionals quickly find the most suitable data for any analytical or business purposes.

A data catalog uses metadata to create an informative and search-able inventory of all data assets in an organization. These assets can include, but are not limited to, structured and unstructured data, reports and query results, and connections between databases and ML models.

For more information about data catalog, see this IBM Cloud [web page](#).

4.3.1 Contextualize data by using AI

AI is proving to be a necessity for extracting useful insights from information. The increase in the volume of information, along with its diversity and complexity, creates impediments for its manual discovery, identification, organization, or application or regulation. That is why infusing AI into the data catalog is increasingly critical.

Infusing AI changes the nature and benefit that data catalogs historically delivered. As a result, AI-infused data catalogs can democratize data access or present tribal knowledge and automate and enact processes, perform thorough analysis, and contextualize information.

This ability enables self-service management of enterprise data, uses automation, and is backed by active metadata and policy management. Also, it provides different features, such as automated profiling, inventorying assets, and assigning business terms from the business taxonomy to new assets. This intelligence in contextualizing data is brought by IBM Watson Knowledge Catalog.

For more information about bringing automation and intelligence to data catalogs, see this [IBM Journey to AI blog](#).

4.4 Governance with IBM Watson Knowledge Catalog

IBM Watson Knowledge Catalog is a data catalog tool that powers intelligent, self-service discovery of data, models and more. The cloud-based enterprise metadata repository activates information for AI, ML, and deep learning. Access, curate, categorize and share data, knowledge assets and their relationships, wherever they are stored. It provides end-to-end governance, delivery of quality data, accelerated collaboration, protection, and compliance.

For more information about IBM Watson Knowledge Catalog, including use cases, features, and benefits, see this [web page](#).

4.4.1 Overview of CP4D

IBM Cloud Pak for Data is a cloud-native solution that enables working with the data quickly and efficiently. Cloud Pak for Data fosters productivity by enabling users to find data or request access to data. With modern tools that facilitate analytics and remove barriers to collaboration, users can spend less time finding data and more time using it effectively.

The Data Fabric architecture that is provided by Cloud Pak for Data enables your organization to accelerate data analysis for better, faster insights.

With the capabilities of the Cloud Pak for Data, Data Fabric architecture, you can:

- ▶ Simplify and automate access to data, across multi-cloud and on-premises data sources, without moving data.
- ▶ Universally safeguard the use of all data, regardless of source.
- ▶ Provide business users with a self-service experience for finding and using data.
- ▶ Use AI-powered capabilities to automate and orchestrate the data lifecycle.

A Data Fabric that is enabled by using Cloud Pak for Data and the corresponding IBM components that support them feature the following main capabilities:

- ▶ Metadata-based knowledge core

Data is enriched with metadata that informs the semantic search for data and curated into catalogs by using automated discovery and classification. The Watson Knowledge Catalog service in the Cloud Pak for Data is used to classify, discover, and organize data, which enables the use of a metadata-based knowledge core.

Also, the data assets can be enriched by creating and assigning custom governance artifacts, such as business vocabulary. They can also import ready to use collections of metadata from industry-specific Knowledge Accelerators.

- ▶ Data self-service in catalogs

The data catalogs, such as IBM Watson Knowledge Catalog, feature data from across the enterprise. The AI-powered semantic search and copy can be used to copy data assets from a catalog into a project and search the data asset for information. The catalogs can be used by different teams in the organization to collaborate in preparing, analyzing, and modeling the data. The technology that enables this kind of visibility and collaboration are available features in Watson Knowledge Catalog service.

► Automated data integration

After preparing the data consumption and providing the access to it in the data architecture, some of the data needs transforming and cleansing by using the Data Refinery tool. Some data requirements might need visual representation of data and regular publishing of updates to data assets. These automated data integration features and the ability to process the data at its location versus moving the data to the processing platform are enabled by the following components:

- Data Virtualization service
- DataStage service
- Data Refinery tool
- Cloud Pak for Data platform

► Unified data governance, security, and compliance

Data protection rules can be created to automatically enforce uniform data privacy across the platform by using features that are provided by Watson Knowledge Catalog service in Cloud Pak for Data.

They can also protect the sensitive data by using data masking, preventing the need for multiple copies of the data. The Knowledge Accelerators that are available in Cloud Pak for Data can be used to import compliance metadata to enrich data assets.

► Unified lifecycle

The unification of the data lifecycle makes it easier for users to monitor and use different types of data in a consistent way. The Cloud Pak for Data platform offers many features for creating different types of data assets as data sources, data models, and more. The data assets can be categorized in different workspaces and moved across them.

The Cloud Pak for Data landing page is shown in Figure 4-2.

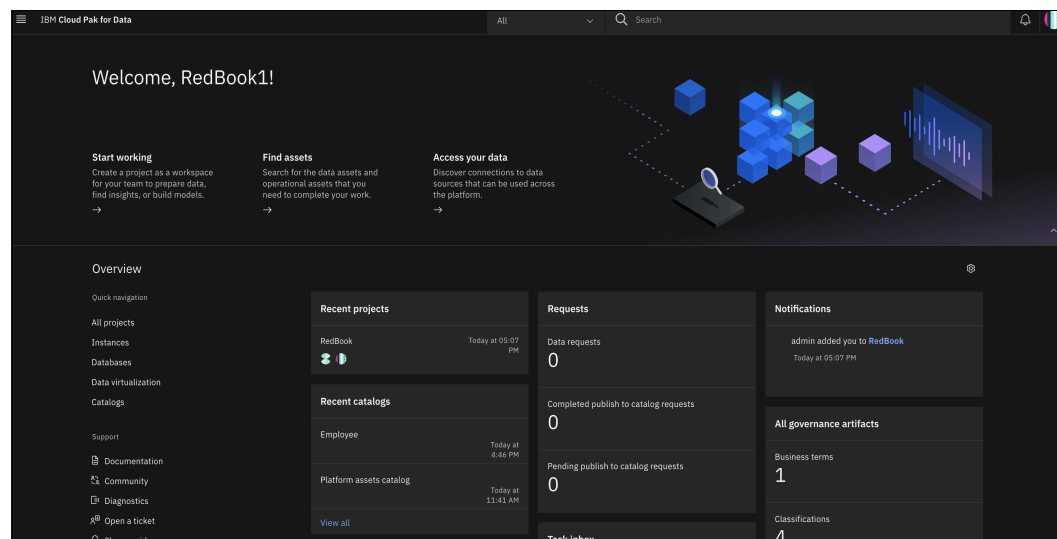


Figure 4-2 Landing page of the Cloud Pak for Data

From this landing page, you can create projects that represent workspaces for organizing the data sources and data models. You also can access data instances and data catalogs, and create assets that can be of the following types:

- ▶ Data assets contain metadata about the data and include how to access the data. The following types of data assets are available:
 - Stored data assets, or data asset from a file, which refer to data that is stored in the current project or workspace and points to a data file uploaded from your local system. For example, a data asset from a file can be an IBM zSystems data file that was previously exported.
 - Connected data assets, which point to a table, file, or folder that can be accessed through a connection to a remote data source to dynamically retrieve data from the source. Cloud Pak for Data provides the technology and services to efficiently facilitate connections to data on IBM zSystems.
 - Connection assets contain the information that is required to establish a connection to your data sources, including on IBM zSystems.
- ▶ Operational assets run code to work with data or other information in projects by using tools for data preparation, analysis, and model building. The following types of operational assets are available with IBM Watson Studio without requiring more services:
 - Jupyter notebooks
 - Data Refinery flows
 - Python scripts
 - Metadata imports
 - Metadata enrichments
- ▶ Configuration assets to configure other assets or jobs with reusable templates. The following types of configuration assets are available with the DataStage service:
 - DataStage subflows
 - Data definitions
 - Parameter sets
- ▶ Business intelligence assets to collect data to create reports and models that provide a business view of that data. You can create the following types of business intelligence assets with the metadata import service:
 - Reports
 - Report queries
 - Report query items

In the project, you can create an asset by clicking **Add to project +** as shown in Figure 4-3.

For more information about the asset types and properties that are available in Cloud Pak for Data, see this [IBM Documentation web page](#).

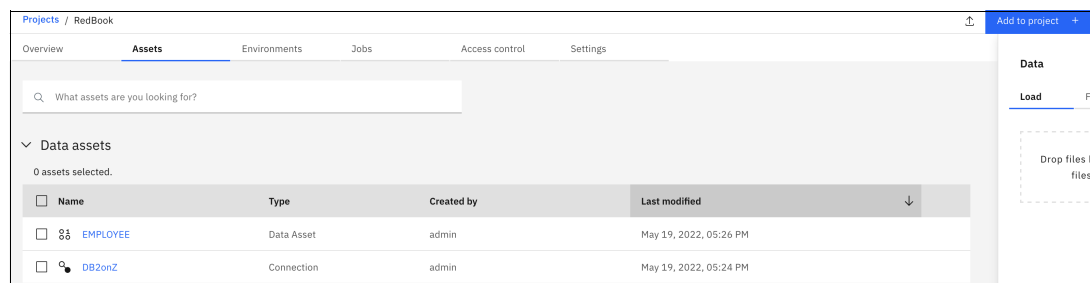


Figure 4-3 Landing page inside a project in CP4D showing the project's assets

Figure 4-4 shows some of the data assets that are available on IBM Cloud Pak for Data.

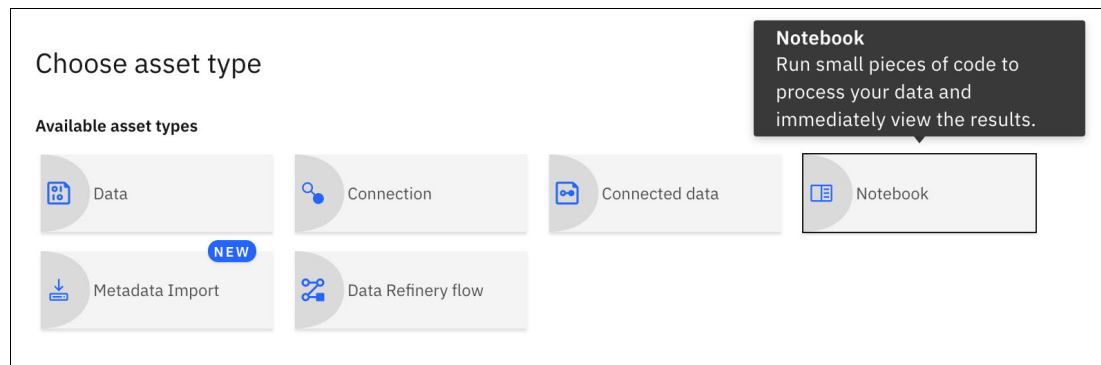


Figure 4-4 Types of Data Assets in IBM Cloud Pak for Data's project

After entering the name of the notebook and selecting the runtime environment as IBM run time on Python (see Figure 4-5) and an optional description, click **Create**.

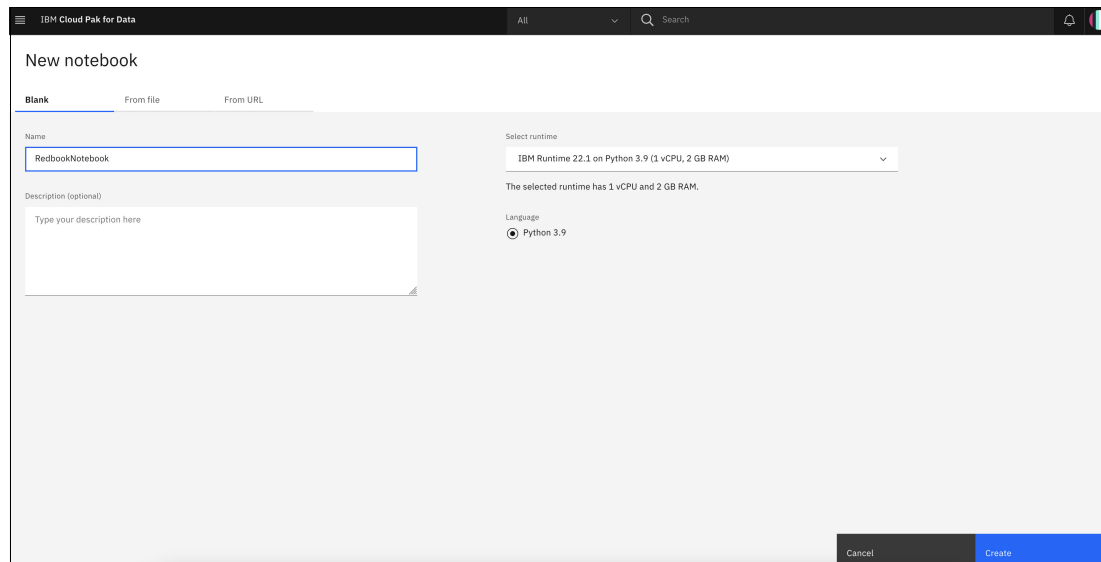


Figure 4-5 Landing page for creating a new notebook

The notebook's environment start instantiating, as shown in Figure 4-6.

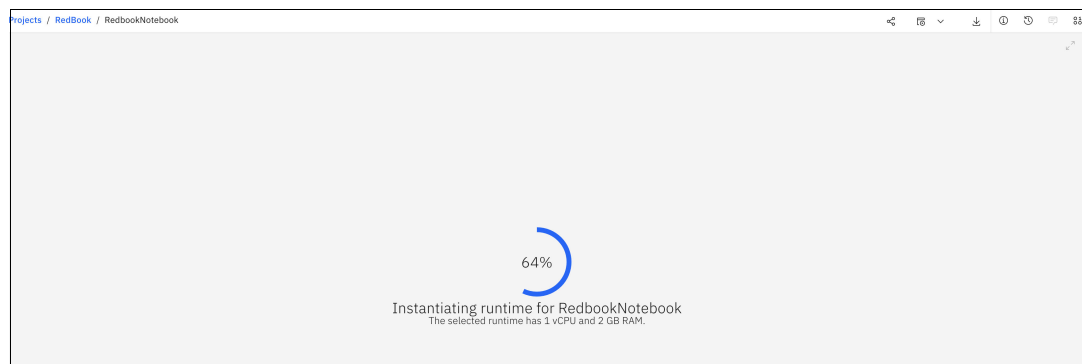


Figure 4-6 Instantiating runtime environment for notebook

From the icon in the upper right, the Connections window appears on the right side, as shown in Figure 4-7.

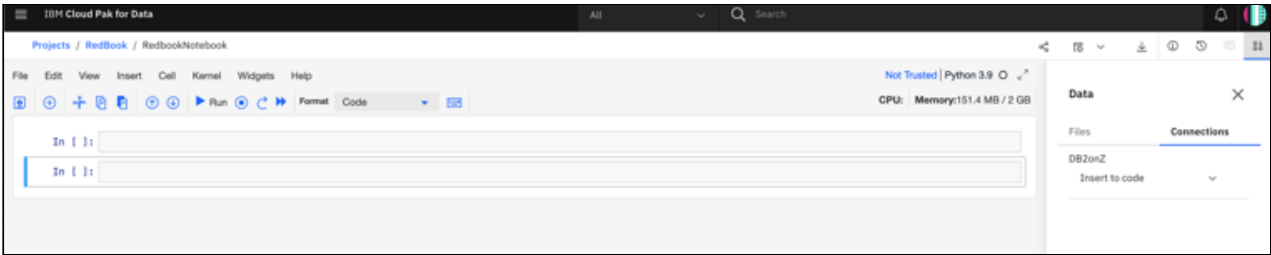


Figure 4-7 Landing page of the Notebook after its creation

From the Connections drop-down, select **pandas DataFrame**. Then, select the data source (**DSN81210**), and table (**DEPT**), as shown in Figure 4-8.

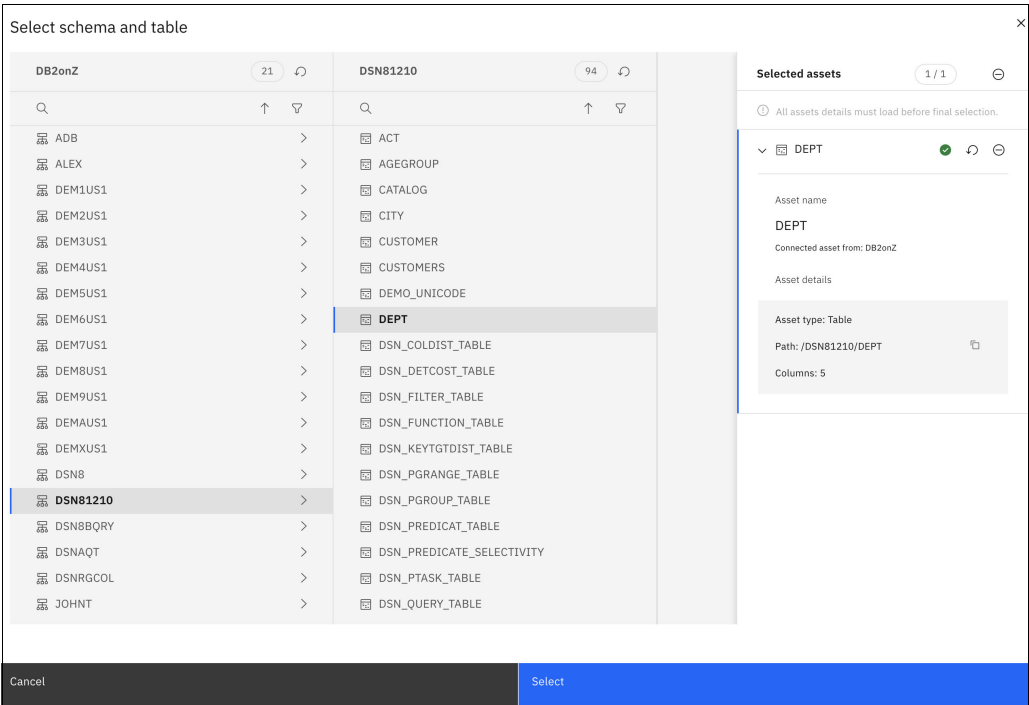


Figure 4-8 Selecting the target data for processing inside the notebook

After loading the data into the notebook, use the code editor in which you can write python code to process your data, Figure 4-9 shows sketching the first 5000 rows from the selected table in the selected data source.

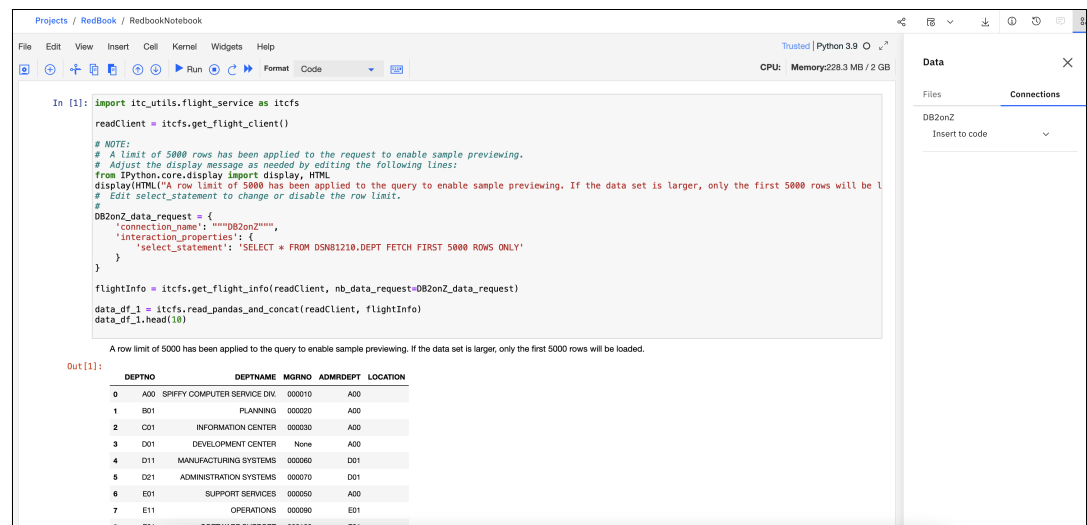


Figure 4-9 Sample Python code and its execution inside the notebook

For more information about data loading in the notebook, see this IBM Documentation [web page](#).

4.4.2 IBM Watson Knowledge Catalog

IBM Watson Knowledge Catalog is an extension of the Information Governance Catalog that provides self-service access to data assets for knowledge workers who must use those data assets to gain insights.

After you create glossary assets and profile, classify, and curate information assets with the Information Governance Catalog, you use IBM Watson Knowledge Catalog to protect and display data assets in a self-service catalog where users can find and prepare data assets.

Figure 4-10 shows how IBM Watson Knowledge Catalog and Information Governance Catalog work together to perform complimentary tasks.

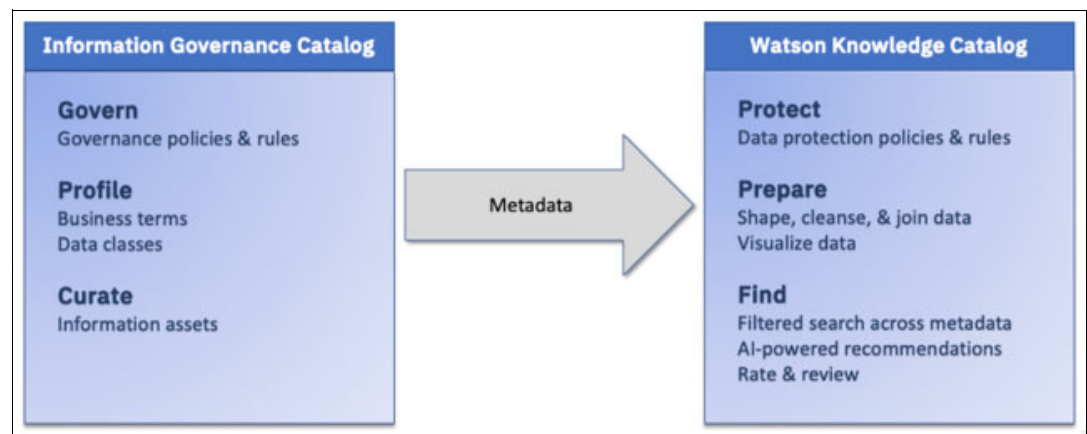


Figure 4-10 IBM Watson Knowledge Catalog and IGC

IBM Watson Knowledge Catalog for IBM Cloud Pak for Data

IBM Watson Knowledge Catalog on IBM Cloud Pak for Data provides a full breadth of capabilities as part of an open, extensible data and AI platform that runs on any cloud.

As part of the foundational set of cloud-native services, IBM Watson Knowledge Catalog provides the following benefits:

- ▶ Unlimited number of catalogs and assets.
- ▶ Access to all connection types.
- ▶ An entry point to support the IBM InfoSphere Information Regulatory Accelerator for IBM Cloud Pak for Data and IBM Watson Knowledge Catalog InstaScan services.
- ▶ Availability of all functions, including data quality, lineage, refinery, reference data management, and advanced workflows.

IBM Watson Knowledge Catalog in data governance

IBM Watson Knowledge Catalog provides a modern (ML catalog for data discovery, cataloging, quality, and governance. Within this framework lies a central Knowledge Catalog that serves as a single source of truth for data engineers, data stewards, data scientists, and business analysts to gain self-service access to enterprise data they can trust and use with confidence.

With incorporated [DataOps](#) automation and methods, IBM Watson Knowledge Catalog enables organizations to make fast and informed business decisions and quickly deliver business-ready data while maximizing the ROI of their data governance initiatives.

For more information about the significance of the use of a modern ML data catalog, see this IBM Journey to AI blog [Data governance: The importance of a modern machine learning knowledge catalog](#).

4.5 IBM Watson Machine Learning

[IBM Watson Machine Learning](#) is a full-service on IBM Cloud that makes it easy for developers and data scientists to work together to integrate predictive capabilities with their applications. The ML service is a set of REST APIs that you can call from any programming language to develop applications that make smarter decisions, solve tough problems, and improve user outcomes.

IBM Watson Machine Learning service provides ML technology that enables the user to manage different ML models. Users can select from various ML frameworks options. In addition to offering an array of interfaces, Watson Machine Learning provides a REST API to deploy models, create and monitor jobs, and get solutions. These features extend the AI capabilities of the applications.

IBM Watson Machine Learning for z/OS (WMLz) is an end-to-end enterprise ML platform that helps you create, train, and deploy ML models to extract value from your mission-critical data on IBM zSystems, while keeping the data where it is stored.

[IBM Z Operational Log and Data Analytics](#) accelerates hybrid incident identification with near real-time operational analytics by having near real-time IBM zSystems operational data streaming to an enterprise analytics platform, or the included log analysis platform, or both. The included insights help to quickly visualize and search operational data to reduce the mean time to identify the cause of operational issues.

4.5.1 WMLz use cases

Different use cases are available for IBM Watson Machine Learning on IBM zSystems. These cases focus mostly on fraud detection, improving business processes and operational productivity:

- ▶ Fraud detection in money transfers for a global bank was embedded in IBM zSystems transactions in addition to providing flexible platform to which data can be imported, exported, or transferred to another platform. By using ML and automation, the minimization of manual processes led to increasing the accuracy.
- ▶ A card issuer of prepaid credit cards in Europe in which tens of millions of Euros were lost to fraud, the WMLz provided an automated and smart solution that detected the fraudulent transaction 30X and adapted to new patterns with a less than 5 milliseconds SLA.
- ▶ Improving business by using WMLz was applied with different clients where AI was used to improve the loan process for a multinational financial services company. This improvement resulted in an accelerated credit approval process and improved customer service and profitability.
- ▶ A financial services organization used WMLz to improve its business through building engaging applications and a predictive model to identify claims that exceed \$300K.
- ▶ A large banking organization in the US used WMLz for operational productivity and the optimization of IBM zSystems resiliency. WMLz provided the ability of playing back the system performance to identify root causes of system problems and anticipating potential system issues to optimize Db2 and IBM zSystems resiliency.

Table 4-1 on page 66 lists several use cases for real-time analytics of different business challenges and an overview of the proof of concept that was built to address the business challenge by using WMLz.

Table 4-1 Previous use cases for real-time analytics

Use case	Business challenge	Proof of concept
Money transfer fraud	Current rules-based process the bank uses for monitoring fraudulent transactions for money transfer is manual and resource intensive.	▶ IBM worked with the bank's IMS data to build a model that scored transactions while minimizing the rate of false positives. ▶ The bank wanted a solution that was colocated with the data on IBM Z.
Credit card fraud	Need a more flexible and scalable option to deploy current and future credit card fraud detection models that were built by their data scientists.	▶ MLz tuned PMML scoring service to meet the bank's strict SLAs (<5 MS end-to-end) ▶ MLz and IzODA together provide the bank with a flexible platform that embraces open standards.
Real-time business insights for operational performance	▶ Need to use more data sources on IBM Z, and reflect real-time changes in their business environment. ▶ Need for an enterprise approach to model deployment and life cycle management.	▶ They used more data sources by accessing IBM Z data in-place and in real-time by using Db2 Analytics Accelerator (IDAA) with MLz. ▶ With IDAA, they processed queries and accessed data in under 5 seconds that used to take several hours.
Loan approval	▶ Loan underwriting is time-consuming, and errors are expensive. ▶ Most loan applications required a skilled underwriter.	▶ Joint development of a binary classification model with loan characteristics as input. The output was an approved (declined) score with a greater than 90% accuracy rate. ▶ The credit union wanted their deployment to be colocated with data on IBM Z.
Real-time business insights for operational performance	▶ Need to deliver greater value to their customers by using technology and analytics. ▶ Achieve the best possible outcomes by getting the most suitable care that is delivered at the best price.	▶ Reward diabetic patients by adjusting their co-pay for taking actions to manage disease. ▶ Because most of their claims data is in Db2 for z/OS, they needed a solution to enable real-time access to IBM Z data.



Security and compliance with automated data governance

Among the top priorities of business leaders today is to reduce the risk of costly breaches and fines by implementing strong data governance practices to achieve better data security and compliance.

In this chapter, we expand on the capabilities that are provided in Cloud Pak for Data to simplify data governance and help maintain privacy, and practices to create a trusted, automated, business-ready data foundation.

This chapter includes the following topics:

- ▶ 5.1, “Data governance and IBM Cloud Pak for Data” on page 68
- ▶ 5.2, “Security capabilities in IBM Cloud Pak for Data” on page 70
- ▶ 5.3, “Data privacy” on page 70
- ▶ 5.4, “Automated governance in an enterprise catalog” on page 71
- ▶ 5.5, “Controlling access to assets in the catalog” on page 78
- ▶ 5.6, “Auditing sensitive data” on page 80
- ▶ 5.7, “Workflow for governance artifacts” on page 81
- ▶ 5.8, “IBM Knowledge Accelerators” on page 83
- ▶ 5.9, “Industry use cases” on page 84

5.1 Data governance and IBM Cloud Pak for Data

As enterprises continue modernizing to a hybrid work environment, the concern around data protection and privacy grows larger and more pressing than ever. This problem of connecting and sharing data in a distributed environment brought about new challenges in the way you must govern your enterprise's data and reduce risk by remaining compliant with internal and external regulations.

As a result, your Data Fabric must include solid data governance practices and solutions that enable secure data collection, preparation, integration, access, and more throughout the data lifecycle. Data governance can be described as enterprise-level management of data availability, relevance, usability, integrity, and security.

As an enabling data governance solution to help enterprises understand and govern their enterprise data, mitigating risk and accelerating insights, IBM offers the IBM Cloud Pak for Data product. IBM Cloud Pak for Data provides a highly securable and self-serving Data Fabric platform that provides an automated method to manage data trust, protection, and compliance with active, end-to-end policy management. This Cloud Pak for Data uses IBM Z technology to bring valuable IBM Z data into a modern analytics and AI platform.

IBM Data Fabric framework enables organizations to securely connect the right data to the right users, regardless of where the data is stored. IBM leads in mature data privacy solutions to get businesses ready for emerging regulatory compliance requirements by various governmental agencies, such as US California Consumer Privacy Act (CCPA), EU General Protection Regulation (GDPR), and other data protection acts.

Enterprises are collecting data from diverse platforms and devices at a faster pace than ever and coupled with unparalleled computing capacity and Enterprises around the world, are facing a growing number of regulations as consumer interest in data privacy increases.

Fines for breaches and noncompliance can be heavy and compromise customer trust in the business. Establishing a governance foundation is a critical step for an organization to transition to a data-centric business.

To become fully data-driven requires enterprises to use an integrated data strategy and security architecture that overcomes data complexity and access challenges.

Automated Metadata Management and Data protection is shown in Figure 5-1.

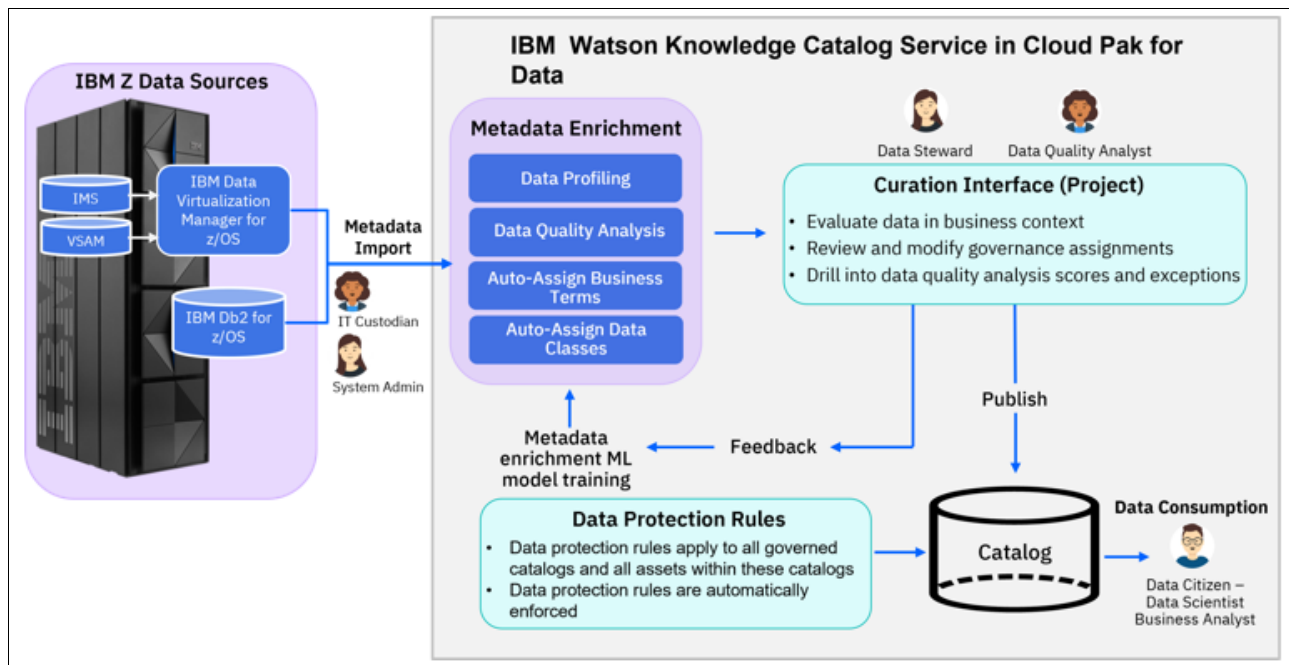


Figure 5-1 Automated Metadata Management and Data protection

The Watson Knowledge Catalog service in Cloud Pak for Data automatically profiles data and enforces data protection policies that are based on the profiles of the data fields. It also masks personal information that is based on the automatic detection of fields, such as addresses, gender, and marital status.

5.2 Security capabilities in IBM Cloud Pak for Data

IBM Cloud Pak for Data follows IBM Security® and Privacy by Design (SPbD). SPbD at IBM is a set of focused security and privacy practices, including the following examples:

- ▶ Vulnerability management
- ▶ Threat modeling
- ▶ Penetration testing
- ▶ Privacy assessments
- ▶ Security testing
- ▶ Patch management

When IBM Cloud Pak for Data is used to facilitate a Data Fabric, you can improve your security and privacy practices by using the following CP4D capabilities:

- ▶ Exporting audit records: Users can configure IBM Cloud Pak for data to forward audit records to a security information and event management (SIEM) solution, such as Splunk, LogDNA, or IBM QRadar®.
- ▶ Encryption: IBM Cloud Pak for Data supports protection of data at-rest and in-transit. It supports Federal Information Processing Standard (FIPS)-compliant encryption for all encryption needs. Ensure that your data in IBM Cloud Pak for Data is stored securely, you can encrypt your storage partition.
- ▶ Audit logging: Provides accountability, traceability, and regulatory compliance. The regulatory compliance must be set in a way that it allows access to and modification of data.

5.3 Data privacy

Critical enterprise data can contain protected information, such as personally identifiable information (PII), that can result in massive fines and a damaged reputation if accessed by unauthorized personnel. Because of government and industry regulations, it is imperative that enterprises enforce and monitor the protection of this information and allow only as-needed access to protected information by business users.

A Data Fabric can help your enterprise address these data privacy challenges by establishing a unified privacy framework that enables the following data privacy practices:

- ▶ Compliance and automated enforcement of data privacy regulatory requirements through defined data protection rules that reflect data privacy regulations.
- ▶ Automated detection and protection of sensitive data by using various, advanced identification methods.
- ▶ Use of advanced data privacy methods; for example, redaction and substitution masking, which is driven by business rules.

Deploying a unified privacy framework through a Data Fabric to enhance data privacy means enabling access to necessary data with reduced risk for costly regulatory compliance violations. Data privacy in a Data Fabric uses autonomous governance policy enforcement to provide automatic decisions on masking and protecting data, which enables safe data usage throughout the lifecycle of the Data Fabric.

Achieving advanced data privacy in a Data Fabric includes the following key aspects:

- ▶ Integrate and use trusted and protected data throughout your Data Fabric.
Enable better understanding of data by all who need it through a business glossary and common business vocabulary. You also can use immediately available regulatory and industry glossaries to automatically provide business context.
- ▶ Automated enforcement of data protection rules where data is stored and when sensitive data is used.
IBM uses AI to intelligently automate the identification, monitoring, and enforcement of policies on sensitive data across organizations with auto-privacy capabilities on IBM Cloud Pak for Data.
- ▶ Protect personal, sensitive, and confidential information.
Align concepts from industry regulations and standards with your business data to accelerate regulatory compliance with IBM Knowledge Accelerators.
- ▶ Use data privacy techniques that preserve data formats and lower exposure risk, without eliminating use of that data due to privacy concerns or risks.
Sensitive data is masked quickly while data utility for insights is maintained.
The de-identification of data is automated with centrally enforced data protection rules, format preserving tokenization, and encryption.

5.4 Automated governance in an enterprise catalog

A governance framework is the foundation that provides the capabilities to govern and enrich your data through the implementation of governance artifacts that are organized into categories for collaboration. Categories are formatted in a hierarchical structure and can be used to define the ownership of artifacts, control authorship of artifacts, and control artifact visibility. These governance artifacts can be used as metadata to add meaning to assets and control access of assets. They also can be used as information to identify assets, and used in data quality analysis.

Governance artifacts are created and assigned to data assets by users who are collaborators in categories. The processes for creating, updating, and deleting governance artifacts are controlled by governance artifact workflow configurations.

With the IBM Watson Knowledge Catalog service in CP4D, you can create AI-augmented catalogs of curated assets that are supported by a governance framework. IBM Watson Knowledge Catalog provides an enterprise catalog management platform that connects business users to the data, which enables sharing assets across your enterprise.

Assets that are in a catalog consist of metadata that describes the data and its lineage. For example, this metadata can include information about how to access the data, its format, the asset classification, collaborator roles with access to the data, and more.

The first step in implementing a governance framework is preparing for governance and that process includes a user with the administrator role who is responsible for the following tasks:

- ▶ Creating governance categories
- ▶ Assigning roles to users
- ▶ Adding users to categories
- ▶ Setting up workflow configurations

Workflow configurations establish the sequence of tasks to be completed before an artifact is published or deleted.

Automated governance enables the use of active policy management to track how policies and rules are enforced. It uses key capabilities, such as dynamic masking of sensitive data, to help protect data and ensure compliance and audit-readiness, but most importantly, helps maintain client trust.

The data governance framework is composed of governance artifacts that enrich data assets and protect sensitive data from unauthorized access. Sensitive information is protected from unauthorized access by creating policies and data protection rules that deny access or mask data values in data assets. Policies and data protection rules are applied to all catalogs that have protection that is enabled.

For more information about categories and governance artifacts, including types of artifacts and their descriptions, see this [IBM Documentation web page](#).

5.4.1 Governance policies

A *policy* is a natural-language description of a governance subject area, and governance policies describe and document how your organization manages and protects data assets. Policies can consist of one or more rules, subpolicies, and can be organized in a hierarchical manner based on meaning and relationships.

Also, various types of rules are available that can comprise a policy; for example, governance rules, data protection rules, or data location rules. These policies are critical to remaining compliant with industry and government regulatory policies to protect confidential, personal, and sensitive information and to adhere to data quality, retention, and archival definitions that are set forth by the business.

Consider the following points when you are working with policies in IBM Watson Knowledge Catalog:

- ▶ Organize policies in categories and hierarchies.
- ▶ Add subpolicies.
- ▶ Add governance, data protection, and data location rules.
- ▶ Relate governance artifacts to the policy.
- ▶ View and edit custom policy attributes.

Creating policies and editing or deleting existing policies requires specific permissions in IBM Watson Knowledge Catalog. The following components are involved with authoring new policies in IBM Watson Knowledge Catalog (see Figure 5-2):

- ▶ **Parent policy:** Select a published policy as the parent for your new policy.
- ▶ **Details:** Edit custom attributes on the policy to provide more information.
- ▶ **Subpolicies:** Assign published policies as subpolicies to the new policy.
- ▶ **Related artifacts:** Add relationships between a policy and other governance artifacts, assign rules, and assign classification and business terms artifacts to the policy.

The screenshot displays the IBM Cloud Pak for Data interface for configuring a policy. The main title is "Protect sensitive personal information(SPI)" with a "Published" status badge. The interface is divided into several sections:

- General:**
 - Description:** Protecting sensitive personal information(SPI) from unauthorized users using data protection rules.
 - Primary category:** Employee
 - Secondary categories:** No secondary category added yet.
 - Parent policy:** No parent policy added yet.
- Subpolicies:** A section for adding subpolicies.
- Rules:** A table with columns "Name" and "Description". It contains one rule: "Hide sensitive information" with categories "Categories / Employee".
- Data protection rules:** A table with columns "Name" and "Description". It contains two rules: "Mask Data of birth" (Masking Date of Birth) and "Hide Salary" (Hide Salary).
- Right-hand sidebar:**
 - About this policy:** A section for policy details.
 - Stewards:** Joe (Data Steward).
 - Effective dates:** Start date: May 12, 2022 5:38 PM CDT; End date: --.
 - Tags:** No tags added yet.
 - Created by:** RedBook1, May 12, 2022.
 - Modified by:** admin, May 16, 2022.

Figure 5-2 Sample Sensitive Personal Information Policy configuration

Governance rules

Governance rules provide the business description of the required behavior or actions to be taken to implement a specific governance policy. They also describe the criteria to determine whether assets are compliant with business objectives.

Because governance rules are business descriptive, they cannot be enforced (unlike other rules that are defined in IBM Watson Knowledge Catalog, such as data protection and data location rules). Governance rules are based on governance subject areas that are important to your organization; for example, information security or regulatory compliance.

The following assets in IBM Watson Knowledge Catalog where governance rules can be enforced are listed in Table 5-1.

Table 5-1 Enforceable catalog assets

Governance artifact	Description	Enforce rules
Categories	Categories organize governance artifacts in a hierarchical structure. Category roles are used for defining ownership of the artifacts, control their authoring, and restrict the visibility.	No
Business terms	Business terms implement a common enterprise-wide vocabulary to describe the meaning of data. This implementation ensures the clarity and compatibility among departments, projects, or products within the organization. Business terms can be used in governance rules and enforcing the rules to identify the affected data	Yes
Data classes	Data classes classify data based on the structure, format, and range of the values of the data. They are automatically assigned to matching data columns during profiling and metadata enrichments.	Yes
Classifications	Classification describes specific characteristics of the meaning of data. Predefined classification describes the sensitivity of the data. You can use classifications to construct governance policies and rules.	Yes
Policies	Policies describe how to manage and protect data assets. You create policies by combining rules and sub policies. Governance rules in the policies are used to document standards and procedures.	Yes
Governance rules	Governance rules describe how to apply policy.	No
Data protection rules	Data protection rules control access to data based on users, asset properties and assigned governance artifacts. Data protection rules control who can see what data.	Yes

Data protection rules

Data protection rules provide attribute-based access control of data and consist of two main components: criteria and an action block. As shown in Figure 5-3, criteria specifies which data to control through identifying one or more conditions for enforcing the data protection rule. Conditions are composed of one or more predicates, which describe properties of data assets or identify users.

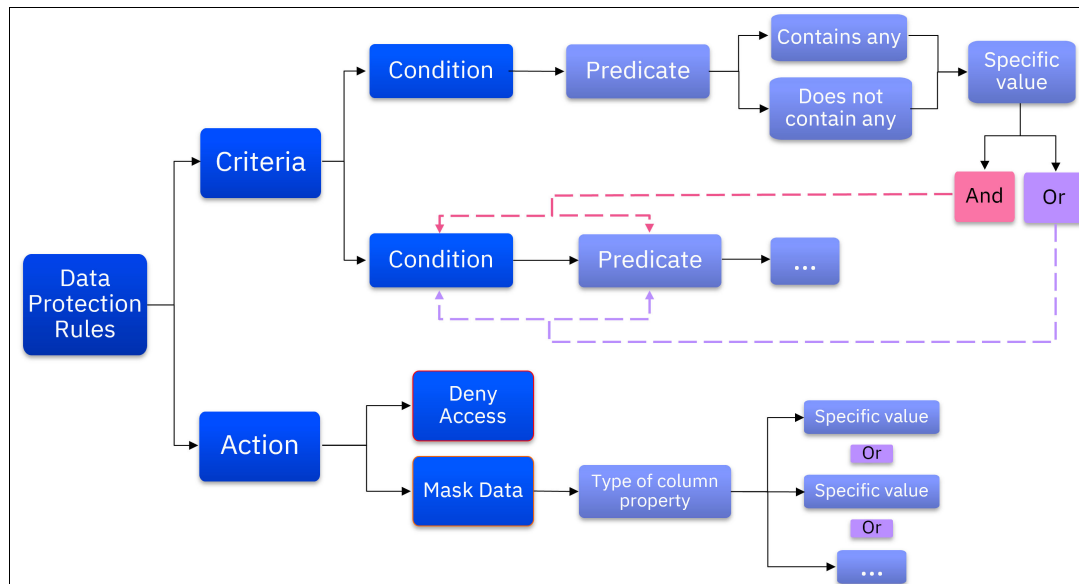


Figure 5-3 Data protection rules flow

The types of predicates include the following examples:

- ▶ Asset owner: The email address of the user who owns the asset.
- ▶ Business term: A business term that is assigned to the asset or to the column.
- ▶ Data classes: Data classes classify data that is based on the structure, format, and range of the values of the data. are automatically assigned to matching data columns during profiling and metadata enrichments.
- ▶ Tag: Metadata that is associated with assets.
- ▶ Username: Name or email address of a user.
- ▶ User group: One or more groups to which a user belongs and is requesting access to an asset.
- ▶ Classification: The type of sensitive information in the asset; for example, sensitive personal information, personally identifiable information, and confidential.
- ▶ Column name: The name of a column in an asset.

After you select the type of predicate for the condition, you must select the operator (contains any or does not contain any) and then, specify a specific value of the predicate.

Also, you can join conditions and predicates within a condition by using the Boolean operators AND or OR to create nested logical structure with precise criteria.

The action block in data protection rules defines the type of enforcement action to take as specified by the conditions; that is, deny access to all data that is within the asset or mask parts of the data. When masking data, you must define masking criteria, such as which columns to mask by selecting the type of column property and specifying one or more specific values of the property. Specific values in masking criteria can be combined by using the OR operator.

Data protection rules apply to governed catalogs and all assets within the governed catalogs. To prevent unauthorized access of sensitive assets in governed catalogs, data protection rules are automatically enforced when catalog members attempt to view or act on the data assets.

Without any data protection rules, access to a data asset in a catalog is controlled by the asset's privacy setting within the catalog, which limits its access to users who are collaborators in the catalog.

An example of creating a data protection rule by using the IBM Cloud Pak for Data IBM Watson Knowledge Catalog service is shown in Figure 5-4.

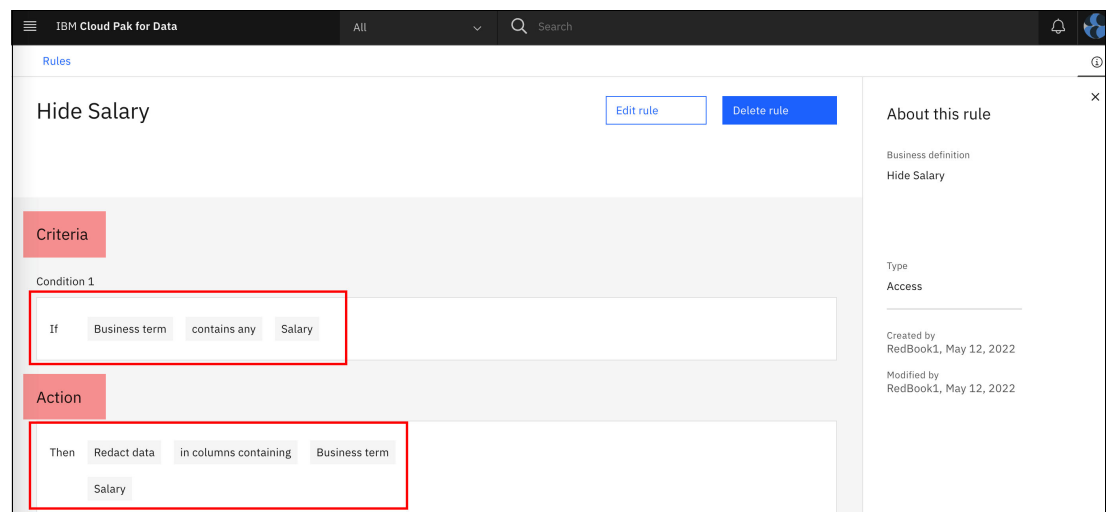


Figure 5-4 Sample data protection rule setup

For more information about creating data protection rules in IBM Watson Knowledge Catalog, see this [IBM Documentation web page](#).

5.4.2 Automating policy enforcement

To apply data policies that are stored in a catalog to the assets within that catalog, you first must set the option for that catalog by using one of the following methods:

- ▶ When you create a catalog, select the **Enforce data policies** option when entering the catalog information.
- ▶ Open the catalog, browse to the **Settings** tab and then, select the **Enforce data protection rules** option, as shown in Figure 5-5.

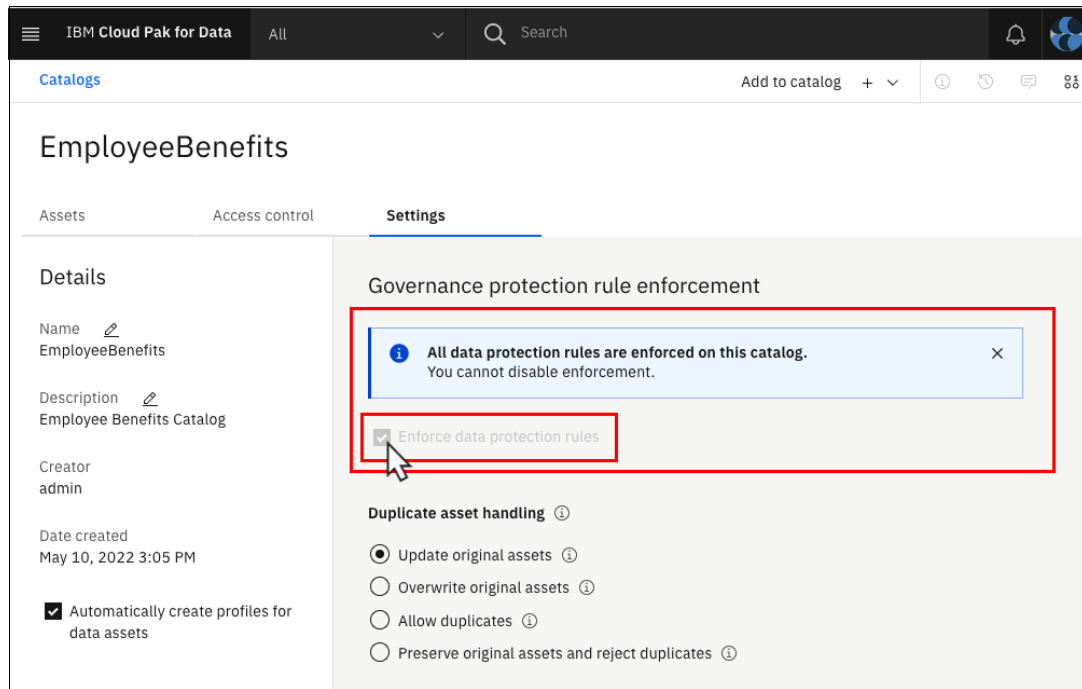


Figure 5-5 Enforce data protection rules on Catalog

For more information about data protection rules enforcement in IBM Watson Knowledge Catalog, see this [IBM Documentation web page](#).

5.5 Controlling access to assets in the catalog

The catalog administrator manages the access to the catalog. Collaborators are added with specific roles that determine their permission to perform the action.

You can control access to a catalog asset through the combination of the privacy level and members of the assets from the Access control tab within a catalog, as shown in Figure 5-6. When data protection rules are enforced on the catalog, the asset owner cannot be blocked from viewing the asset overview without masking or the use of the asset. All other users (including asset members) are subject to the enforcement of data protection rules on the asset.

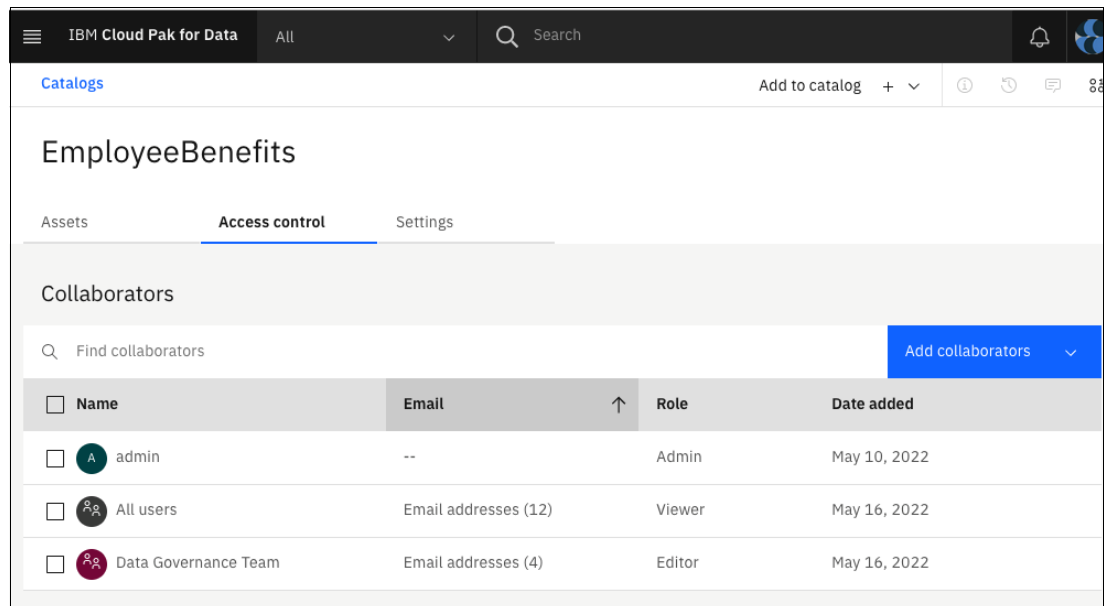


Figure 5-6 Configuring catalog access control

5.5.1 Global search

IBM's Data Fabric framework that uses IBM Cloud Pak for Data enables the self-serving method of finding the trusted data by using global search. Users search for data that is published in the catalogs.

Global search is used to search for assets across all of the projects, catalogs, and deployment spaces to which you can access. Global search results show you only the governance artifacts that you have permissions to view.

To search for an asset or artifact, you can enter one or more words in the global search field. The search results are matches from the following assets and artifacts properties:

- ▶ Name
- ▶ Description
- ▶ Tags
- ▶ Synonyms of business terms
- ▶ Column names
- ▶ Column descriptions

- ▶ Assigned governance artifacts
- ▶ Assigned categories
- ▶ Custom field

An example of a search result is shown in Figure 5-7.

IBM Cloud Pak for Data

All

Employee

New data request

Search results for Employee

Filter by:

Any asset type

Any tag

Steward/Owner

Modified by

Clear all

Showing 14 of 14 items

Sort by: Most relevant

Name	Type	Tags	Modified by	Modified on	
Employee Categories > [uncategorized] Employee Category	Category		admin	May 12, 2022	:
EMPLOYEE All projects > RedBook Project	Data	DEM1...	System	May 12, 2022	:
EMPLOYEE All catalogs > Default Catalog	Data	info_a... DEM1...	System	May 12, 2022	:
Employee Protect confidential information Categories > Employee	Policy		RedBook1	May 12, 2022	:
Salary Categories > Employee	Business term		RedBook1	May 12, 2022	:
Hide sensitive information Categories > Employee	Rule		RedBook1	May 12, 2022	:
Protect personal identifiable information(PII) Categories > Employee	Policy		RedBook1	May 12, 2022	:
Protect sensitive personal information(SPI) Categories > Employee Protecting sensitive personal information(SPI) from unauthorized users using data ...	Policy		admin	May 16, 2022	:
Personal Information Categories > [uncategorized] Personal Information (PI) is defined as any data relating to an identified or identifia...	Classification		admin	May 09, 2022	:
US Employer Identification Number Categories > [uncategorized] Nine digit number to identify US employer, typically in nn-nnnnnn format with das...	Data class		admin	May 09, 2022	:
Protect confidential information Categories > Utilities Payment Risk Prediction	Policy		RedBook1	May 12, 2022	:
Protect personal identifiable information (PII)	Policy		RedBook1	May 12, 2022	:

Figure 5-7 Global Search results

The highest scoring results are for matches to the name of the asset or artifact. Multiple assets and artifacts can have the same name. However, the name of the project, catalog, deployment space, or category is displayed below the asset or artifact name so that you can determine which result is the one you want.

Results are prioritized in the following order:

- ▶ Exact matches of complete words
- ▶ Partial matches of complete words
- ▶ Fuzzy matches

5.5.2 Data requests

Data requests are used when the data is not found by using global search. This request is sent to the administrator and data engineer to informally provide the data access to the users. The content of the data request form is shown in Figure 5-8.

The screenshot shows the 'New data request' form in the IBM Cloud Pak for Data interface. The form is titled 'New data request' and has three tabs: 'Overview', 'Attributes', and 'Source'. The 'Overview' tab is selected. Below the tabs, there is a instruction: 'Use this form to request access to data. If you need specific data, provide information about what you need on the Attributes tab.' The form contains several input fields: 'Request name' (with a placeholder 'Enter a display name for the request'), 'Add to project (optional)' (a dropdown menu with 'Select a project'), 'Issue link (optional)' (with a placeholder 'Enter the URL of an existing issue'), 'Priority (optional)' (a dropdown menu with 'Select a priority'), 'Description (optional)' (with a placeholder 'Enter a brief description of your request' and a character count '0/100'), and 'Data intent (optional)' (with a placeholder 'Briefly explain how you plan to use the data' and a character count '0/100'). At the top right of the form, there are three buttons: 'Cancel', 'Save as draft', and 'Submit'.

Figure 5-8 Data Requests form

5.6 Auditing sensitive data

Auditing is the process of recording the activity that occurs on databases or applications. Auditing helps detect and prioritize threats and data breaches. To support the protection and compliance of sensitive data, Watson Knowledge Catalog service in Cloud Pak for Data supports integration with IBM Guardium® to provide auditing capabilities. IBM Guardium safeguards sensitive information by auditing what is happening in your sensitive-data environments and can audit any assets when connected to Cloud Pak for Data.

IBM Guardium External S-TAP is the component of IBM Guardium that works with your databases for compliance monitoring and data security. The component intercepts TCP/IP traffic between Cloud Pak for Data users and database services and sends the traffic to the IBM Guardium collector for parsing, policy enforcement, logging, and reporting (see Figure 5-9).

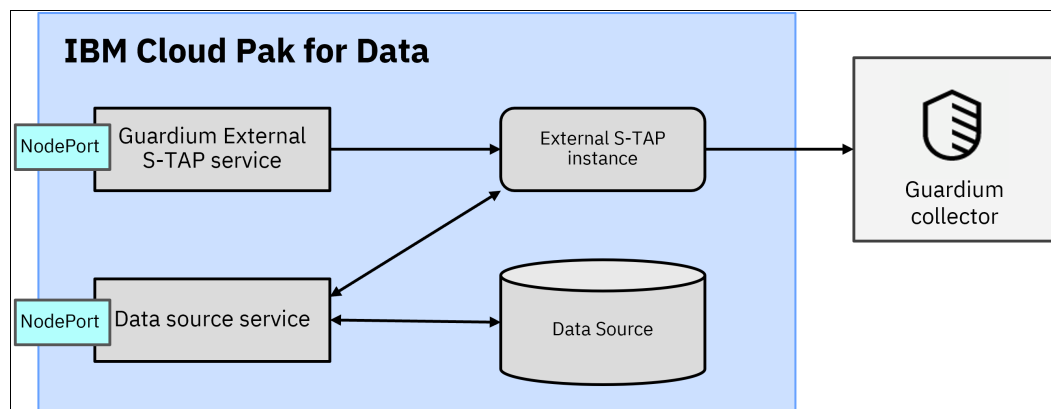


Figure 5-9 Integrating the CPD with IBM Guardium

Additionally, you can use the External S-TAP service to monitor any data source that is supported by the Guardium External S-TAP. IBM Cloud Pak for Data provides an end-to-end experience that is rooted in metadata and active policy management. It also automates the management of data trust, protection, and compliance.

If a data security breach occurs, it is important to ensure that an audit trail exists so that you know who accessed the data, when they accessed it, where they accessed it, and what data they accessed.

You can use the data governance features in IBM Cloud Pak for Data to identify sensitive data, including data that is on Hadoop systems.

When you integrate IBM Guardium and IBM Cloud Pak for Data, users with manage catalog permissions can specify which assets are audited by IBM Guardium.

Numerous regulations determine how businesses must handle PII data, including the US General Data Protection Regulation (GDPR), the Health Insurance Portability and Accountability Act (HIPAA), and the Family Educational Right and Privacy Act (FERPA).

5.7 Workflow for governance artifacts

A successful enterprise includes well-implemented information architecture for data governance. To create a suitable catalog of governed data, the process that is used to create it must be streamlined.

The workflows feature of IBM Watson Knowledge Catalog enables businesses to streamline the processes of creating, deleting, and updating of governance artifacts. It also helps define the sequence of tasks that must be completed before an artifact is published.

A workflow can control one or more governance artifact types for one or more categories.

Shown in Figure 5-10 on page 82 is the New workflow configuration window where you specify the details of a new workflow configuration.

The screenshot shows a 'New workflow configuration' dialog box. The 'Name' field is filled with 'Governance Workflow for Automatic publishing of Business terms'. The 'Description (optional)' field is empty. The 'Choose a workflow template' section has three options: 'One approval step and one review step', 'Automatic publishing' (which is selected), and 'Two approval steps and one review step'. The 'Create' button is highlighted in blue.

Figure 5-10 Creating workflow for governance artifacts

Complete the following steps to create a workflow configuration:

1. Click **Administrator** → **Workflow configuration** and select **Governance artifact management**.
2. Select the **Workflow configurations** tab and then, select **New configuration**.
3. Enter a name for the workflow (Governance Workflow for Automatic publishing of Business terms) and optionally enter a description.
4. Select a template or import a custom template.
5. Click **Create**.
6. For the Workflow conditions step, specify the conditions or replace the default workflow configuration.
7. For each of the other steps, add assignees and notifications, as needed.
8. Click **Save**.
9. Click **Activate** and then, **Save and activate**.

After the workflow is enabled for a business term and a business term is created, the workflow is triggered and adds the requirement of approval of the authorized user from the data governance team before the business term is published.

5.8 IBM Knowledge Accelerators

Data governance analysts have use manual processes to provision, manage, and govern data. However, this approach is not efficient, does not scale, and is a lengthy iterative process.

IBM Knowledge Accelerators offer extensive, curated glossaries to improve data classification, regulatory compliance, self-service analytics, and other governance operations.

IBM Knowledge Accelerators help organize data along a common and known business vocabulary, in addition to automatically providing business context and definitions during the onboarding of regulatory and industry data content within IBM Watson Knowledge Catalog. An extensive business vocabulary accelerates making data understandable and elevates your data catalog investment.

Building a business vocabulary is more than creating a word list. It takes time to create a usable business vocabulary with definitions and business context. It also requires in-depth industry experience and a breadth of operational business knowledge that usually is beyond a single person's capability.

By using IBM Knowledge Accelerators, scaling a business vocabulary quickly and comprehensively is possible immediately for the following industries:

- ▶ Healthcare
- ▶ Energy and utilities
- ▶ Insurance
- ▶ Financial services
- ▶ Cross industry

The use of the IBM Knowledge Accelerators as part of your data governance practices enables the following benefits:

- ▶ Speed up time to value: Saves your organization time across planning, creation, research, and analysis workflows to establish enterprise-wide common vocabulary.
- ▶ Help reduce data risk and oversight: With thousands of predefined terms and KPIs customized for your industry, manual process error is decreased.
- ▶ Elevate your DataOps investments: Works within Watson Knowledge Catalog, which is an intelligent enterprise data catalog to ensure a strong foundation for data quality, governance, and self-service analytics.

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- ▶ Help reduce data risk and oversight: With thousands of predefined terms and KPIs customized for your industry, manual process error is decreased.
- ▶ Elevate your DataOps investments: Works within Watson Knowledge Catalog, which is an intelligent enterprise data catalog to ensure a strong foundation for data quality, governance, and self-service analytics.

IBM Knowledge Accelerators are represented as collections of business terms that consist of the following components:

- ▶ **Business Core Vocabulary:** Defines industry concepts in plain business language. These terms include a term hierarchy and are organized by logical business categories. Clearly defined business terms help standardization and communication within a company and are a foundation for governance of enterprise information assets.
- ▶ **Business Performance Indicators:** Terms that describe industry key performance indicators (KPIs); that is, business measures. The business measures are grouped by the Performance Analysis areas. These measure terms also are related to terms in the Business Core Vocabulary and allow business users to fully articulate the requirements for analysis by using their own business terminology. Business Performance indicators can be used for defining and meeting reporting requirements for the organization.
- ▶ **Business Scopes:** Provides groupings of related Business Core Vocabulary terms that address a specific business use case. These business scopes are typically at a finer grain than the core vocabulary categories and help business users to focus on the terms that are required to support a specific business need.
- ▶ **Industry Alignment Vocabularies:** These terms align key elements from industry regulations and standards to the terms within the Business Core Vocabulary for assisting with regulatory coverage and compliance for the healthcare organization. The supported alignment vocabularies include FHIR, HIPAA, IFRS, SII, GDPR, and CCPA.
- ▶ **Reference Data:** The quality and consistency of this data are key within an organization. The Knowledge Accelerators include industry-specific reference data that is mapped to the core business vocabulary. The reference data sets can also be extended or customized with values from data sources. IBM Watson Knowledge Catalog can use this reference data with Data Classes to profile the data.

5.9 Industry use cases

The IBM Cloud Pak for Data can help your enterprise tackle industry challenges by using AI-infused data governance, risk management, and regulatory compliance.

Cloud Pak for data provides the capabilities that are needed to implement automated enforcement of universal privacy and usage policies across the system of sources and users. Also, it provides auto privacy capabilities to ensure those users see only the data that they are supposed to see.

Auto Privacy supercharges the data discovery and ensures enforcement of governance policies across many sources and user. Some of the use cases include the following examples:

- ▶ Regulatory Compliance (CRO)
- ▶ Data anonymization and monetization
- ▶ Financial crimes and fraud
- ▶ Risk management



Consuming data from a Data Fabric

After you establish a Data Fabric architecture, consider the best practices that are presented in this chapter. Consider also the use cases that you can pursue with IBM Z data as part of your Data Fabric architecture.

This chapter includes the following topics:

- ▶ 6.1, “Best practices for AI application development and deployment” on page 86
- ▶ 6.2, “Machine learning operations” on page 88
- ▶ 6.3, “Planning and implementing user authorization” on page 90
- ▶ 6.4, “Planning and implementing application access” on page 92

6.1 Best practices for AI application development and deployment

At a high level, two principles are enabled by a well-governed Data Fabric:

- ▶ The use of data across your enterprise can be democratized. Business users need not be experts in data engineering or data science to take advantage of well-categorized data sets for the construction and deployment of new models.
- ▶ The Data Fabric architecture enables an improved agile framework, with democratized access to data allowing for quick experimentation across the DataOps and MLOps life cycles.

The act of connecting, securing, and granting access to your various forms of enterprise data allows you to be more nimble and responsive to emerging threats, changing demands, and challenging opportunities.

6.1.1 DataOps

Broadly defined, *DataOps* is the set of practices and processes that is used to enable the flow of enterprise data across an organization.

One of the key benefits of improving DataOps in a Data Fabric architecture is the pressure that is removed from data scientists and data engineers. As organizations seek to infuse more planning and processes with data, it is difficult for specialists to dedicate extensive time to all efforts, especially because so much time is spent wrangling and manipulating information from disparate data sources.

The governance rules that manage your Data Fabric reduce a great portion of this up-front work while also enabling users across your business to manage the process as nonexperts.

Watson Studio in IBM Cloud Pak for Data offers a suite of capabilities that can be used by nonexperts to view and use enterprise data. Data resources from across a hybrid architecture can be collected, organized, and analyzed in a standard way. This process ensures that the organization at-large can become more proactive through data-infused insights, which results in boosted productivity and increased savings.

6.1.2 AutoSQL and AutoAI

Two key features can be found in Watson Studio that allow for the proliferation of AI-driven insights across your organization:

- ▶ AutoSQL

With AutoSQL, users can easily query and integrate data resources, regardless of where they are stored. Where data is traditionally trapped in silos across varying parts of the enterprise, IBM Cloud Pak for Data users can unify data resources without the need to move or replicate it.

- ▶ AutoAI

With AutoAI, automation is enabled throughout the AI lifecycle. It provides nonexperts with insights about the best-fit model to use given the type of data they are working with, and the best-fit columns to set as predictors for a specific use case.

With the user in the loop, it also tests the model tuning and ranks the candidate pipelines to quickly derive accurate, precise results. The ensuing model then can be pushed to production to begin delivering business insights.

AutoAI requires specific formats of data as input:

- Tabular data (in the form of CSV files)
- Connected data from a Networked File System (NFS)
- Data from other supported data connections (see this [IBM Documentation web page](#))

These two capabilities offer the nonexpert the ability to fully operationalize their enterprise data. Without requiring extensive training, and without leaning on the time and resources of their data science counterparts, they can understand their available data resources and use them in various use cases.

Next, we discuss the process of model development, and how it is enabled through a Data Fabric architecture.

6.1.3 Building and managing models

Without a Data Fabric architecture, organizations can encounter many challenges in the planning, development, and deployment of machine learning models. Especially in the context of IBM Z, if the models are trained in data sets that are replicated away from core runtimes, you run the risk of inaccurate predictions after they are deployed into production.

With a Data Fabric built around IBM Z, you can account for the constant movement of your enterprise data. The training data can be supplied directly from critical runtimes, which allows the deployed model to reflect more accurately the state of the systems that it is built to predict.

6.1.4 Metadata and self-service

One of the key features of a Data Fabric with relation to DataOps is the opportunity to catalog and organize data at scale. Metadata enables business users to easily contextualize the data they are accessing and more easily conduct self-service analysis.

As your enterprise data grows (which it inevitably will), it can be fitted against the existing metadata. The result is a system of operations that is more sustainable, compared to the mess of structured, unstructured, and semi-structured data that was housed in disparate repositories and used different categorical structures.

Good metadata features benefits beyond the use of data for the development of new models. It also can allow for more easily managing data in a compliance context, with data more easily collected for reporting purposes, or more easily withheld to protect sensitive information/PII when data sets are shared.

6.1.5 Data fabric benefits

A Data Fabric architecture includes the following benefits:

- ▶ The establishment of a self-service model so that nonexperts across your business can securely and reliably access well-organized data.
- ▶ The automated governance and protection of data, so that new inputs are fitted against your framework, and all data can be easily managed for regulatory reporting and action.
- ▶ The integration of data across a hybrid cloud architecture, which is made more possible with good metadata that allows significant information to be more easily detected and integrated where suitable.

6.2 Machine learning operations

Beyond improved DataOps, a well-governed Data Fabric also allows for improved machine learning operations (MLOps).

6.2.1 MLOps benefits

Broadly defined, MLOps is a repeatable, shared process for an organization to plan, train, deploy, manage, and monitor its machine learning models, especially in a continuous integration/continuous development (CI/CD) paradigm.

MLOps features a few key benefits, including the following examples:

- ▶ A more efficient pipeline for model planning-to-production
- ▶ Scalability for many models to be managed by using the same governing rules
- ▶ Reproducibility and collaboration across business users
- ▶ Greater transparency in the processes and data that are used for machine learning

With a well-governed Data Fabric, there is no need to plan for an entirely separate machine learning management platform to train your models. Instead, the Data Fabric can use the production data that co-exists in the fabric.

With colocation comes more accurate models that, in the case of IBM Z, can be more rapidly built for the analysis and scoring of business-critical transactions. Therefore, use cases from fraud detection to loan approval processing can be achieved and scaled as never before.

6.2.2 MLOps Process

The MLOps process includes the following steps:

1. Access the data.

With well-established metadata, this process is quick and securely governed, which ensures that business users can quickly identify the data sets (or portions within data sets) that are most useful to their purpose.

In the context of running IBM Cloud Pak for Data on IBM Z, this process involves incorporating IBM Z data assets with the data catalog that you are making available to business users.

For example, you might make Db2 for z/OS data available through a direct connection, or non-Db2 data available through data virtualization. Before this data can be accessed by your business users, you can apply the governance and protection policies that are required to ensure the security of sensitive customer data.

2. Develop the model.

With transparent visibility and access to previously deployed models, business users can easily identify templates for their work to operate in a standard way.

In the context of IBM Cloud Pak for Data, you have visibility to the projects and experiments that are conducted by collaborators in your organization. With this visibility, tools (such as AutoAI) become increasingly useful. When conducting an experiment by using AutoAI, the tool automatically suggests the type of model that is best-suited for data sampling.

3. Validate the model.

By sourcing transactional data directly from runtimes, validation can be a rapid process that proves the accuracy of new models. Then, before those models are deployed, they can be equipped with the same governance rules to ensure their protection within the larger paradigm of the Data Fabric.

Validation is easily achieved by using AutoAI, which presents a leader board of available model pipelines with their corresponding accuracy scores and build times.

4. Infuse insights into an application.

After the model is ready for deployment, it can be called or deployed to the runtime by using various methods, including many APIs for IBM Z.

For example, regardless of the framework and platform that are used for training, a model can be deployed to Watson Machine Learning for z/OS on IBM Z where you can start model scoring against your runtime with CICS APIs, Java APIs, or REST APIs, depending on the specific runtime that is used. This flexibility allows you to gain the unique advantage of inferencing in-transaction, in real time to deliver significant value in critical, time-sensitive business areas, such as fraud detection.

5. Monitor the model.

With the governance rules that are established, the model can be continually evaluated. This evaluation process allows for iterative action as needed and governance to secure or report against the data that is used by the model.

Watson OpenScale is an available feature within IBM Cloud Pak for Data with which you can monitor and analyze deployed AI models with trust and transparency and understand the decision logic used by the model, which also promotes trustworthy AI.

For example, you can configure your deployments so that they are viewable through Watson OpenScale, which features a dashboard that explains the performance and accuracy drift of models to detect and mitigate model bias and drift.

6.3 Planning and implementing user authorization

Achieving the many benefits of DataOps and MLOps in a Data Fabric architecture is made possible only by allowing business users to access data with the suitable set of authorizations. In planning your Data Fabric, it is critical that you collect the needs of these users and plan for how they can be incorporated into your platform.

With IBM Cloud Pak for Data on IBM Z, you can orchestrate the accessibility of your data and analytics services with users across your business. It allows for collaboration through workflows, with automation and governance that is infused throughout to ensure consistent and secure self-service access.

6.3.1 Administrator responsibilities

As an administrator on IBM Cloud Pak for Data, you control the permissions that are provided to other platform users. When those users must be added, it is up to your discretion to decide the best approach.

We recommend that you make the following series of checks before you begin adding users to IBM Cloud Pak for Data:

- ▶ By default, IBM Cloud Pak for Data user records are stored in an internal repository. We recommend that you plan for an enterprise-grade password management solution; for example, SSO or LDAP.
- ▶ IBM Cloud Pak foundational services offer an Identity and Access Management (IAM) Service that can be used for this purpose.
- ▶ Whether you opt for the IAM Service, it is also worth considering whether you want to add all users in a specific LDAP group to a user group on IBM Cloud Pak for Data.
- ▶ To that end, you also need to ask whether the default roles meet your business requirements, and whether you want to orchestrate similar access requirements for a specific user group.

As an administrator, you have some flexibility in planning and adapting IBM Cloud Pak for Data for your specific requirements.

6.3.2 IBM Cloud Pak for Data planning roles

The role of a user in IBM Cloud Pak for Data defines their permissions, which vary based on the services that are installed on top of Cloud Pak for Data. Roles can also be assigned to groups of users. Users and groups can have multiple roles.

The following default roles are available:

- ▶ Administrator
- ▶ Business analyst
- ▶ Data engineer
- ▶ Data quality analyst
- ▶ Data scientist
- ▶ Data steward
- ▶ Developer
- ▶ Reporting administrator
- ▶ User

For more information about the permissions that are granted to each default role, and the services contributing to that permission, see this [IBM Documentation web page](#).

As an administrator, you can edit the permissions that are granted to default roles, or even create custom roles if the default settings do not align with your needs.

If you create a custom role, consider the following categories of permissions that are available on different services of the IBM Cloud Pak for Data product:

- ▶ Catalogs
- ▶ Data curation
- ▶ Deployments
- ▶ Governance artifacts
- ▶ Platform administration
- ▶ Projects
- ▶ Service instances
- ▶ User administration
- ▶ Vaults
- ▶ Workflows

For more information about the specific permissions that are available in each category, and the services contributing to that permission, see this [IBM Documentation web page](#).

A specific set of users in your organization often is assigned the same roles and levels of permissions. In that case, you can plan to create user groups to simplify the overall process. This is ideal for groups of users who will plan to collaborate and require the same level of access and visibility across each other's work.

When adding users to the platform, a record is generated for that user. You can manually create records for new users, or you can add LDAP users to a user group, which automatically generates a record for them.

You can also grant an entire LDAP group access to the platform by adding them as a user group, in which case records are automatically generated for each user.

After they are assigned a role in IBM Cloud Pak for Data, the user can access a catalog of available data and search for the assets that best fit their purpose.

6.4 Planning and implementing application access

Your Data Fabric architecture can be a boon for AI productivity. It improves AI/ML models that are based on newly connected data sources, accelerates project times, provides self-service access to key enterprise resources, and improves end-to-end orchestration of the AI lifecycle.

In this section, methods are described for incorporating your IBM Z data with applications and runtimes that are operating in a Data Fabric architecture.

6.4.1 Incorporating IBM Z data with applications and runtimes

Your IBM Z system might be producing a combination of structured, unstructured, and semi-structured data. In a well-governed Data Fabric, these data artifacts might exist alongside the various forms of distributed data that is produced by customer-facing applications, IoT devices, and edge computing resources.

Considering the inherent data gravity of IBM Z, a significant advantage can be gained in producing models that live as close to your transactions as possible, which allows you to achieve real-time insights that can create beneficial ripples across your technical system.

This section describes various ways to approach incorporating IBM Z data with your AI applications. For more information, see this [web page](#).

Deploying and running models on IBM Z without moving data

If you correctly delineated roles and responsibilities among your organization within IBM Cloud Pak for Data, the Data Steward (or some custom-made equivalent) is responsible for contributing IBM Z data to your data catalog, which can then be used by other members of the organization.

Two methods are available for connecting your IBM Z data for organizational consumption without moving it from the IBM Z platform (and thus risking the security of sensitive information):

- Your Data Steward (or some equivalent custom role) can enable connections to Db2 for z/OS and add it as an asset within the shared organizational catalog. This ability can be enabled with Db2 for z/OS Data Gate, or ODBC/JDBC connectors.

When the Data Stewards add an asset, they can enable the governing rules and protections that are necessary to disguise sensitive data from viewing or use. Then, business users can connect to these sources through the shared catalog, preview them, and use them in model development; for example, in the creation of a model with AutoAI.

Note: Db2 for z/OS is not a supported connector for quick scan or auto-discovery for use in auto classification and auto term assignments.

- If your organization must access IBM Z data that is *not* housed in a Db2 database, consider the use of Data Virtualization Manager. This manager allows the Data Steward to construct a virtualized table of IBM Z data, which (similar to the Db2 case) can be constructed with specific governing policies and protections to ensure security.

You might also connect distributed data resources into this virtualized table to plan for a model that requires data from outside the core runtimes. This virtualized table then can be accessed by business users in the planning and construction of their models.

Exposing IBM Z data through APIs

With IBM Z running in your Data Fabric architecture, opportunities might be available to use IBM Z data as exposed through APIs with Cloud Pak for Data.

When you want to access a resource that is in a z/OS subsystem, consider the use of REST APIs. This process is done by creating a service that defines the request or response mappings to the resource and designing a REST API to define the relevant HTTP actions.

z/OS Connect Enterprise Edition provides an API toolkit, with which you can design and create REST APIs for your services.



A

Unlock IBM Z data for your AI journey

This appendix is designed to quickly direct you toward helpful materials about how to unlock your IBM Z data to further your AI journey under the following framework:

- ▶ Keep the data where it originates and use IBM Z (that is, use AI accelerator on IBM z16 and Data Virtualization on z/OS)
- ▶ Use the Data Fabric architecture in CP4D to protect and govern IBM Z data

This appendix includes the following topics:

- ▶ “IBM Z data sources and Data Fabric credentials” on page 96
- ▶ “Creating an IBM DVM connection” on page 97
- ▶ “Discovering assets” on page 97
- ▶ “Adding users to the catalog with IBM Watson Knowledge Catalog” on page 98
- ▶ “IBM Watson Knowledge Catalog business terms” on page 98
- ▶ “Accessing data as a data scientist” on page 100
- ▶ “IBM Data Virtualization Manager for z/OS” on page 101
- ▶ “Build Model” on page 102
- ▶ “Deploying the model in production” on page 103

IBM Z data sources and Data Fabric credentials

This section provides an overview of the different ways of accessing data sources on IBM Z to integrate into a Data Fabric by using IBM Cloud Pak for Data.

Obtaining access credentials for IBM Data Virtualization Manager for z/OS

IBM Data Virtualization Manager for z/OS (DVM) technology provides various methods for client application to access data (see Table 1).

Table 1 Common Client access methods

Client types	Access method
C, C++ based applications	ODBC driver to DVM Server
Java applications	▶ JDBC driver to DVM Server ▶ JDBC driver to DVM Studio to DVM Server
▶ Mainframe applications ▶ Peer DVM instance Servers	IDF that uses DRDA
Mainframe applications	IDF that uses DRDA
Web browser (HTTP and HTTPS)	IBM z/OS Connect EE to DVM WOLA service provider
Web browser (HTTP)	JDBC Gateway to DRDA
ISPF	DVM Server
▶ RDBMS (DRDA Supported) ▶ Db2 distributed family ▶ Oracle (DRDA) ▶ SQL Server ▶ IBM Informix®	DRDA to DrDA AR to DVM Server
▶ RDBMS (non-DRDA) ▶ PostgreSQL ▶ Oracle ▶ Apache Hive ▶ SQL Server ▶ MySQL	▶ JDBC driver to JDBC Gateway Server ▶ JDBC Gateway Server to DVM DS Client ▶ DS Client to mainframe applications or Db2 UDTF

These access methods allow external applications to seamlessly make requests for READ or WRITE operations of underlying data, regardless of location or format.

For more information about access methods that are provided by IBM DVM, see this [IBM Support web page](#).

Administrator access to Cloud Pak for Data web console

After you access IBM Cloud Pak for Data as an administrator, several features and tasks are available to you; for example, managing users, managing secrets and vaults, and customizing the web client to meet specific organizational needs. For more information, see this [IBM Documentation web page](#).

For more information about the initialization and use of IBM Cloud Pak for Data as an administrator, see this [IBM Documentation web page](#).

Creating an IBM DVM connection

In IBM Cloud Pak for Data, complete the following steps to create an IBM DVM connection:

1. Browse to the **Connections** tab on the Platform connections page in the Cloud Pak for Data interface and then, select **New connection**.
2. Select **Data Virtualization Manager for z/OS** from the list.
3. Enter the name, description, hostname and IP address and port number that are associated with your instance of IBM DVM.

For more information about this process, see this [IBM Support web page](#).

Note: For more information about the process to establish a DVM connection, see “Creating a DVM connection in IBM Cloud Pak for Data” on page 37.

Adding data connections on IBM Cloud Pak for Data

Various data sources can be connected to your IBM Cloud Pak for Data Instance and made available for use in model development.

For information about this process, see Chapter 3, “Securely connect a Data Fabric to any data source, anywhere” on page 31.

Discovering assets

Data discovery is a key factor in enabling data governance. It includes analyzing metadata and transforming it into different forms for various use cases.

For more information, see Chapter 4, “Data governance and discovery” on page 51.

Navigating to discover assets

With IBM Watson Knowledge Catalog available on IBM Cloud Pak for Data, data assets can be discovered by using quick scans of tables or files to automatically uncover results, including a data quality score, data classes, and data quality information. This process is suitable for quickly analyzing many tables or files.

Assets also can be discovered by using automated discovery, which imports metadata and analysis results into your default catalog. This process is suitable for a smaller number of tables/files.

For more information, see this [IBM Documentation web page](#).

Adding users to the catalog with IBM Watson Knowledge Catalog

A key component of a successful Data Fabric is the enablement of a self-service model. By adding your organization's users as collaborators on IBM Watson Knowledge Catalog, you can drive a more democratized, self-sufficient analysis system.

For more information, see this [IBM Documentation web page](#).

Granting catalog permission

Various actions can be enabled for your collaborators on IBM Watson Knowledge Catalog, ranging from finding and viewing assets, to modifying, publishing, and deleting assets.

For more information, see this [IBM Documentation web page](#).

IBM Watson Knowledge Catalog business terms

In this section, we present common actions that you can consider when business terms are created in IBM Watson Knowledge Catalog.

Implementing a business glossary

You create business terms to standardize business concept definitions across your enterprise.

Business terms can describe the contents of the data, the sensitivity of the data, or other aspects of the data, such as the subject or purpose of the data. You can assign one or more business terms to individual columns in relational data sets, other governance artifacts, or data assets.

Table A-1 shows a snippet of a business glossary for a team that is working on InfoSphere data management.

Table A-1 Example business terms on InfoSphere Master Data Management Team

Key term	Definition
Data steward	The data management role that is tasked with maintaining and implementing the data governance policies that were agreed upon by the company. Data stewards are often the individuals that use the InfoSphere Implement MDM Inspector application to resolve data issues.
Data profiling	Enables users to view various aspects about the data (attributes) that are stored in their source's systems to better understand and improve the quality of the data. Specifically, you can determine which sources have or are creating bad or incomplete data, which ultimately affects your overall business operational and financial success.
Data manager	The data management role that is responsible for the project management aspects of a Customer Data Integration (CDI) or MDM implementation at an enterprise. This role manages stakeholder expectations, the team of data stewards, and the successes of the project from a cost and time perspective.
Data files	Files that are written by message broker interfaces that contain messages that are relevant to processing HL7 or XML messages; for example, input.dat, success.dat, and reject.dat.

For more information about business terms in IBM Watson Knowledge Catalog, see this IBM Documentation [web page](#).

Downloading business glossaries

Included business glossaries are provided by the Knowledge Accelerators. The vocabulary includes thousands of business terms in industry-specific vocabularies for the energy and utilities, financial services, healthcare, and insurance industries.

For more information, see this [IBM Documentation web page](#).

Importing categories

You can import categories into the IBM Watson Knowledge Catalog from a file that is in a CSV format. You can also export categories to a file that is in a CSV format. Then, this file can be modified outside of the catalog and later imported.

For more information, see this [IBM Documentation web page](#).

Importing business terms

You can use the business terms from the Knowledge Accelerators and import them in IBM Watson Knowledge Catalog. You also can import your predefined business terms in IBM Watson Knowledge Catalog from a CSV file format or create them directly on IBM Watson Knowledge Catalog and authorize different users to edit, create, or delete business terms from IBM Watson Knowledge Catalog.

For more information, see this [IBM Documentation web page](#).

Creating a rule

In IBM Watson Knowledge Catalog, you can create data protection rules or governance rules:

- ▶ A *governance rule* is a natural-language description of the criteria that are used to determine whether information assets are compliant with business objectives.
- ▶ A *data protection* rule specifies what data to control and how to control it.

For more information about data protection rules, see this [IBM Documentation web page](#).

For more information about governance rules, see this [IBM Documentation web page](#).

Creating a policy

A *policy* in IBM Watson Knowledge Catalog describes how data is to be used. It is based on the rules that you define.

You can design, structure, and establish policies separately and independently. Policies are artifacts that you can create, view, edit, rename, publish, delete, or import according to the assigned user permissions.

For more information, see this [IBM Documentation web page](#).

Running Automated Discovery

When planning to source data through Automated Discovery, it is essential to first understand which types of connections support this feature.

For more information, see this [IBM Documentation web page](#).

Adding rules to metadata

With IBM Cloud Pak for Data, data stewards can enrich their data assets with specific business rules; for example, to document ownership, quality standards, or reviews. In this way, data assets can be equipped for optimal use and protected as it is used across the model development process.

For more information, see this [IBM Cloud Pak for Data web page](#).

Accessing data as a data scientist

The following section describes several common actions that you can consider taking as a data scientist that is operating in a Data Fabric architecture.

Creating your AI project

In IBM Cloud Pak for Data, you can create analytics projects by using Watson Studio and IBM Watson Knowledge Catalog. This feature allows you to work with available data from your catalog and can be started as:

- ▶ An empty project that is created from scratch
- ▶ A project that uses sample data
- ▶ A previously exported project that you can import into the tool

For more information about this process, see this [IBM Documentation web page](#).

Importing assets from glossary

You can use imported glossary assets and data classes as governance artifacts in IBM Cloud Pak for Data.

For more information about adding cataloged assets to your analytics project, see this [IBM Documentation web page](#).

Checking asset details

You can check the details of your information assets in IBM Watson Knowledge Catalog.

For more information about this process, see this [IBM Documentation web page](#).

IBM Data Virtualization Manager for z/OS

With IBM DVM, you can make your IBM Z data available through IBM Cloud Pak for Data to authorized users in your organization.

In this section, we describe some key considerations and actions you can take in this process.

Mapping IBM Z data sources on IBM DVM

Several data sources are available that can be virtualized by using IBM DVM, including virtual storage access manager (VSAM), adaptable database system (ADABAS), IBM IMS/ESA® Database Manager, IBM Db2 for z/OS, and IBM System Management Facility (SMF).

For more information about how these resources can be mapped on IBM DVM, see *IBM Data Virtualization Manager for z/OS*, [SG24-8514](#).

Adding a data source by using IBM DVM connection

In IBM Cloud Pak for Data, adding a data source by using IBM DVM can be done by completing the following steps:

1. Access the Platform connection within the main menu drop-down menu.
2. Select **Data Virtualization Manager for z/OS**.
3. Configure the connection service name, hostname, port, and access credentials.

For more information, see 6.8.1 in *IBM Data Virtualization Manager for z/OS*, [SG24-8514](#).

Adding virtual tables or views to catalog

Your data assets can be connected to IBM Watson Knowledge Catalog for organizational consumption. Connections can be based on a relational table or view, a set of partitioned data files, or a file that is accessed through your data connection.

For more information about this process, see this [IBM Documentation web page](#).

Publishing virtualized IBM Z tables

After the connection is made, you can publish your virtualized IBM Z tables in IBM Watson Knowledge Catalog for use by authorized users in your organization. You might discover and publish one or multiple assets from a single connected data source.

For more information about this process, see this [IBM Documentation web page](#).

Accessing information by virtual tables

To access data, a virtual table or virtual view is created that maps to your source data and matches the definition of the source data structure on the mainframe.

From the virtual table or virtual view, you generate the SQL that is used to read and access the mapped data from the mainframe. The virtual tables can be created as part of IBM Open Data Analytics for z/OS.

For more information, see this [IBM Documentation web page](#).

Build Model

With IBM Watson Machine Learning for z/OS and Watson Studio, services and tools are available for the model building process.

AutoAI can be used to automatically select the suitable algorithms, produce pipelines candidates, train models and then, evaluate them by performance. The resulting product can be deployed or exported.

Alternatively, you can use notebooks and scripts to create your own feature engineering and model training process (by using Python or R), which allows you more control of the entire process.

IBM SPSS® Modeler flows can also be used for training, evaluation, and scoring models.

For more information, see this [IBM Documentation web page](#).

Creating a deployment space

A deployment space is used to deploy and manage your model deployments. By using such spaces, you can view and manage the activity and assets for your deployments.

A deployment space is *not* associated to a project, and assets from multiple projects can be deployed to a space. Likewise, you can deploy assets to more than one space.

Your deployment space can be used for different use cases: for example, one space to conduct deployment testing, and another to run production models.

To create a deployment space, complete the following steps:

1. Browse to **Deployments** in the main menu of IBM Cloud Pak for Data.
2. Click **New deployment space** and choose whether to create a space or import a space that was saved as a .zip file.

For more information, see this [IBM Documentation web page](#).

Creating a notebook

As an administrator or editor in IBM Cloud Pak for Data, the following options are available for creating notebooks for your projects:

- ▶ Creating a notebook file
- ▶ Copying a sample notebook from the Gallery
- ▶ Adding a notebook from the catalog

For more information about these methods, see this [IBM Documentation web page](#).

Training the model

The process of training a machine learning model involves providing the model with training data from which to learn.

The IBM Cloud Pak for Data platform provides an integration environment to access trusted data from a catalog for use in training machine learning models. IBM Watson Studio is an integrated platform that organizes your assets throughout this process, such as data sets, models, notebooks, and collaborators in a project.

For more information about this process, see this [IBM Documentation web page](#).

Deploying the model in production

Deploying models into a production environment is one of the final stages of the machine learning model development lifecycle. Watson Machine Learning for z/OS (WMLz) provides the tools that are needed to transition assets into deployment spaces (see Figure 1).

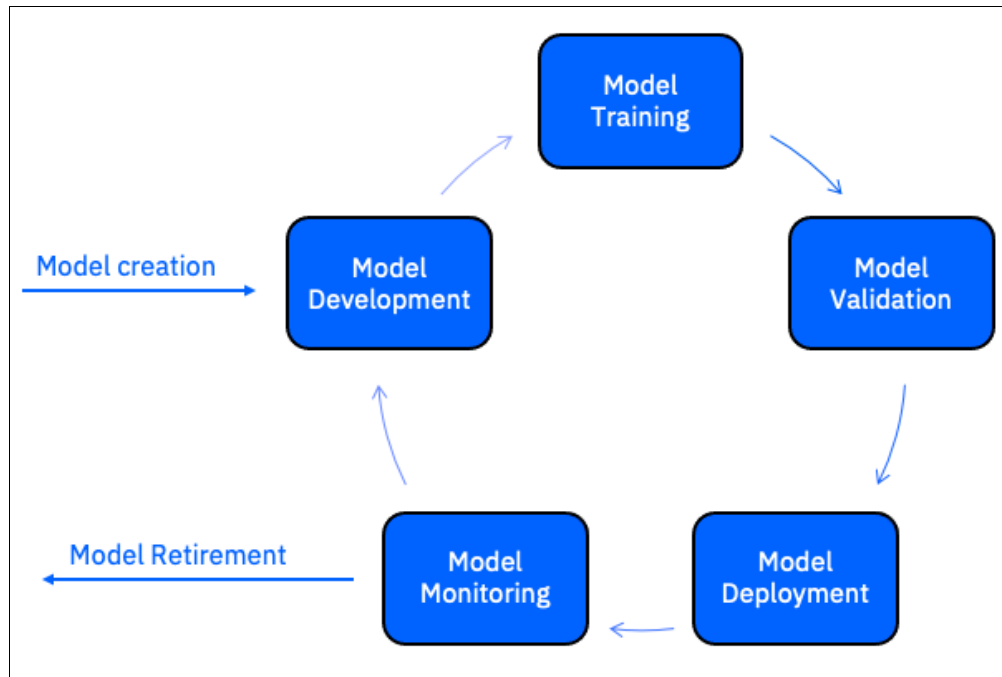


Figure 1 AI/ML Model lifecycle

Trusted AI requires the continuous evaluation and monitoring of the model for bias, drift, accuracy, and quality.

IBM offers a model risk management solution with IBM Watson OpenScale to monitor and measure the outcome of AI models across its lifecycle, which aids in the overall validation of models. It also helps detect and mitigate bias and drift in the deployed models to increase the accuracy of predictions, which fuels the adoption of trusted AI across enterprise applications.

For more information, see this [IBM Documentation web page](#).

Related publications

The publications that are listed in this section are considered particularly suitable for a more detailed discussion of the topics that are covered in this paper.

IBM Redbooks

The following IBM Redbooks publications provide more information about the topic in this document. Note that some publications referenced in this list might be available in softcopy only:

- ▶ *IBM Data Virtualization Manager for z/OS*, SG24-8514
- ▶ *Optimized Inferencing and Integration with AI on IBM Z Introduction, Methodology, and Use Cases*, REDP-5661
- ▶ *Demystifying Data with AI on IBM Z*, REDP-5633

You can search for, view, download, or order these documents and other Redbooks, Redpapers, Web Docs, draft, and additional materials, at the following website:

ibm.com/redbooks

Other publications

The following publications are also relevant as further information sources:

- ▶ Summary Translation: Quick Answer: Are Data Fabric and Data Mesh the Same or Different?:
<https://www.gartner.com/en/documents/4008708>
- ▶ Eliminate data silos: Query many systems as one, Data virtualization in IBM Cloud Pak for Data:
<https://www.ibm.com/downloads/cas/97AJPYNN>
- ▶ IBM Support White Paper, *Introduction to IBM Data Virtualization Manager for z/OS*:
<https://www.ibm.com/support/pages/introduction-ibm-data-virtualization-manager-zos>
- ▶ The AI Ladder - Demystifying AI Challenges:
<https://www.ibm.com/downloads/cas/01VADKY2>
- ▶ Enterprise Strategy Group white paper: *Intelligent Data Governance for a Trusted and Business-Ready Data Lake: Best Practices to Ensure Data Quality, Accessibility, and Security as a Foundation to an AI-centric Data Architecture*:
<https://www.ibm.com/downloads/cas/QL2MLX0V>
- ▶ Data Fabric and Z Series-Part 3: Fuel your Machine Learning models with data on zSystems:
https://medium.com/@wy_catherine/data-fabric-and-z-series-part-3-fuel-your-machine-learning-models-with-data-on-zsystems-9face3f18cf1

Online resources

The following websites are also relevant as further information sources:

- ▶ IBM Cloud Learn Hub - What is ETL (Extract, Transform, Load)?:
<https://www.ibm.com/cloud/learn/etl>
- ▶ Db2 Data Gate blog series Part 2: IBM Db2 for z/OS Data Gate: IBM Watson Knowledge Catalog integration:
<https://medium.com/icp-for-data/db2-data-gate-blog-series-part-2-ibm-db2-for-zo-s-data-gate-ibm-watson-knowledge-catalog-55b208a5c0f1>
- ▶ Product pages:
 - IBM Cloud Pak for Data:
<https://www.ibm.com/products/cloud-pak-for-data>
 - IBM Db2 Analytics Accelerator for z/OS:
<https://www.ibm.com/products/db2-analytics-accelerator>
 - IBM Db2 for z/OS Data Gate 2.1:
<https://www.ibm.com/products/db2-for-zos-data-gate>
 - IBM Data Virtualization Manager for z/OS:
<https://www.ibm.com/products/data-virtualization-manager-for-zos>
 - IBM Watson Studio:
<https://www.ibm.com/cloud/watson-studio>
 - IBM Watson Machine Learning for z/OS:
<https://www.ibm.com/products/machine-learning-for-zos>
 - IBM InfoSphere Classic Federation Server for z/OS:
<https://www.ibm.com/products/infosphere-classic-federation-server>
 - IBM InfoSphere Classic Change Data Capture for z/OS:
<https://www.ibm.com/products/infosphere-classic-change-data-capture>
 - IBM Watson Knowledge Catalog:
<https://www.ibm.com/cloud/watson-knowledge-catalog>
 - IBM Z Operational Log and Data Analytics:
<https://www.ibm.com/products/z-log-and-data-analytics>
- ▶ IBM Journey to AI Blog:
 - Augmented data management: Data fabric versus data mesh:
<https://www.ibm.com/blogs/journey-to-ai/2022/04/augmented-data-management-data-fabric-versus-data-mesh/>
 - Bringing automation and intelligence to data catalogs:
<https://www.ibm.com/blogs/journey-to-ai/2019/03/bringing-automation-and-intelligence-to-data-catalogs/>
 - Data governance: The importance of a modern machine learning knowledge catalog:
<https://www.ibm.com/blogs/journey-to-ai/2020/06/data-governance-the-importance-of-a-modern-machine-learning-knowledge-catalog/>

- ▶ IBM Support: IBM Data Replication Change Data Capture (CDC) Best Practices:
<https://www.ibm.com/support/pages/ibm-data-replication-change-data-capture-cdc-best-practices>
- ▶ IBM Db2 12 for z/OS product documentation: Introduction to Db2 data sharing:
<https://www.ibm.com/docs/en/db2-for-zos/12?topic=db2-data-sharing>
- ▶ IBM Db2 11.5 product documentation: High availability through clustering:
<https://www.ibm.com/docs/en/db2/11.5?topic=strategies-clustering>
- ▶ IBM Db2 Analytics Accelerator Leader for z/OS product documentation: Overview:
<https://www.ibm.com/docs/en/accel-loader/2.1.0?topic=welcome-overview>
- ▶ IBM Data Virtualization Manager for z/OS product documentation:
 - Supported data sources:
<https://www.ibm.com/docs/en/dvm/1.1.0?topic=overview-supported-data-sources>
 - The SQL solution:
https://www.ibm.com/docs/en/dvm/1.1.0?topic=SS4NKG_1.1.0/havuga10/topics/dvs_sql_con_configuring_SQL.html
 - The JDBC driver V3.1:
<https://www.ibm.com/docs/en/dvm/1.1.0?topic=applications-jdbc-driver-v31>
 - Connecting to a data source using JDBC:
<https://www.ibm.com/docs/en/dvm/1.1.0?topic=v31-connecting-data-source-using-jdbc>
- ▶ IBM Z and Cloud Modernization Center:
 - Improve risk and compliance management:
<https://www.ibm.com/community/z-and-cloud/value-driver/risk-and-compliance/>
 - AI and Analytics: Business Insight:
<https://www.ibm.com/community/z-and-cloud/use-case/ai-and-analytics-business-insights/>
- ▶ IBM Cloud Pak for Data 4.5.x product documentation:
 - Services:
<https://www.ibm.com/docs/en/cloud-paks/cp-data/4.5.x?topic=integrations-services>
 - Creating a Db2 Data Gate instance:
<https://www.ibm.com/docs/en/cloud-paks/cp-data/4.5.x?topic=setup-creating-db2-data-gate-instance>
 - Encrypting outbound network traffic from Db2 for z/OS to Db2 Data Gate:
<https://www.ibm.com/docs/en/cloud-paks/cp-data/4.5.x?topic=z-encrypting-outbound-network-traffic>
 - Discovering assets (Watson Knowledge Catalog):
<https://www.ibm.com/docs/en/cloud-paks/cp-data/4.5.x?topic=data-discovering-assets>
 - Running automated discovery (Watson Knowledge Catalog):
<https://www.ibm.com/docs/en/cloud-paks/cp-data/4.5.x?topic=assets-running-automated-discovery>

- Working with the discovered assets (Watson Knowledge Catalog):
<https://www.ibm.com/docs/en/cloud-paks/cp-data/4.0?topic=data-working-discovered-assets>
- Data load support:
<https://www.ibm.com/docs/en/cloud-paks/cp-data/4.5.x?topic=notebook-support-loading-data-in>
- Governance artifacts (Watson Knowledge Catalog):
<https://www.ibm.com/docs/en/cloud-paks/cp-data/4.5.x?topic=catalog-governance-artifacts>
- Designing data protection rules (Watson Knowledge Catalog):
<https://www.ibm.com/docs/en/cloud-paks/cp-data/4.5.x?topic=rules-designing-data-protection>
- Data protection rules enforcement (Watson Knowledge Catalog):
<https://www.ibm.com/docs/en/cloud-paks/cp-data/4.5.x?topic=rules-data-protection-enforcement>
- AutoAI Overview (Watson Machine Learning):
<https://www.ibm.com/docs/en/cloud-paks/cp-data/4.5.x?topic=models-autoai>
- Predefined roles and permissions in Cloud Pak for Data:
<https://www.ibm.com/docs/en/cloud-paks/cp-data/4.5.x?topic=users-predefined-roles-permissions-in-cloud-pak-data>
- IBM Architecture Center:
 - Data, Analytics and AI architecture:
<https://www.ibm.com/cloud/architecture/architectures/dataAIArchitecture>
 - Replicate IBM Z data:
<https://www.ibm.com/cloud/architecture/architectures/z-replicate-data-pattern/>
 - Cloud data lake:
<https://www.ibm.com/cloud/architecture/architectures/cloud-data-lake/>
- IBM Cloud Pak for Data Cloud docs:
 - Integrating with other cloud platforms:
<https://dataplatfom.cloud.ibm.com/docs/content/wsj/admin/int-cloud.html?context=analytics&audience=wdp>
 - Securing connections:
<https://dataplatfom.cloud.ibm.com/docs/content/wsj/manage-data/securingconn.html?audience=wdp>
 - Connection types:
https://dataplatfom.cloud.ibm.com/docs/content/wsj/manage-data/conn_types.html?context=analytics&audience=wdp
 - Connecting to data sources in Watson Query:
https://dataplatfom.cloud.ibm.com/docs/content/dvaas/data_source_configuration.html?context=analytics&audience=wdp

- Adding connections to projects:
<https://dataplatform.cloud.ibm.com/docs/content/wsj/manage-data/create-conn.html?audience=wdp>
- Adding data from a connection to a project:
<https://dataplatform.cloud.ibm.com/docs/content/wsj/manage-data/connected-data.html?audience=wdp>
- The Cloud Pak for Data as a Service data fabric solution:
<https://dataplatform.cloud.ibm.com/docs/content/wsj/getting-started/overview-cpdaas.html#data-fabric>
- ▶ Confidential computing on IBM Cloud:
<https://www.ibm.com/cloud/confidential-computing>
- ▶ IBM Blog - Go Beyond Confidential Computing with IBM Cloud Hyper Protect Services:
<https://www.ibm.com/cloud/blog/go-beyond-confidential-computing-with-ibm-cloud-hyper-protect-services>
- ▶ IBM Analytics:
 - What is a data lake?:
<https://www.ibm.com/topics/data-lake>
 - Data governance tools and solutions:
<https://www.ibm.com/analytics/data-governance>
 - DataOps platform solutions:
https://www.ibm.com/analytics/dataops?utm_content=SRCWW&p1=Search&p4=43700068092268571&p5=p&gclid=Cj0KCQjw1emWBhDUARIsAFp1rLUW_Nr5h9d8wemD0xkuwe0YwWkLQ7bj90Dzu94ktsYgKuZ4n1YhqaAaSH5EALw_wcB&gclidsrc=aw.ds
- ▶ IBM Db2 Big SQL 7.1 - Data governance:
<https://www.ibm.com/docs/en/db2-big-sql/7.1?topic=governance-data>
- ▶ IBM InfoSphere Information Server 11.7:
 - Metadata:
<https://www.ibm.com/docs/en/iis/11.7?topic=jobs-metadata>
 - Running data discovery:
<https://www.ibm.com/docs/en/iis/11.7?topic=data-running-discovery>
- ▶ IBM Cloud - Data Catalog:
<https://www.ibm.com/topics/data-catalog>

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